女娲补天视觉检测方案

RGB-D 球与柱识别、定位和仿真评测

数据重测时间：2026-10-11T04:05:22.393765+00:00；检测器 SHA-256：43c770b325bc8917。

评测结果

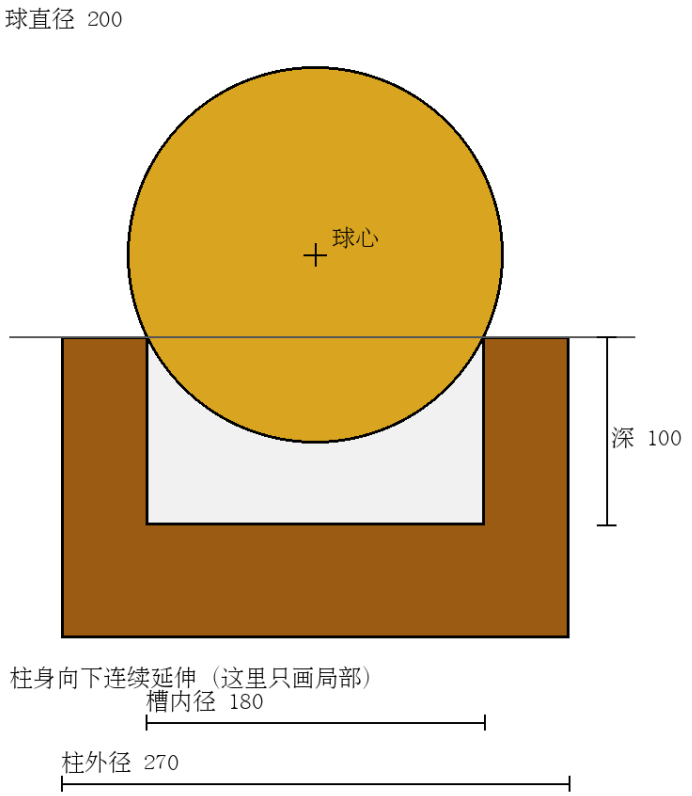
测试集	样本	球结果	定位精度	柱结果
Blender 正样本	60	60/60	中位 0.55 cm；≤5 cm 59/60	60/60；柱心中位 0.47 cm
负样本	81	无球误检 0/54	无柱球检出 27/27	无柱误检 0/51
鲁棒性集	192	179/192	中位 0.67 cm；≤5 cm 178/192	187/192

深度数据来自 Blender 场景射线求交，不是 D435i 实测。Gazebo 只有 60 张 RGB 视角图，因此不计入 RGB-D 成功率。

1 规则尺寸与输出坐标

对象/参数	定义
五色石球	直径 200 mm、半径 100 mm
中央柱体	外径 270 mm、外半径 135 mm
顶部凹槽	直径 180 mm、深 100 mm；柱身和槽沿连续
球心理论高度	杯口上方 √(100 − 90)=43.59 mm
输出	球心 XYZ；柱轴截面中心 X/Y 和杯口 Z，单位 m

球、柱与凹槽局部剖面（杯口 z=0；尺寸单位 mm）



接触几何

球半径 R=100；杯口半径 r=90

球心高于杯口：sqrt(R − r)=43.59

球底低于杯口：56.41；槽深：100

槽沿宽：(270 − 180)/2=45

霍夫圆与定位

球轮廓完整、接近圆：霍夫圆给候选

遮挡或投影变形：补充颜色、椭圆候选

球的最终位置：对齐深度与球面几何

柱的最终位置：竖直圆柱点云拟合

柱顶椭圆只能描述投影，不能代替柱轴

2 检测方法

RGB 生成多类候选：HSV
金黄色区域、连通域、局部椭圆和多阈值霍夫圆（param2=12/18/24）。柱位置不使用霍夫圆。候选最多保留 32 个；为颜色/椭圆候选保留最多 8
席，霍夫候选先按不同阈值的支持票数、再按边缘与颜色得分排列，后续逐个按深度和尺寸确认。

对候选中心读取 5×5 有效深度中值，按球半径 100 mm 得到中心射线距离，并用 fx×R/Z
计算物理投影半径。球颜色、有效深度和固定半径球面点集都在该圆盘内检查。有效点至少 20；固定球面 RMSE≤80 mm；自由拟合半径为 85 至 125 mm（靠中央杯口的可见球面下界放宽至 75 mm）；物理圆盘内有效深度率≥20%。

宽颜色掩膜 HSV 为 H=8 至 32、S≥65；高置信金黄色票为 H=18 至 35、S≥110、V≥20，物理圆盘内比例至少 8%。橙色传递区会通过宽掩膜，但高置信色相门控用来排除这类场地块。

中央柱以棕色候选 ROI 限定后，在场地坐标拟合竖直圆柱，外半径接近 135 mm、可见高度至少 220 mm。原始柱掩膜拟合失败时，以 9×9 中值滤波后的 RGB 重建候选并复用原几何门控。球体与柱体按规则尺寸做实体碰撞门控。颜色、边缘和霍夫中心不取平均；通过硬门控的候选按拟合残差、颜色支持、物理尺寸差及霍夫重复票数排序。

通过硬门控后，以较小的 S 为优：S = RMSE + 0.02×e_r + 0.0003×e_p − 0.002×c − 0.003×c_s − 0.0008×min(v,4)。RMSE 单位 m；e_r 是候选半径与物理投影半径的相对差；e_p 是球面拟合中心的重投影误差，单位 px；c、c_s 是宽颜色和高置信金黄色覆盖率；v
是霍夫支持票数。该分数只在通过门控的候选之间排序，不能抵消任何拒绝条件。球心采用自适应融合：RMSE≤8 mm 且自由拟合半径距 100 mm 不超过 20 mm 时，球面拟合中心占 70%、中心射线估计占 30%；其他合格候选分别占 10%、90%。在相机坐标中融合后再通过 TF 转到场地坐标。

状态判定：球心到柱轴水平误差≤35 mm，球底与杯口间隙相对理论值负 56.41 mm 的误差≤45 mm，输出 ON_COLUMN。检测到球但几何不符合为 OFF_COLUMN；深度/柱拟合/TF 不足为 UNKNOWN。

3 分视角与高度结果

球 3D 误差和柱 XY 误差的单位均为 cm；分母为该组全部图像。

视角	n	球检出	像素中位 px	球 3D 中位 cm	⩽5 cm	柱检出	柱 XY 中位 cm
level	12	12/12 (100.0%)	3.5	0.83	12/12	12/12	0.47
near	12	12/12 (100.0%)	2.5	0.42	12/12	12/12	0.31
oblique	12	12/12 (100.0%)	4.1	1.03	11/12	12/12	0.54
top	12	12/12 (100.0%)	9.3	0.80	12/12	12/12	1.06
upward	12	12/12 (100.0%)	1.7	0.51	12/12	12/12	0.41

高度 m	n	球检出	像素中位 px	球 3D 中位 cm	⩽5 cm	柱检出	柱 XY 中位 cm
0.85	12	12/12 (100.0%)	1.7	0.51	12/12	12/12	0.41
1.15	12	12/12 (100.0%)	2.5	0.42	12/12	12/12	0.31
1.5	12	12/12 (100.0%)	3.5	0.83	12/12	12/12	0.47
2.2	12	12/12 (100.0%)	4.1	1.03	11/12	12/12	0.54
3.5	12	12/12 (100.0%)	9.3	0.80	12/12	12/12	1.06

相机高度是光心相对世界地面的高度。当前 12 张/组的仿真结果只用于比较视角和高度，不代表 D435i 的实机最佳安装高度。

4 负样本和鲁棒性结果

no_ball：无球有柱 30 张；no_column：有球无柱 27 张；empty：球柱都无 24 张。no_column 中球仍放在中央柱名义位置上方，属于隐藏柱体的控制测试，不代表球在场地其他位置的实拍/仿真验证。

类别	总数	有球检出	无球误检	有柱检出	无柱误检
empty	24	0/0	0/24 (0.0%)	0/0	0/24 (0.0%)
no_ball	30	0/0	0/30 (0.0%)	30/30	0/0 (—)
no_column	27	27/27	0/0 (—)	0/0	0/27 (0.0%)

鲁棒条件	n	球检出	球 3D 中位 cm	⩽5 cm	柱检出
blur	20	19/20 (95.0%)	0.74	19/20	20/20
bright	20	20/20 (100.0%)	0.58	20/20	20/20
dark	12	12/12 (100.0%)	0.65	12/12	11/12
gamma	20	20/20 (100.0%)	0.51	20/20	20/20
glare	20	20/20 (100.0%)	0.68	20/20	20/20
noise	20	20/20 (100.0%)	0.67	20/20	20/20
normal	20	20/20 (100.0%)	0.51	20/20	20/20
occlusion	20	8/20 (40.0%)	0.47	8/20	16/20
side_crop	20	20/20 (100.0%)	1.95	20/20	20/20
tilt	20	20/20 (100.0%)	0.54	19/20	20/20

鲁棒视角	n	球检出	球 3D 中位 cm	⩽5 cm	柱检出
level	38	33/38 (86.8%)	0.69	32/38	38/38
near	36	36/36 (100.0%)	0.42	36/36	36/36
oblique	39	35/39 (89.7%)	1.07	35/39	39/39
top	40	36/40 (90.0%)	1.34	36/40	39/40
upward	39	39/39 (100.0%)	0.67	39/39	35/39

鲁棒高度 m	n	球检出	球 3D 中位 cm	⩽5 cm	柱检出
0.85	39	39/39 (100.0%)	0.67	39/39	35/39
1.15	36	36/36 (100.0%)	0.42	36/36	36/36
1.5	38	33/38 (86.8%)	0.69	32/38	38/38
2.2	39	35/39 (89.7%)	1.07	35/39	39/39
3.5	40	36/40 (90.0%)	1.34	36/40	39/40

5 视频坐标与连续帧处理

60 帧；球坐标有效 60 帧；球误差中位 0.61 cm，≤5 cm 59 帧；柱坐标有效 60 帧；速度门控拒绝 0 帧。坐标系：field。

序列由不同固定相机位姿按顺序组成，不是机器人连续运动录制。ON、OFF、UNK 分别对应 ON_COLUMN、OFF_COLUMN、UNKNOWN。下面分别统计视角和相机高度；逐帧表包含相机位置、球心估计、球心真值和柱截面圆心。漏检和不确定帧的 ROI/掩膜合图及分析表位于 output/media_runs/rgbd_video_verified/error_analysis。

视角	帧	坐标有效	球误差中位 cm	均值 cm	≤5 cm	柱 XY 中位 cm	速度拒绝
level	12	12/12 (100.0%)	0.83	0.95	12/12	0.48	0
near	12	12/12 (100.0%)	0.38	0.44	12/12	0.30	0
oblique	12	12/12 (100.0%)	0.55	1.32	11/12	0.58	0
top	12	12/12 (100.0%)	0.98	1.11	12/12	0.97	0
upward	12	12/12 (100.0%)	0.59	0.71	12/12	0.41	0

相机高度 m	帧	坐标有效	球误差中位 cm	均值 cm	≤5 cm	柱 XY 中位 cm	速度拒绝
0.85	12	12/12 (100.0%)	0.59	0.71	12/12	0.41	0
1.15	12	12/12 (100.0%)	0.38	0.44	12/12	0.30	0
1.5	12	12/12 (100.0%)	0.83	0.95	12/12	0.48	0
2.2	12	12/12 (100.0%)	0.55	1.32	11/12	0.58	0
3.5	12	12/12 (100.0%)	0.98	1.11	12/12	0.97	0

帧	来源/视角/高度/相机 XYZ m	状态	检出	球心 field XYZ m	球真值 XYZ m	柱心 field XYZ m	误差 m	门控/原因	
0	001_top_h3.50.png	top / h=3.50 m / cam=[4.50, 0.00, 3.50]	ON	是	[-0.008, -0.006, 1.745]	[0.000, 0.000, 1.744]	[0.008, 0.005, 1.700]	0.010	通过：rgbd_geometry_ok
1	002_top_h3.50.png	top / h=3.50 m / cam=[3.90, 2.25, 3.50]	ON	是	[0.002, -0.004, 1.745]	[0.000, 0.000, 1.744]	[0.005, 0.008, 1.700]	0.005	通过：rgbd_geometry_ok
2	003_top_h3.50.png	top / h=3.50 m / cam=[2.25, 3.90, 3.50]	ON	是	[0.001, -0.001, 1.745]	[0.000, 0.000, 1.744]	[-0.000, 0.009, 1.700]	0.002	通过：rgbd_geometry_ok
3	004_top_h3.50.png	top / h=3.50 m / cam=[0.00, 4.50, 3.50]	ON	是	[-0.029, 0.001, 1.738]	[0.000, 0.000, 1.744]	[-0.005, 0.009, 1.700]	0.029	通过：rgbd_geometry_ok
4	005_top_h3.50.png	top / h=3.50 m / cam=[-2.25, 3.90, 3.50]	ON	是	[0.001, 0.000, 1.745]	[0.000, 0.000, 1.744]	[-0.008, 0.005, 1.700]	0.002	通过：rgbd_geometry_ok
5	006_top_h3.50.png	top / h=3.50 m / cam=[-3.90, 2.25, 3.50]	ON	是	[-0.013, -0.010, 1.729]	[0.000, 0.000, 1.744]	[-0.010, 0.000, 1.700]	0.022	通过：rgbd_geometry_ok
6	007_top_h3.50.png	top / h=3.50 m / cam=[-4.50, 0.00, 3.50]	ON	是	[0.007, 0.002, 1.750]	[0.000, 0.000, 1.744]	[-0.009, -0.005, 1.700]	0.010	通过：rgbd_geometry_ok
7	008_top_h3.50.png	top / h=3.50 m / cam=[-3.90, -2.25, 3.50]	ON	是	[-0.009, 0.008, 1.732]	[0.000, 0.000, 1.744]	[-0.006, -0.008, 1.700]	0.016	通过：rgbd_geometry_ok
8	009_top_h3.50.png	top / h=3.50 m / cam=[-2.25, -3.90, 3.50]	ON	是	[-0.002, -0.001, 1.743]	[0.000, 0.000, 1.744]	[-0.000, -0.010, 1.700]	0.003	通过：rgbd_geometry_ok
9	010_top_h3.50.png	top / h=3.50 m / cam=[-0.00, -4.50, 3.50]	ON	是	[0.001, 0.009, 1.754]	[0.000, 0.000, 1.744]	[0.005, -0.008, 1.700]	0.014	通过：rgbd_geometry_ok
10	011_top_h3.50.png	top / h=3.50 m / cam=[2.25, -3.90, 3.50]	ON	是	[-0.002, 0.001, 1.748]	[0.000, 0.000, 1.744]	[0.008, -0.005, 1.700]	0.005	通过：rgbd_geometry_ok

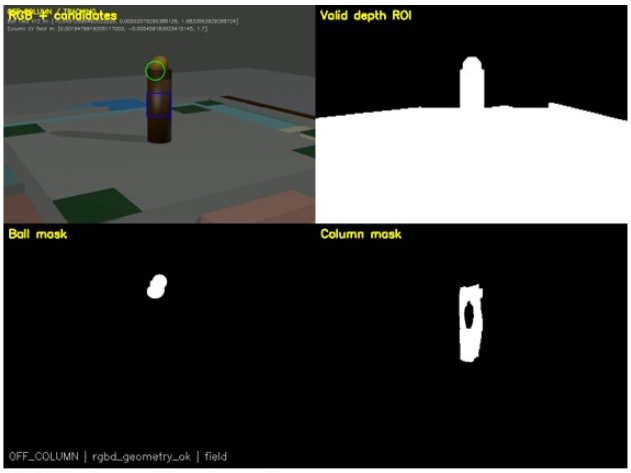
帧	来源/视角/高度/相机 XYZ m	状态	检出	球心 field XYZ m	球真值 XYZ m	柱心 field XYZ m	误差 m	门控/原因	
11	012_top_h3.50.png	top / h=3.50 m / cam=[3.90, -2.25, 3.50]	ON	是	[-0.005, 0.008, 1.755]	[0.000, 0.000, 1.744]	[0.009, 0.000, 1.700]	0.014	通过: rgbd_geometry_ok
12	013_oblique_h2.20.png	oblique / h=2.20 m / cam=[4.50, 0.00, 2.20]	ON	是	[0.004, -0.001, 1.741]	[0.000, 0.000, 1.744]	[0.003, 0.004, 1.700]	0.005	通过: rgbd_geometry_ok
13	014_oblique_h2.20.png	oblique / h=2.20 m / cam=[3.90, 2.25, 2.20]	ON	是	[0.004, 0.001, 1.741]	[0.000, 0.000, 1.744]	[0.000, 0.005, 1.700]	0.005	通过: rgbd_geometry_ok
14	015_oblique_h2.20.png	oblique / h=2.20 m / cam=[2.25, 3.90, 2.20]	ON	是	[0.000, 0.004, 1.737]	[0.000, 0.000, 1.744]	[-0.002, 0.005, 1.700]	0.008	通过: rgbd_geometry_ok
15	016_oblique_h2.20.png	oblique / h=2.20 m / cam=[0.00, 4.50, 2.20]	ON	是	[-0.002, 0.004, 1.744]	[0.000, 0.000, 1.744]	[-0.004, 0.004, 1.700]	0.004	通过: rgbd_geometry_ok
16	017_oblique_h2.20.png	oblique / h=2.20 m / cam=[-2.25, 3.90, 2.20]	ON	是	[-0.004, 0.002, 1.744]	[0.000, 0.000, 1.744]	[-0.006, 0.001, 1.700]	0.004	通过: rgbd_geometry_ok
17	018_oblique_h2.20.png	oblique / h=2.20 m / cam=[-3.90, 2.25, 2.20]	ON	是	[-0.003, 0.003, 1.737]	[0.000, 0.000, 1.744]	[-0.005, -0.002, 1.700]	0.008	通过: rgbd_geometry_ok
18	019_oblique_h2.20.png	oblique / h=2.20 m / cam=[-4.50, 0.00, 2.20]	ON	是	[-0.004, -0.002, 1.741]	[0.000, 0.000, 1.744]	[-0.004, -0.004, 1.700]	0.006	通过: rgbd_geometry_ok
19	020_oblique_h2.20.png	oblique / h=2.20 m / cam=[-3.90, -2.25, 2.20]	ON	是	[-0.004, -0.001, 1.744]	[0.000, 0.000, 1.744]	[-0.001, -0.006, 1.700]	0.004	通过: rgbd_geometry_ok
20	021_oblique_h2.20.png	oblique / h=2.20 m / cam=[-2.25, -3.90, 2.20]	OFF	是	[-0.049, 0.005, 1.683]	[0.000, 0.000, 1.744]	[0.002, -0.005, 1.700]	0.078	通过: rgbd_geometry_ok
21	022_oblique_h2.20.png	oblique / h=2.20 m / cam=[-0.00, -4.50, 2.20]	ON	是	[-0.001, 0.000, 1.759]	[0.000, 0.000, 1.744]	[0.004, -0.004, 1.700]	0.016	通过: rgbd_geometry_ok
22	023_oblique_h2.20.png	oblique / h=2.20 m / cam=[2.25, -3.90, 2.20]	ON	是	[0.001, -0.004, 1.741]	[0.000, 0.000, 1.744]	[0.006, -0.001, 1.700]	0.005	通过: rgbd_geometry_ok
23	024_oblique_h2.20.png	oblique / h=2.20 m / cam=[3.90, -2.25, 2.20]	ON	是	[-0.004, -0.007, 1.756]	[0.000, 0.000, 1.744]	[0.005, 0.002, 1.700]	0.015	通过: rgbd_geometry_ok
24	025_level_h1.50.png	level / h=1.50 m / cam=[4.50, 0.00, 1.50]	ON	是	[0.005, -0.001, 1.738]	[0.000, 0.000, 1.744]	[0.002, 0.005, 1.700]	0.007	通过: rgbd_geometry_ok
25	026_level_h1.50.png	level / h=1.50 m / cam=[3.90, 2.25, 1.50]	ON	是	[0.005, 0.001, 1.738]	[0.000, 0.000, 1.744]	[-0.001, 0.005, 1.700]	0.007	通过: rgbd_geometry_ok
26	027_level_h1.50.png	level / h=1.50 m / cam=[2.25, 3.90, 1.50]	ON	是	[0.001, 0.005, 1.738]	[0.000, 0.000, 1.744]	[-0.003, 0.004, 1.700]	0.007	通过: rgbd_geometry_ok

帧	来源/视角/高度/相机 XYZ m	状态	检出	球心 field XYZ m	球真值 XYZ m	柱心 field XYZ m	误差 m	门控/原因	
27	028_level_h1.50.png	level / h=1.50 m / cam=[0.00, 4.50, 1.50]	ON	是	[−0.008, 0.004, 1.732]	[0.000, 0.000, 1.744]	[−0.005, 0.002, 1.700]	0.015	通过: rgbd_geometry_ok
28	029_level_h1.50.png	level / h=1.50 m / cam=[−2.25, 3.90, 1.50]	ON	是	[−0.001, 0.005, 1.745]	[0.000, 0.000, 1.744]	[−0.005, −0.001, 1.700]	0.005	通过: rgbd_geometry_ok
29	030_level_h1.50.png	level / h=1.50 m / cam=[−3.90, 2.25, 1.50]	ON	是	[−0.005, 0.001, 1.738]	[0.000, 0.000, 1.744]	[−0.004, −0.003, 1.700]	0.007	通过: rgbd_geometry_ok
30	031_level_h1.50.png	level / h=1.50 m / cam=[−4.50, 0.00, 1.50]	ON	是	[−0.004, 0.001, 1.735]	[0.000, 0.000, 1.744]	[−0.002, −0.004, 1.700]	0.010	通过: rgbd_geometry_ok
31	032_level_h1.50.png	level / h=1.50 m / cam=[−3.90, −2.25, 1.50]	ON	是	[−0.003, −0.004, 1.738]	[0.000, 0.000, 1.744]	[0.001, −0.005, 1.700]	0.007	通过: rgbd_geometry_ok
32	033_level_h1.50.png	level / h=1.50 m / cam=[−2.25, −3.90, 1.50]	ON	是	[−0.003, −0.003, 1.755]	[0.000, 0.000, 1.744]	[0.003, −0.004, 1.700]	0.012	通过: rgbd_geometry_ok
33	034_level_h1.50.png	level / h=1.50 m / cam=[−0.00, −4.50, 1.50]	ON	是	[−0.001, −0.004, 1.758]	[0.000, 0.000, 1.744]	[0.004, −0.002, 1.700]	0.015	通过: rgbd_geometry_ok
34	035_level_h1.50.png	level / h=1.50 m / cam=[2.25, −3.90, 1.50]	ON	是	[−0.002, −0.005, 1.755]	[0.000, 0.000, 1.744]	[0.005, 0.001, 1.700]	0.013	通过: rgbd_geometry_ok
35	036_level_h1.50.png	level / h=1.50 m / cam=[3.90, −2.25, 1.50]	ON	是	[0.001, −0.006, 1.752]	[0.000, 0.000, 1.744]	[0.004, 0.003, 1.700]	0.010	通过: rgbd_geometry_ok
36	037_upward_h0.85.png	upward / h=0.85 m / cam=[3.50, 0.00, 0.85]	ON	是	[0.005, −0.006, 1.746]	[0.000, 0.000, 1.744]	[0.002, 0.004, 1.700]	0.008	通过: rgbd_geometry_ok
37	038_upward_h0.85.png	upward / h=0.85 m / cam=[3.03, 1.75, 0.85]	ON	是	[0.003, 0.001, 1.738]	[0.000, 0.000, 1.744]	[0.000, 0.004, 1.700]	0.007	通过: rgbd_geometry_ok
38	039_upward_h0.85.png	upward / h=0.85 m / cam=[1.75, 3.03, 0.85]	ON	是	[0.003, 0.003, 1.741]	[0.000, 0.000, 1.744]	[−0.002, 0.004, 1.700]	0.005	通过: rgbd_geometry_ok
39	040_upward_h0.85.png	upward / h=0.85 m / cam=[0.00, 3.50, 0.85]	ON	是	[0.001, 0.004, 1.741]	[0.000, 0.000, 1.744]	[−0.004, 0.002, 1.700]	0.005	通过: rgbd_geometry_ok
40	041_upward_h0.85.png	upward / h=0.85 m / cam=[−1.75, 3.03, 0.85]	ON	是	[−0.001, 0.003, 1.738]	[0.000, 0.000, 1.744]	[−0.004, 0.000, 1.700]	0.007	通过: rgbd_geometry_ok
41	042_upward_h0.85.png	upward / h=0.85 m / cam=[−3.03, 1.75, 0.85]	ON	是	[−0.002, 0.008, 1.748]	[0.000, 0.000, 1.744]	[−0.003, −0.002, 1.700]	0.009	通过: rgbd_geometry_ok
42	043_upward_h0.85.png	upward / h=0.85 m / cam=[−3.50, 0.00, 0.85]	ON	是	[−0.001, 0.003, 1.730]	[0.000, 0.000, 1.744]	[−0.002, −0.004, 1.700]	0.014	通过: rgbd_geometry_ok

帧	来源/视角/高度/相机 XYZ m	状态	检出	球心 field XYZ m	球真值 XYZ m	柱心 field XYZ m	误差 m	门控/原因	
43	044_upward_h0.85.png	upward / h=0.85 m / cam=[-3.03, -1.75, 0.85]	ON	是	[-0.003, -0.004, 1.743]	[0.000, 0.000, 1.744]	[0.000, -0.004, 1.700]	0.005	通过: rgbd_geometry_ok
44	045_upward_h0.85.png	upward / h=0.85 m / cam=[-1.75, -3.03, 0.85]	ON	是	[-0.008, -0.002, 1.751]	[0.000, 0.000, 1.744]	[0.002, -0.003, 1.700]	0.011	通过: rgbd_geometry_ok
45	046_upward_h0.85.png	upward / h=0.85 m / cam=[-0.00, -3.50, 0.85]	ON	是	[-0.001, -0.004, 1.741]	[0.000, 0.000, 1.744]	[0.004, -0.002, 1.700]	0.005	通过: rgbd_geometry_ok
46	047_upward_h0.85.png	upward / h=0.85 m / cam=[1.75, -3.03, 0.85]	ON	是	[0.001, -0.004, 1.741]	[0.000, 0.000, 1.744]	[0.004, -0.000, 1.700]	0.005	通过: rgbd_geometry_ok
47	048_upward_h0.85.png	upward / h=0.85 m / cam=[3.03, -1.75, 0.85]	ON	是	[0.003, -0.003, 1.741]	[0.000, 0.000, 1.744]	[0.004, 0.002, 1.700]	0.005	通过: rgbd_geometry_ok
48	049_near_h1.15.png	near / h=1.15 m / cam=[2.50, 0.00, 1.15]	ON	是	[0.003, 0.001, 1.739]	[0.000, 0.000, 1.744]	[0.002, 0.003, 1.700]	0.006	通过: rgbd_geometry_ok
49	050_near_h1.15.png	near / h=1.15 m / cam=[2.17, 1.25, 1.15]	ON	是	[0.003, 0.001, 1.739]	[0.000, 0.000, 1.744]	[0.000, 0.003, 1.700]	0.006	通过: rgbd_geometry_ok
50	051_near_h1.15.png	near / h=1.15 m / cam=[1.25, 2.17, 1.15]	ON	是	[0.002, 0.002, 1.742]	[0.000, 0.000, 1.744]	[-0.001, 0.003, 1.700]	0.003	通过: rgbd_geometry_ok
51	052_near_h1.15.png	near / h=1.15 m / cam=[0.00, 2.50, 1.15]	ON	是	[0.002, 0.003, 1.739]	[0.000, 0.000, 1.744]	[-0.003, 0.002, 1.700]	0.006	通过: rgbd_geometry_ok
52	053_near_h1.15.png	near / h=1.15 m / cam=[-1.25, 2.17, 1.15]	ON	是	[-0.003, 0.002, 1.744]	[0.000, 0.000, 1.744]	[-0.003, 0.000, 1.700]	0.004	通过: rgbd_geometry_ok
53	054_near_h1.15.png	near / h=1.15 m / cam=[-2.17, 1.25, 1.15]	ON	是	[-0.003, 0.001, 1.742]	[0.000, 0.000, 1.744]	[-0.003, -0.001, 1.700]	0.003	通过: rgbd_geometry_ok
54	055_near_h1.15.png	near / h=1.15 m / cam=[-2.50, 0.00, 1.15]	ON	是	[-0.003, -0.005, 1.741]	[0.000, 0.000, 1.744]	[-0.002, -0.003, 1.700]	0.006	通过: rgbd_geometry_ok
55	056_near_h1.15.png	near / h=1.15 m / cam=[-2.17, -1.25, 1.15]	ON	是	[-0.003, -0.001, 1.742]	[0.000, 0.000, 1.744]	[-0.000, -0.003, 1.700]	0.003	通过: rgbd_geometry_ok
56	057_near_h1.15.png	near / h=1.15 m / cam=[-1.25, -2.17, 1.15]	ON	是	[-0.001, -0.003, 1.742]	[0.000, 0.000, 1.744]	[0.001, -0.003, 1.700]	0.003	通过: rgbd_geometry_ok
57	058_near_h1.15.png	near / h=1.15 m / cam=[-0.00, -2.50, 1.15]	ON	是	[-0.001, -0.004, 1.746]	[0.000, 0.000, 1.744]	[0.003, -0.002, 1.700]	0.005	通过: rgbd_geometry_ok
58	059_near_h1.15.png	near / h=1.15 m / cam=[1.25, -2.17, 1.15]	ON	是	[0.001, -0.003, 1.744]	[0.000, 0.000, 1.744]	[0.003, -0.000, 1.700]	0.003	通过: rgbd_geometry_ok

帧	来源/视角/高度/相机 XYZ m	状态	检出	球心 field XYZ m	球真值 XYZ m	柱心 field XYZ m	误差 m	门控/原因	
59	060_near_h1.15.png	near / h=1.15 m / cam=[2.17, -1.25, 1.15]	ON	是	[0.002, -0.004, 1.744]	[0.000, 0.000, 1.744]	[0.003, 0.001, 1.700]	0.004	通过: rgbd_geometry_ok

视频有 1 帧球心误差超过 5 cm，该帧同时误判为 OFF_COLUMN；诊断图和 accuracy_errors.csv 已按定位偏差与状态错误归档。



000020_diagnostic_roi_masks.png

柱掩膜去噪消融

6 种策略分别测试相同的 333 张 RGB-D，共 1998 次柱体拟合。只改变 RGB 柱候选去噪，圆柱几何门控保持一致。原拟合失败才使用中值 9×9 备用候选。

分集合

方法	集合/条件/视角/高度	n	有柱检出	无柱误检	柱 XY 中位 cm
原掩膜	positive	60	60/60	—	0.47
中值3×3	positive	60	60/60	—	0.47
中值5×5	positive	60	60/60	—	0.48
中值7×7	positive	60	60/60	—	0.48
中值9×9	positive	60	60/60	—	0.48
失败后中值9×9	positive	60	60/60	—	0.47
原掩膜	negative	81	30/30	0/51	0.52
中值3×3	negative	81	30/30	0/51	0.52
中值5×5	negative	81	30/30	0/51	0.51
中值7×7	negative	81	30/30	0/51	0.50
中值9×9	negative	81	30/30	0/51	0.50
失败后中值9×9	negative	81	30/30	0/51	0.52
原掩膜	robust	192	167/192	—	0.48
中值3×3	robust	192	167/192	—	0.48
中值5×5	robust	192	173/192	—	0.48
中值7×7	robust	192	183/192	—	0.48
中值9×9	robust	192	187/192	—	0.48
失败后中值9×9	robust	192	187/192	—	0.48

分鲁棒条件

方法	集合/条件/视角/高度	n	有柱检出	无柱误检	柱 XY 中位 cm
原掩膜	robust / blur	20	20/20	—	0.48
中值3×3	robust / blur	20	20/20	—	0.48
中值5×5	robust / blur	20	20/20	—	0.48
中值7×7	robust / blur	20	20/20	—	0.48
中值9×9	robust / blur	20	20/20	—	0.48
失败后中值9×9	robust / blur	20	20/20	—	0.48
原掩膜	robust / bright	20	20/20	—	0.48
中值3×3	robust / bright	20	20/20	—	0.48
中值5×5	robust / bright	20	20/20	—	0.48
中值7×7	robust / bright	20	20/20	—	0.48
中值9×9	robust / bright	20	20/20	—	0.48
失败后中值9×9	robust / bright	20	20/20	—	0.48
原掩膜	robust / dark	12	11/12	—	0.61
中值3×3	robust / dark	12	11/12	—	0.61
中值5×5	robust / dark	12	11/12	—	0.60
中值7×7	robust / dark	12	11/12	—	0.61
中值9×9	robust / dark	12	11/12	—	0.63
失败后中值9×9	robust / dark	12	11/12	—	0.61
原掩膜	robust / gamma	20	20/20	—	0.48
中值3×3	robust / gamma	20	20/20	—	0.48
中值5×5	robust / gamma	20	20/20	—	0.49
中值7×7	robust / gamma	20	20/20	—	0.48
中值9×9	robust / gamma	20	20/20	—	0.48
失败后中值9×9	robust / gamma	20	20/20	—	0.48
原掩膜	robust / glare	20	20/20	—	0.48
中值3×3	robust / glare	20	20/20	—	0.48
中值5×5	robust / glare	20	20/20	—	0.48
中值7×7	robust / glare	20	20/20	—	0.48
中值9×9	robust / glare	20	20/20	—	0.48
失败后中值9×9	robust / glare	20	20/20	—	0.48
原掩膜	robust / noise	20	0/20	—	—
中值3×3	robust / noise	20	0/20	—	—
中值5×5	robust / noise	20	6/20	—	1.03
中值7×7	robust / noise	20	16/20	—	0.52
中值9×9	robust / noise	20	20/20	—	0.51
失败后中值9×9	robust / noise	20	20/20	—	0.51

方法	集合/条件/视角/高度	n	有柱检出	无柱误检	柱 XY 中位 cm
原掩膜	robust / normal	20	20/20	—	0.48
中值3×3	robust / normal	20	20/20	—	0.47
中值5×5	robust / normal	20	20/20	—	0.48
中值7×7	robust / normal	20	20/20	—	0.48
中值9×9	robust / normal	20	20/20	—	0.48
失败后中值9×9	robust / normal	20	20/20	—	0.48
原掩膜	robust / occlusion	20	16/20	—	0.50
中值3×3	robust / occlusion	20	16/20	—	0.49
中值5×5	robust / occlusion	20	16/20	—	0.49
中值7×7	robust / occlusion	20	16/20	—	0.50
中值9×9	robust / occlusion	20	16/20	—	0.50
失败后中值9×9	robust / occlusion	20	16/20	—	0.50
原掩膜	robust / side_crop	20	20/20	—	0.48
中值3×3	robust / side_crop	20	20/20	—	0.48
中值5×5	robust / side_crop	20	20/20	—	0.48
中值7×7	robust / side_crop	20	20/20	—	0.47
中值9×9	robust / side_crop	20	20/20	—	0.47
失败后中值9×9	robust / side_crop	20	20/20	—	0.48
原掩膜	robust / tilt	20	20/20	—	0.47
中值3×3	robust / tilt	20	20/20	—	0.46
中值5×5	robust / tilt	20	20/20	—	0.46
中值7×7	robust / tilt	20	20/20	—	0.46
中值9×9	robust / tilt	20	20/20	—	0.46
失败后中值9×9	robust / tilt	20	20/20	—	0.47

分鲁棒视角

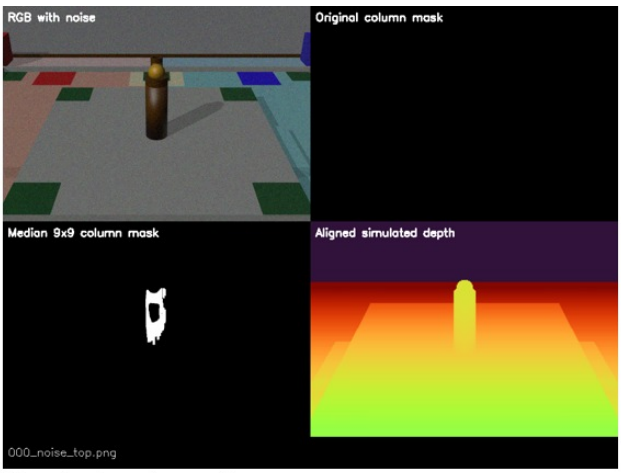
方法	集合/条件/视角/高度	n	有柱检出	无柱误检	柱 XY 中位 cm
原掩膜	robust / top	40	35/40	—	1.05
中值3×3	robust / top	40	35/40	—	1.07
中值5×5	robust / top	40	35/40	—	1.09
中值7×7	robust / top	40	37/40	—	1.12
中值9×9	robust / top	40	39/40	—	1.14
失败后中值9×9	robust / top	40	39/40	—	1.06
原掩膜	robust / oblique	39	35/39	—	0.55
中值3×3	robust / oblique	39	35/39	—	0.55
中值5×5	robust / oblique	39	35/39	—	0.55
中值7×7	robust / oblique	39	37/39	—	0.55
中值9×9	robust / oblique	39	39/39	—	0.57
失败后中值9×9	robust / oblique	39	39/39	—	0.55
原掩膜	robust / level	38	34/38	—	0.48
中值3×3	robust / level	38	34/38	—	0.47
中值5×5	robust / level	38	34/38	—	0.48
中值7×7	robust / level	38	38/38	—	0.48
中值9×9	robust / level	38	38/38	—	0.48
失败后中值9×9	robust / level	38	38/38	—	0.48
原掩膜	robust / upward	39	31/39	—	0.40
中值3×3	robust / upward	39	31/39	—	0.40
中值5×5	robust / upward	39	34/39	—	0.40
中值7×7	robust / upward	39	35/39	—	0.40
中值9×9	robust / upward	39	35/39	—	0.40
失败后中值9×9	robust / upward	39	35/39	—	0.40
原掩膜	robust / near	36	32/36	—	0.29
中值3×3	robust / near	36	32/36	—	0.29
中值5×5	robust / near	36	35/36	—	0.29
中值7×7	robust / near	36	36/36	—	0.29
中值9×9	robust / near	36	36/36	—	0.29
失败后中值9×9	robust / near	36	36/36	—	0.29

分鲁棒高度

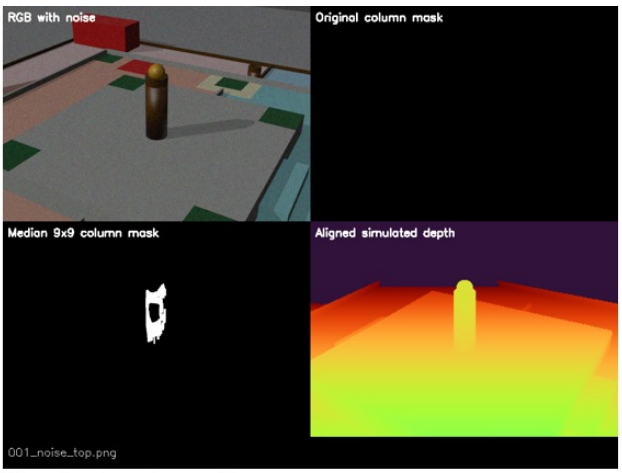
方法	集合/条件/视角/高度	n	有柱检出	无柱误检	柱 XY 中位 cm
原掩膜	robust / 3.5	40	35/40	—	1.05
中值3×3	robust / 3.5	40	35/40	—	1.07

方法	集合/条件/视角/高度	n	有柱检出	无柱误检	柱 XY 中位 cm
中值5×5	robust / 3.5	40	35/40	—	1.09
中值7×7	robust / 3.5	40	37/40	—	1.12
中值9×9	robust / 3.5	40	39/40	—	1.14
失败后中值9×9	robust / 3.5	40	39/40	—	1.06
原掩膜	robust / 2.2	39	35/39	—	0.55
中值3×3	robust / 2.2	39	35/39	—	0.55
中值5×5	robust / 2.2	39	35/39	—	0.55
中值7×7	robust / 2.2	39	37/39	—	0.55
中值9×9	robust / 2.2	39	39/39	—	0.57
失败后中值9×9	robust / 2.2	39	39/39	—	0.55
原掩膜	robust / 1.5	38	34/38	—	0.48
中值3×3	robust / 1.5	38	34/38	—	0.47
中值5×5	robust / 1.5	38	34/38	—	0.48
中值7×7	robust / 1.5	38	38/38	—	0.48
中值9×9	robust / 1.5	38	38/38	—	0.48
失败后中值9×9	robust / 1.5	38	38/38	—	0.48
原掩膜	robust / 0.85	39	31/39	—	0.40
中值3×3	robust / 0.85	39	31/39	—	0.40
中值5×5	robust / 0.85	39	34/39	—	0.40
中值7×7	robust / 0.85	39	35/39	—	0.40
中值9×9	robust / 0.85	39	35/39	—	0.40
失败后中值9×9	robust / 0.85	39	35/39	—	0.40
原掩膜	robust / 1.15	36	32/36	—	0.29
中值3×3	robust / 1.15	36	32/36	—	0.29
中值5×5	robust / 1.15	36	35/36	—	0.29
中值7×7	robust / 1.15	36	36/36	—	0.29
中值9×9	robust / 1.15	36	36/36	—	0.29
失败后中值9×9	robust / 1.15	36	36/36	—	0.29

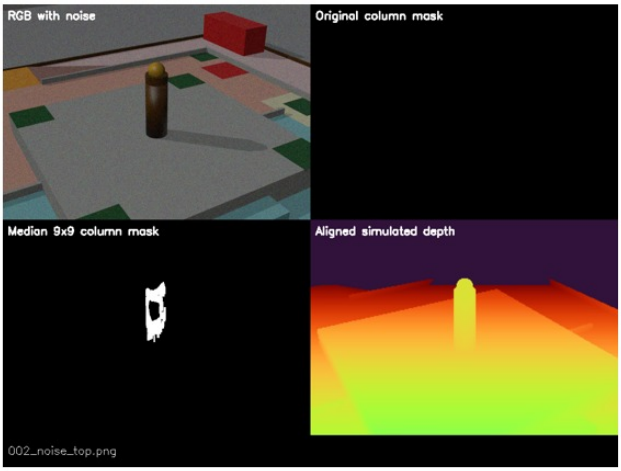
噪声组柱掩膜去噪对照：第 1 页



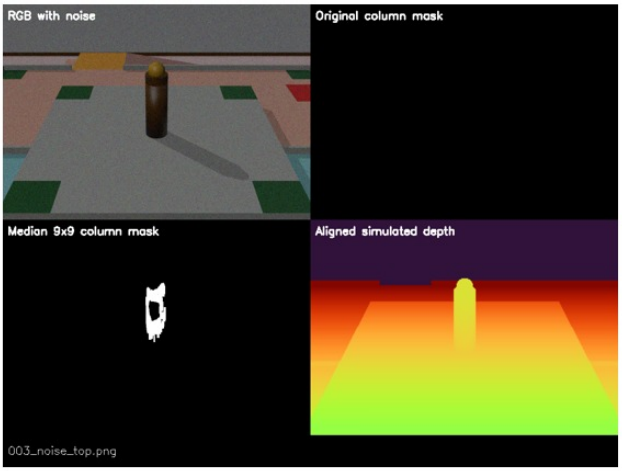
000_noise_top_diagnostic.png



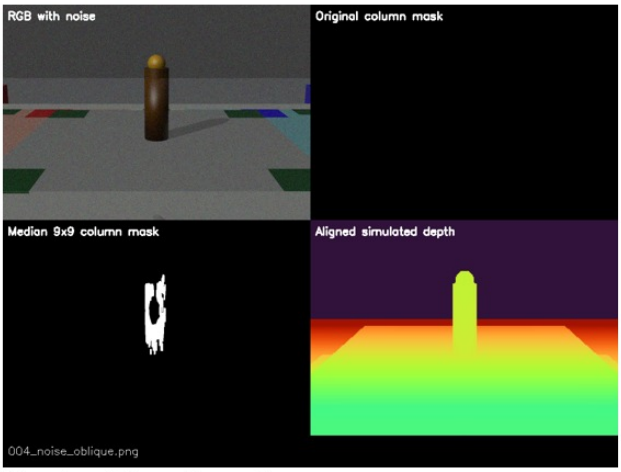
001_noise_top_diagnostic.png



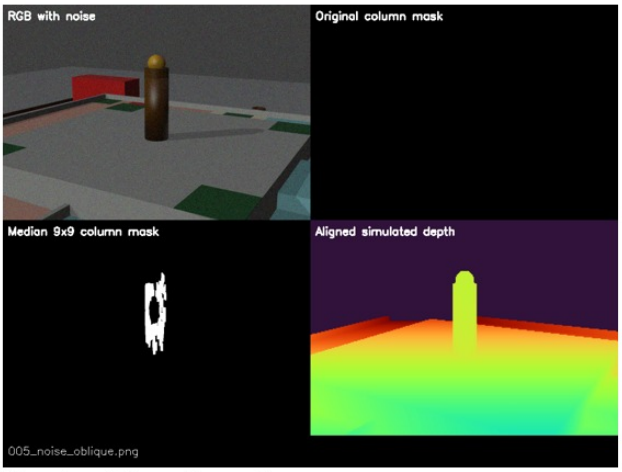
002_noise_top_diagnostic.png



003_noise_top_diagnostic.png

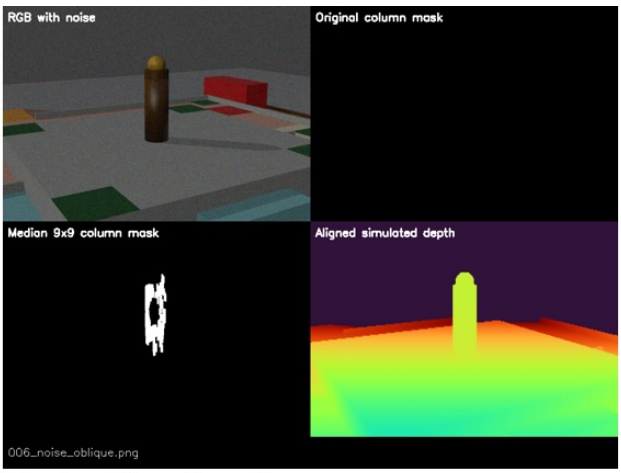


004_noise_oblique_diagnostic.png

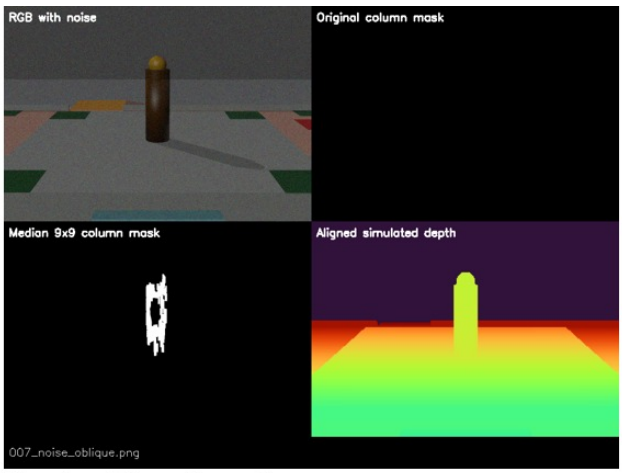


005_noise_oblique_diagnostic.png

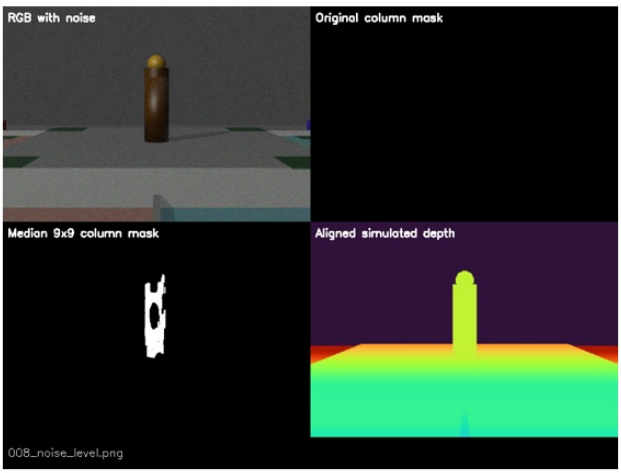
噪声组柱掩膜去噪对照：第 2 页



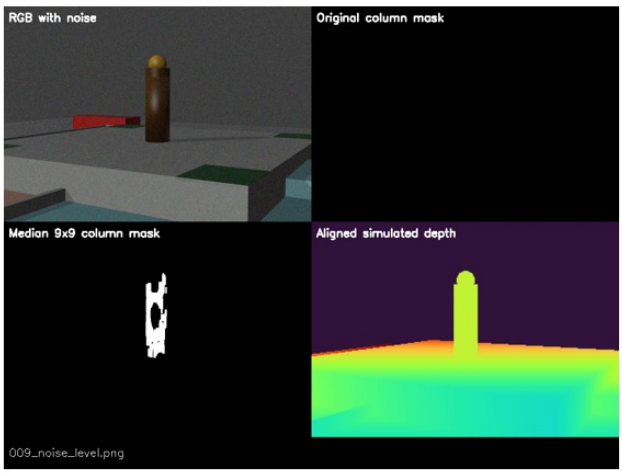
006_noise_oblique_diagnostic.png



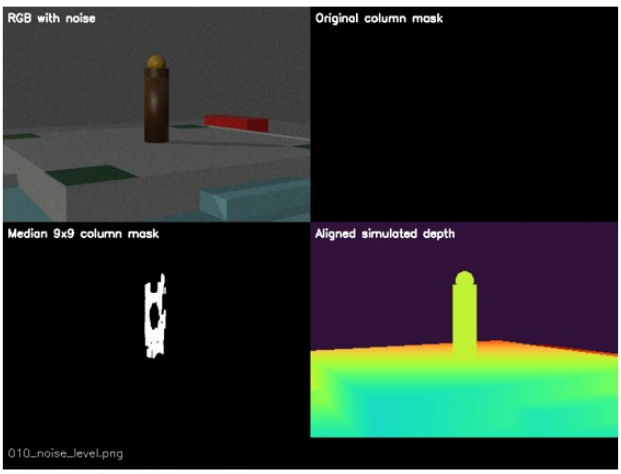
007_noise_oblique_diagnostic.png



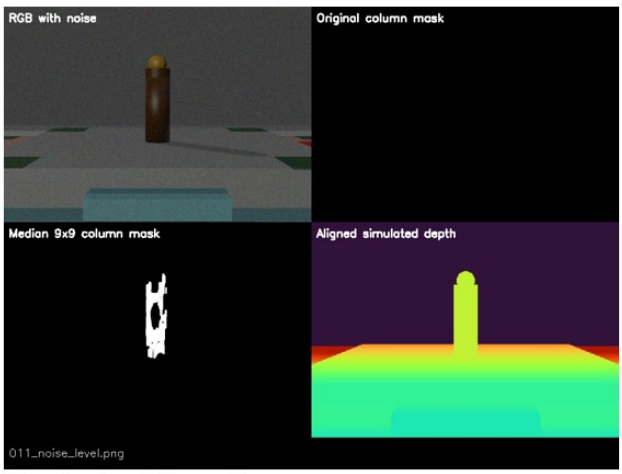
008_noise_level_diagnostic.png



009_noise_level_diagnostic.png

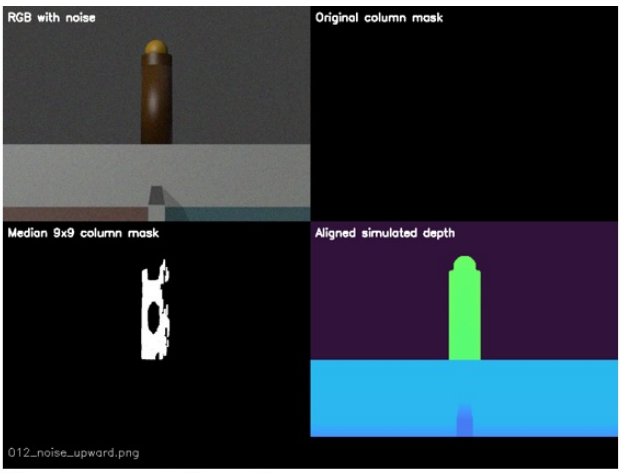


010_noise_level_diagnostic.png

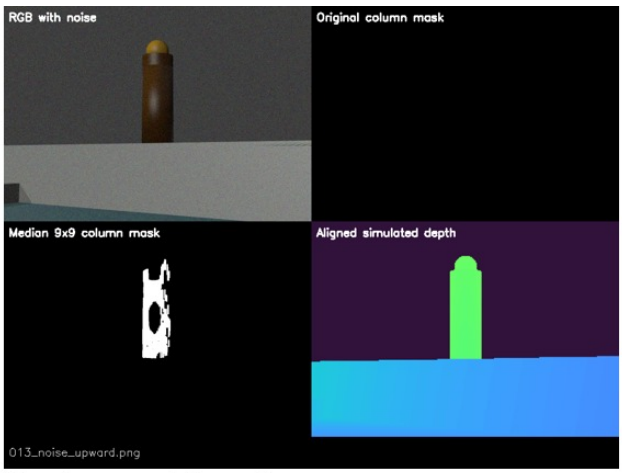


011_noise_level_diagnostic.png

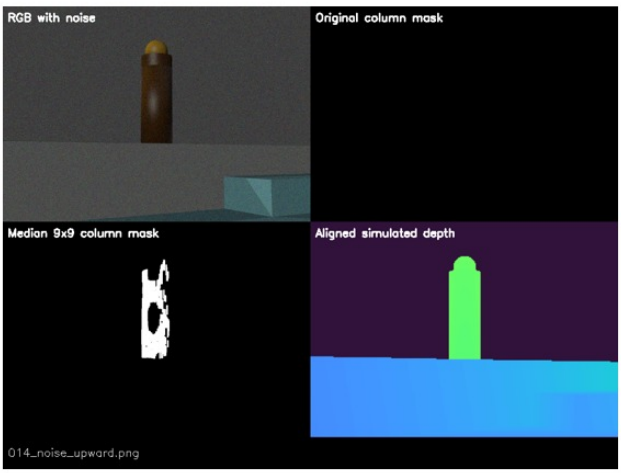
噪声组柱掩膜去噪对照：第 3 页



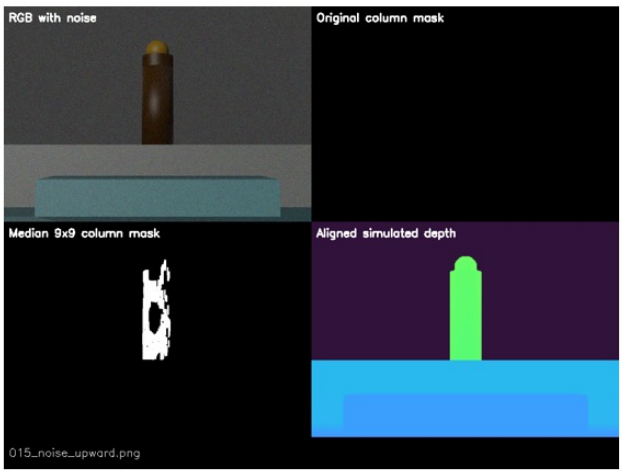
012_noise_upward_diagnostic.png



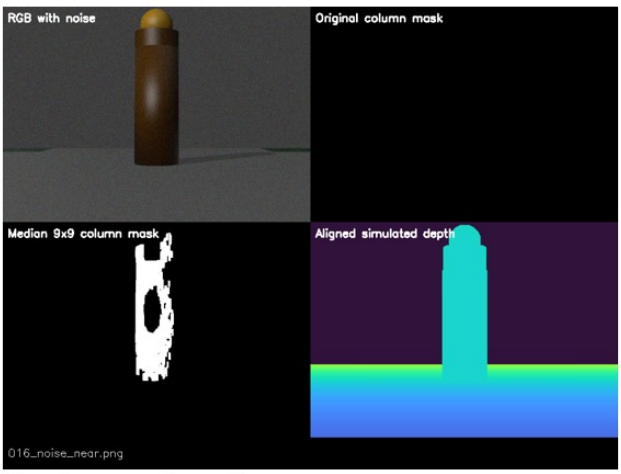
013_noise_upward_diagnostic.png



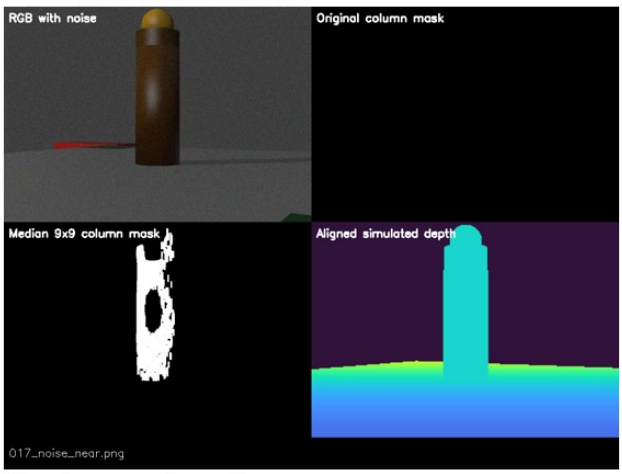
014_noise_upward_diagnostic.png



015_noise_upward_diagnostic.png

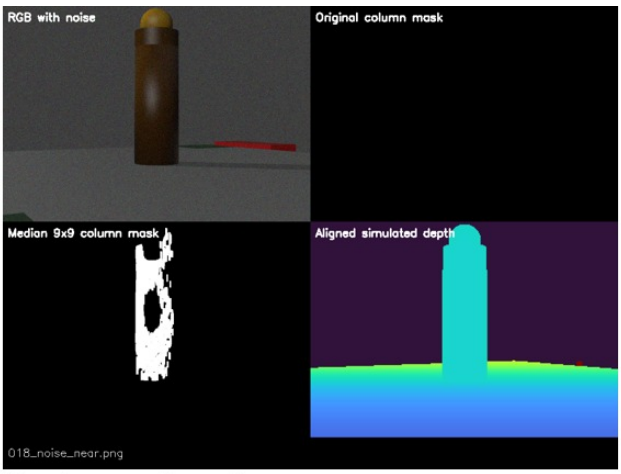


016_noise_near_diagnostic.png

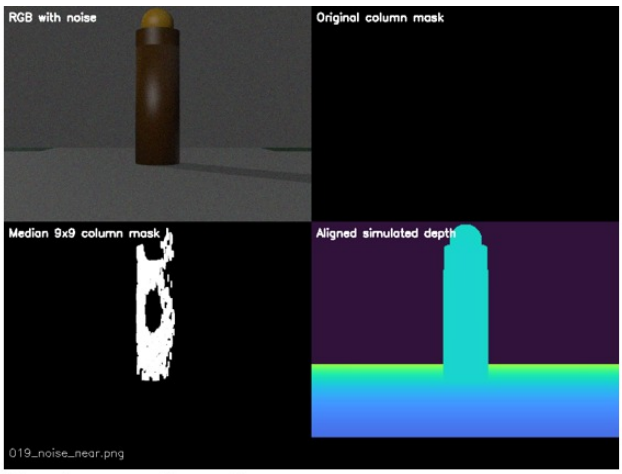


017_noise_near_diagnostic.png

噪声组柱掩膜去噪对照：第 4 页



018_noise_near_diagnostic.png



019_noise_near_diagnostic.png

6 误检、漏检和掩膜分析

每张诊断图都合成 RGB 候选、有效深度
ROI、球掩膜、柱掩膜四个视图。原始图片按误检、漏检、柱误检、柱漏检、定位偏差和不确定结果分类保存。

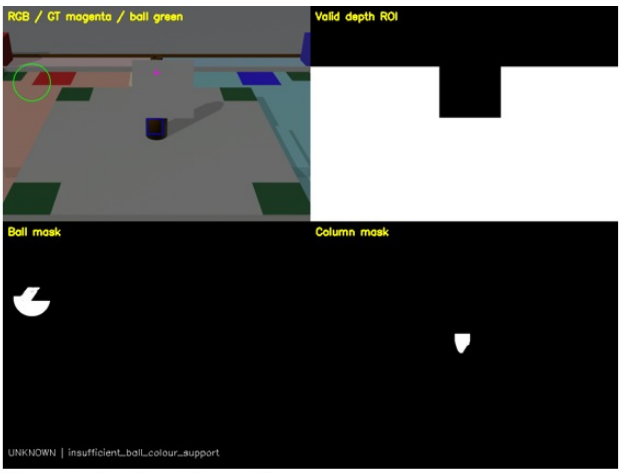
球误检 0 张，球漏检 13 张，定位偏差 2 张，柱漏检 5 张。

UNKNOWN 中，81 张来自缺少球或柱的负样本，符合关系不可判定的定义；18
张来自球柱都存在的样本，属于检出不足。正样本中有 1 张球在柱上却未判为
ON_COLUMN。未定位到柱体时，保留有效球坐标并输出 UNKNOWN，不能据此判定球已离柱。

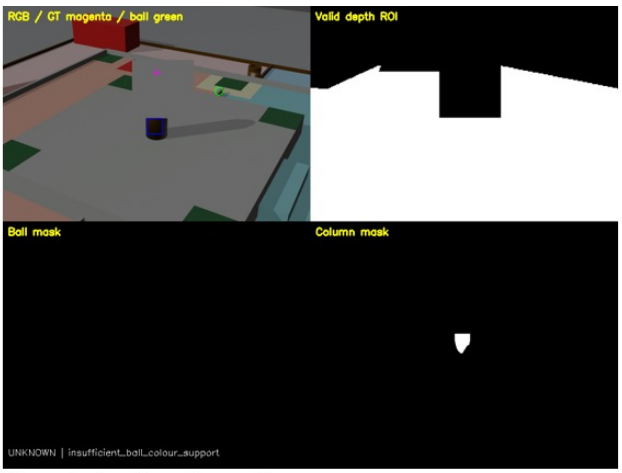
问题	条件	视角	球检测原因	数量
柱漏检	dark	top	column_geometry_unavailable	1
柱漏检	occlusion	upward	column_geometry_unavailable	4
球定位超过5cm	normal	oblique	rgbd_geometry_ok	1
球定位超过5cm	tilt	level	rgbd_geometry_ok	1
球漏检	blur	level	insufficient_ball_colour_support	1
球漏检	occlusion	level	insufficient_ball_colour_support	4
球漏检	occlusion	oblique	insufficient_ball_colour_support	3
球漏检	occlusion	oblique	insufficient_depth_support	1
球漏检	occlusion	top	insufficient_ball_colour_support	4

reason 字段来自球检测，柱漏检需结合柱掩膜和几何条件分析。

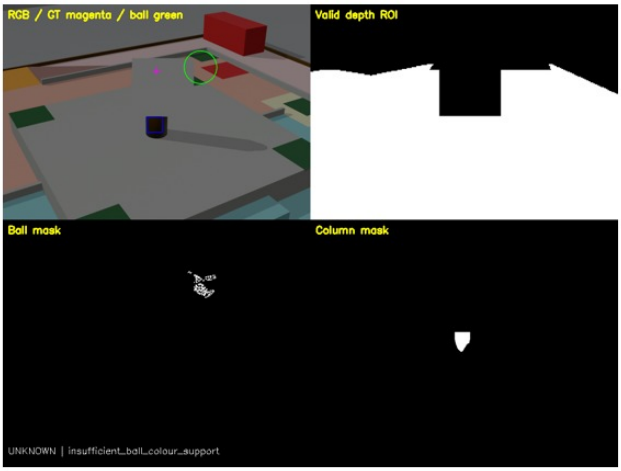
球漏检：第 1 页



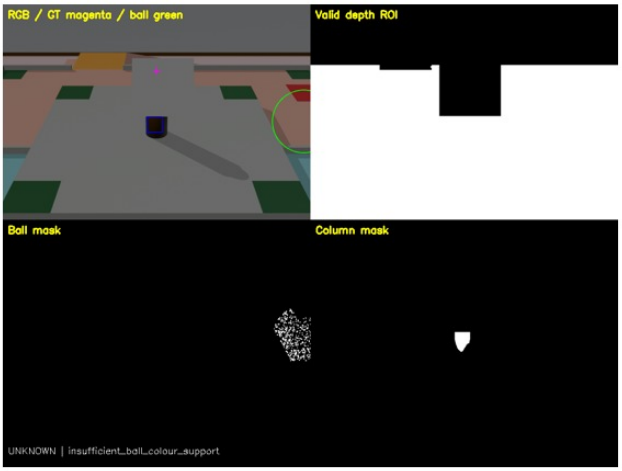
000_occlusion_top_diagnostic.png



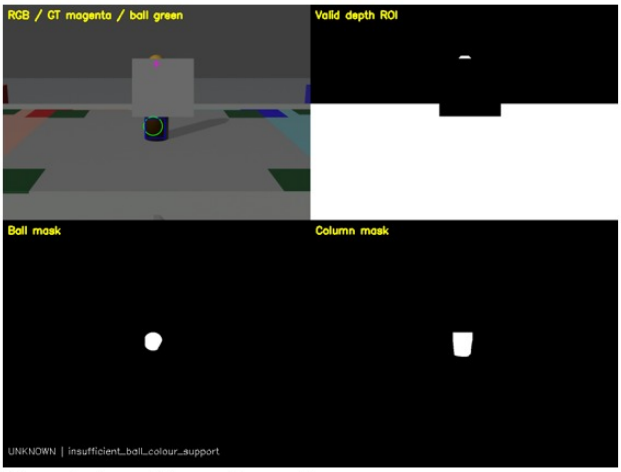
001_occlusion_top_diagnostic.png



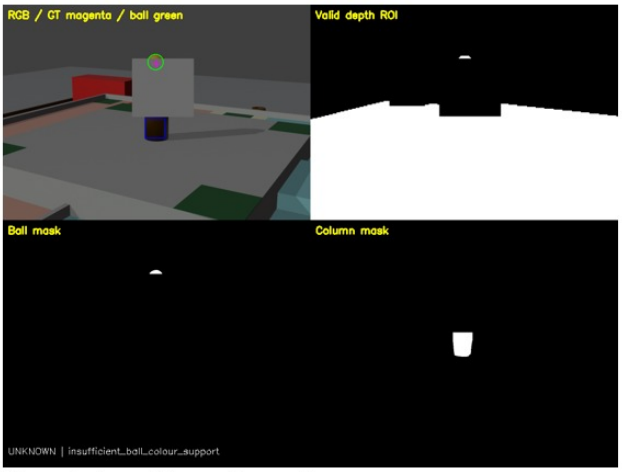
002_occlusion_top_diagnostic.png



003_occlusion_top_diagnostic.png

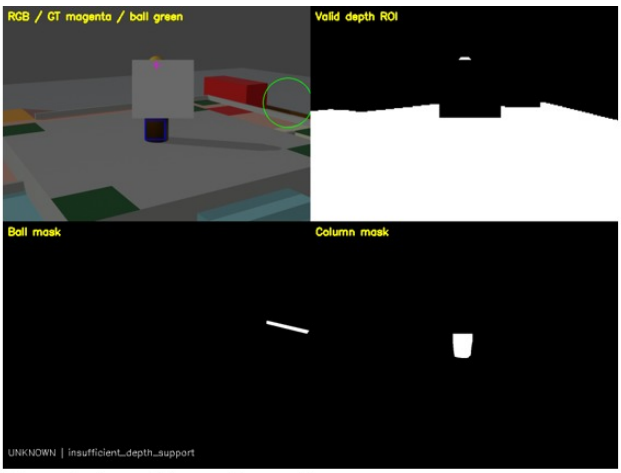


004_occlusion_oblique_diagnostic.png

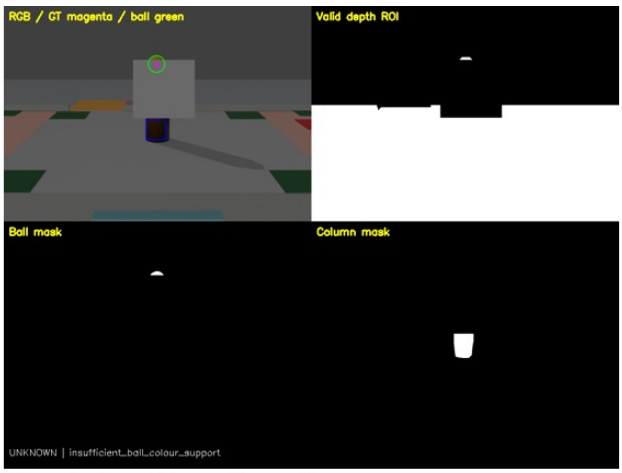


005_occlusion_oblique_diagnostic.png

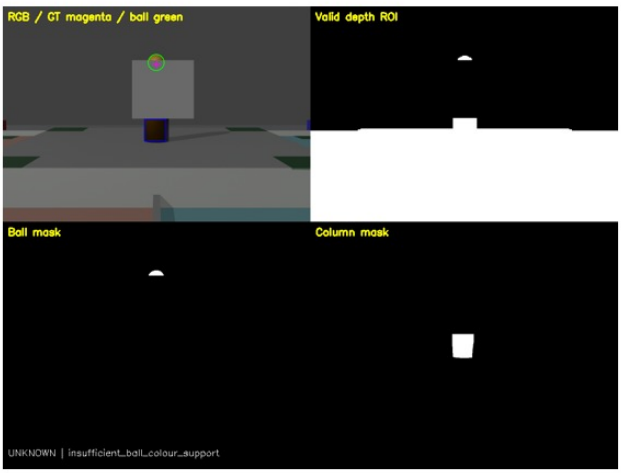
球漏检：第 2 页



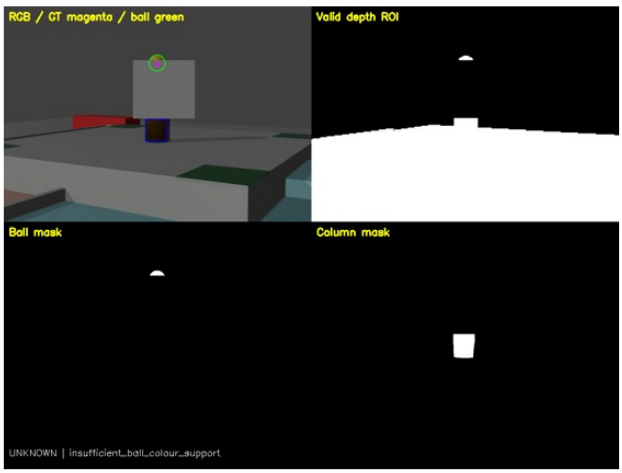
006_occlusion_oblique_diagnostic.png



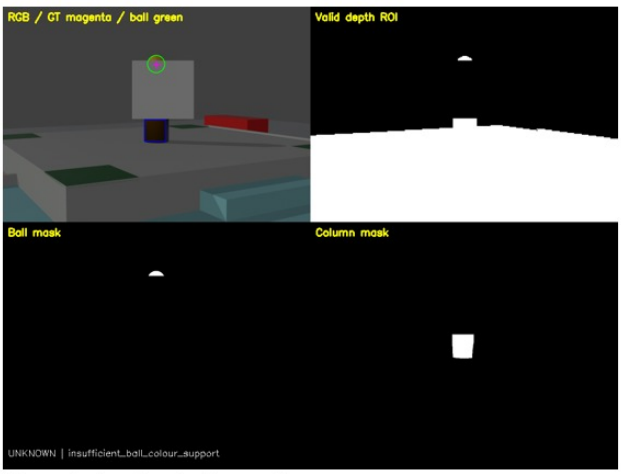
007_occlusion_oblique_diagnostic.png



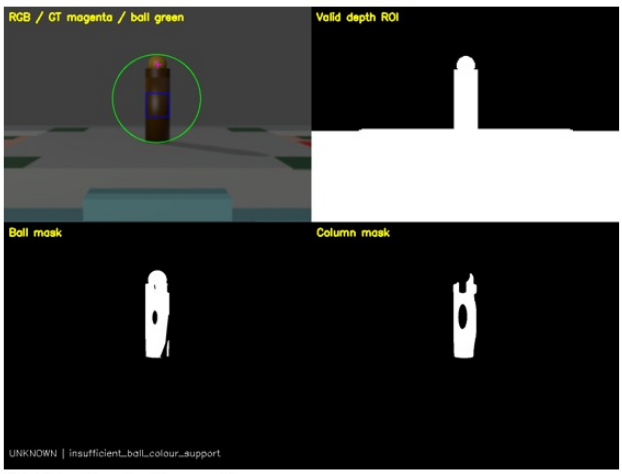
008_occlusion_level_diagnostic.png



009_occlusion_level_diagnostic.png

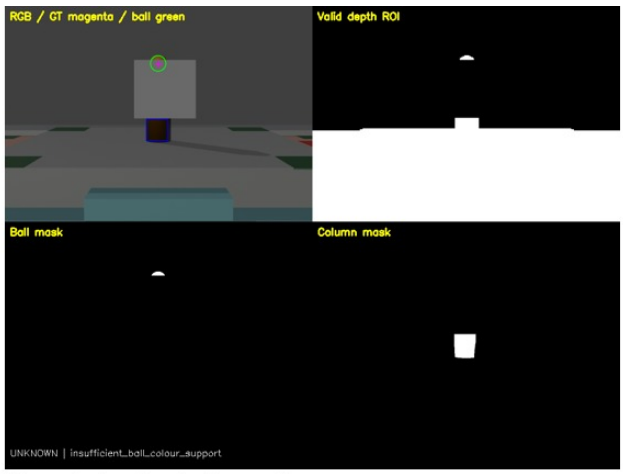


010_occlusion_level_diagnostic.png



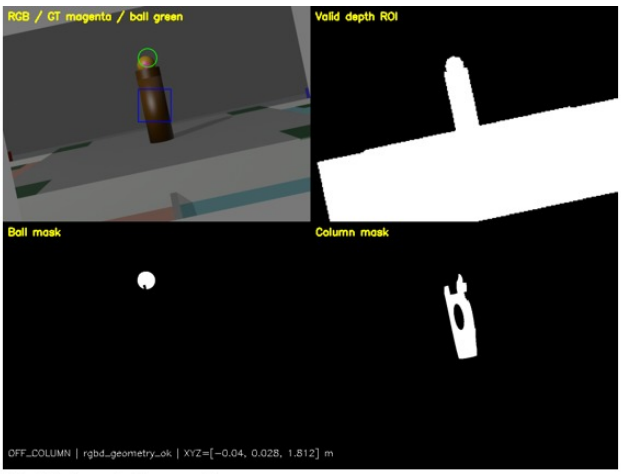
011_blur_level_diagnostic.png

球漏检：第 3 页

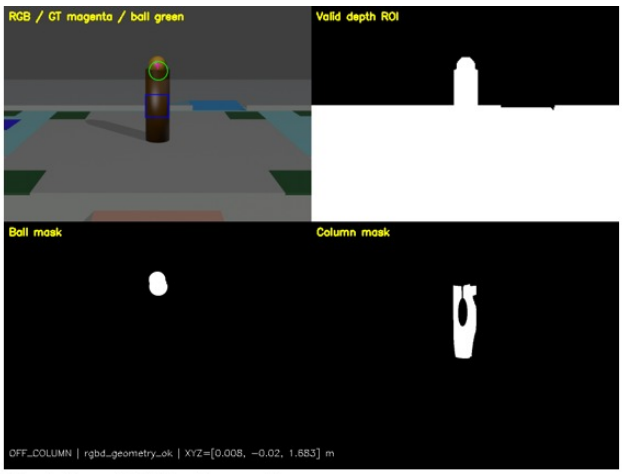


011_occlusion_level_diagnostic.png

定位误差：第 1 页



008_tilt_level_diagnostic.png



022_oblique_h2.20_diagnostic.png

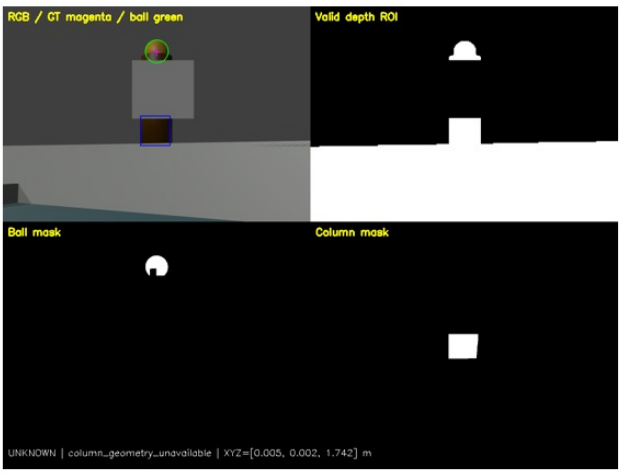
柱漏检：第 1 页



003_dark_top_diagnostic.png



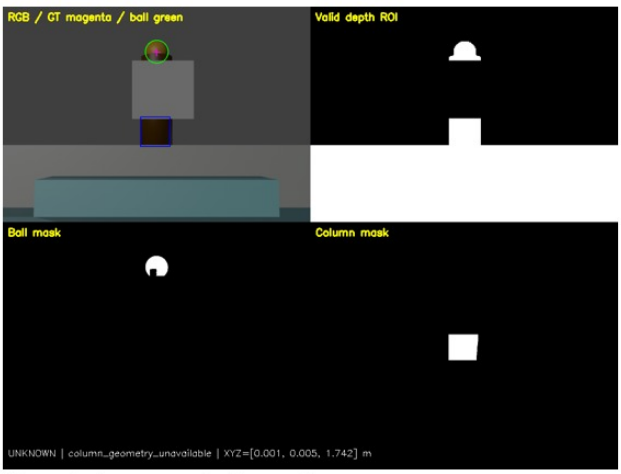
012_occlusion_upward_diagnostic.png



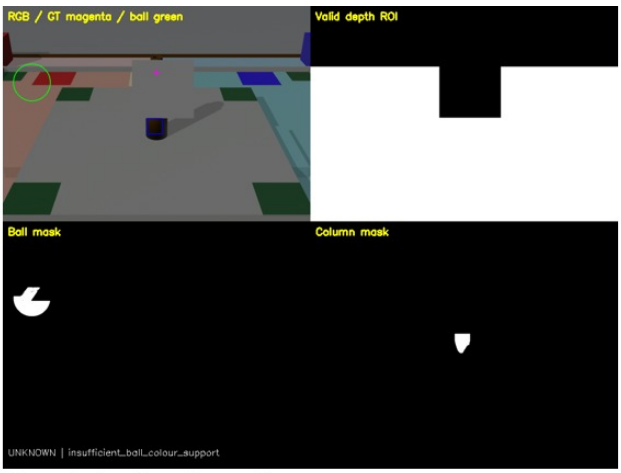
013_occlusion_upward_diagnostic.png



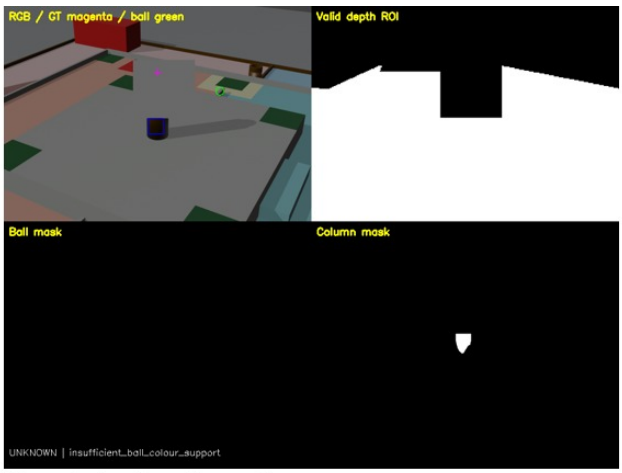
014_occlusion_upward_diagnostic.png



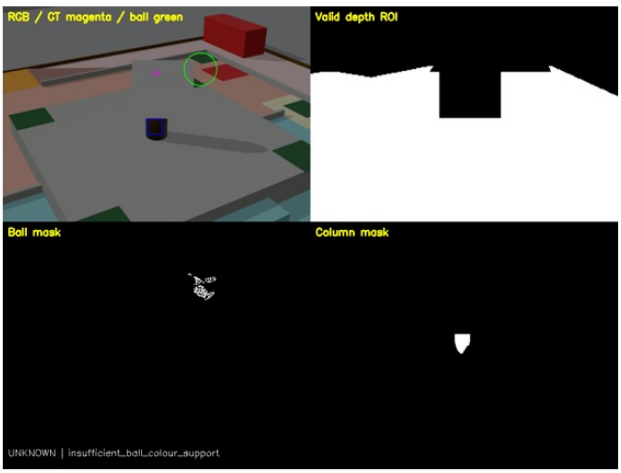
015_occlusion_upward_diagnostic.png



000_occlusion_top_diagnostic.png



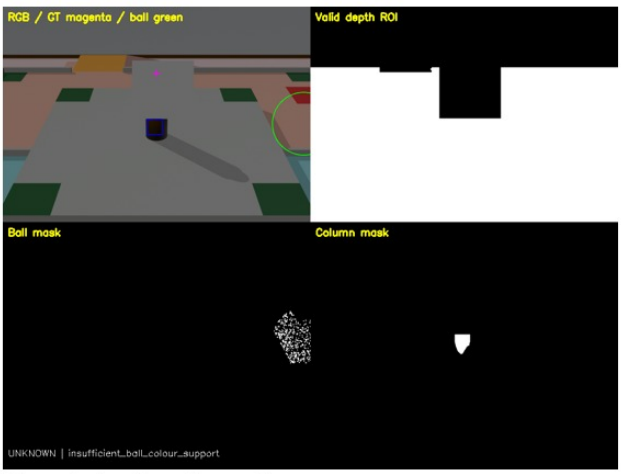
001_occlusion_top_diagnostic.png



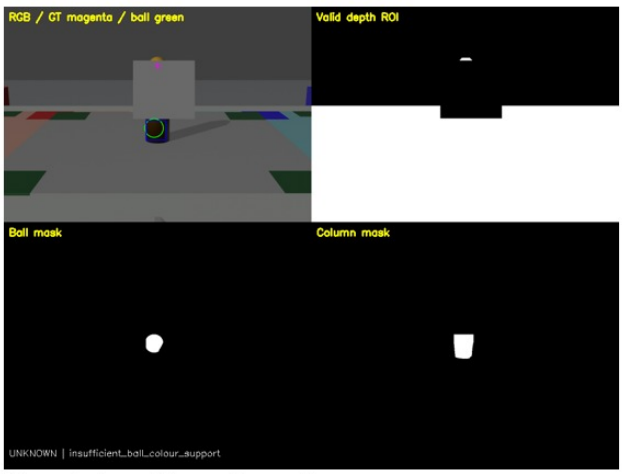
002_occlusion_top_diagnostic.png



003_dark_top_diagnostic.png

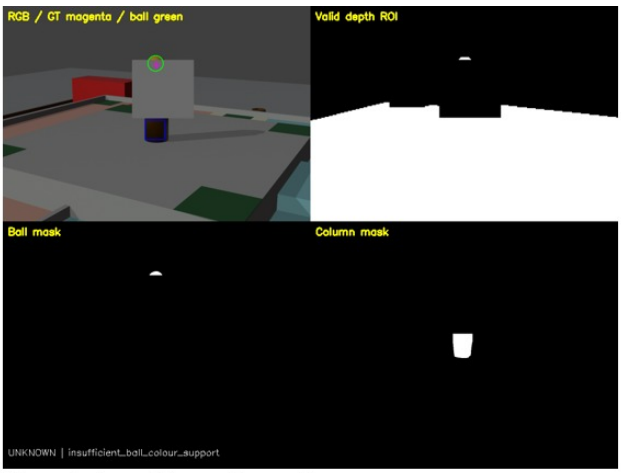


003_occlusion_top_diagnostic.png

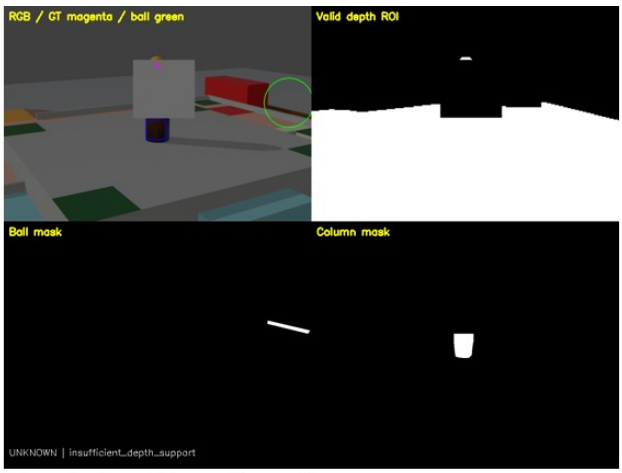


004_occlusion_oblique_diagnostic.png

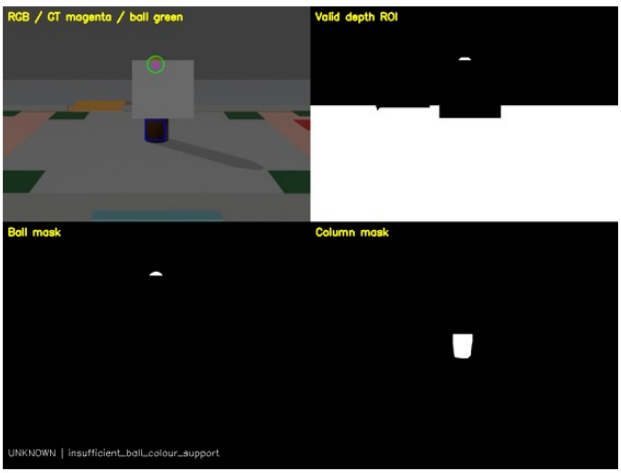
几何不确定：第 2 页



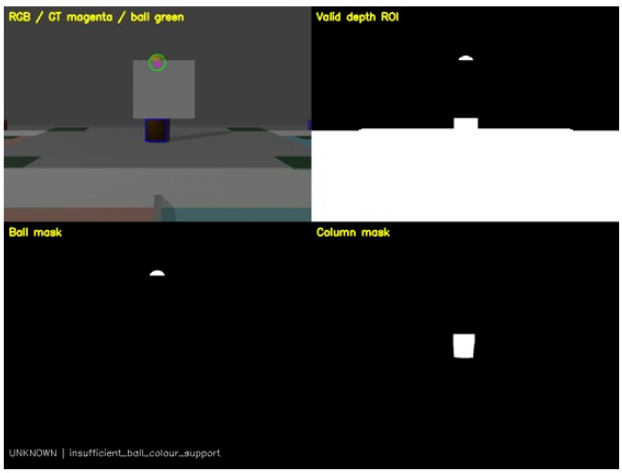
005_occlusion_oblique_diagnostic.png



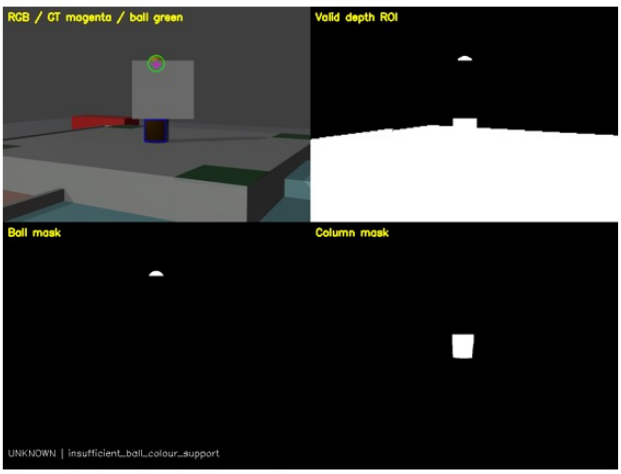
006_occlusion_oblique_diagnostic.png



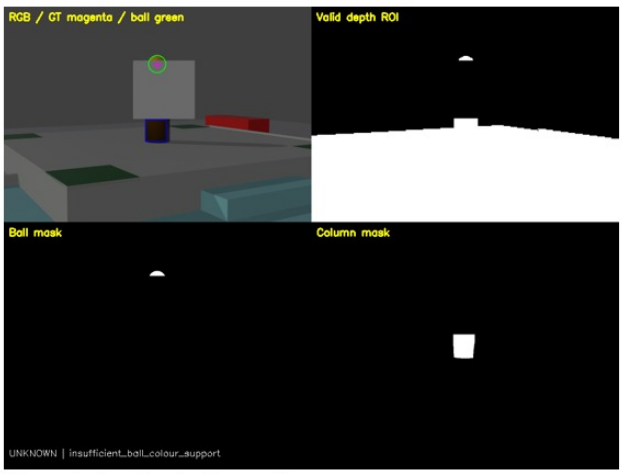
007_occlusion_oblique_diagnostic.png



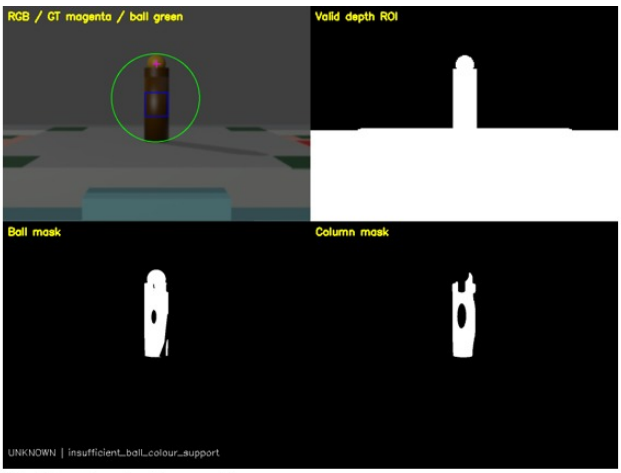
008_occlusion_level_diagnostic.png



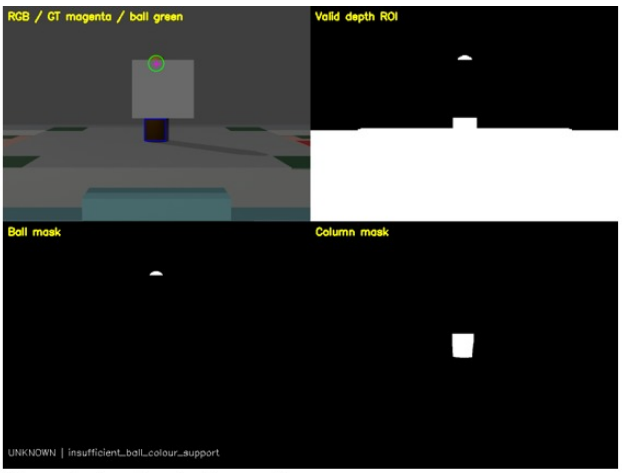
009_occlusion_level_diagnostic.png



010_occlusion_level_diagnostic.png



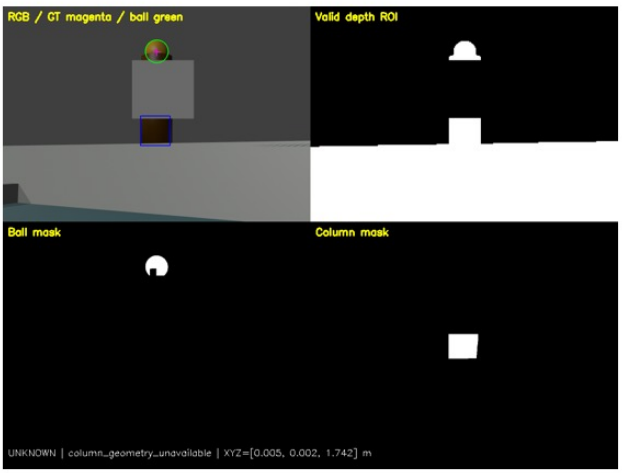
011_blur_level_diagnostic.png



011_occlusion_level_diagnostic.png



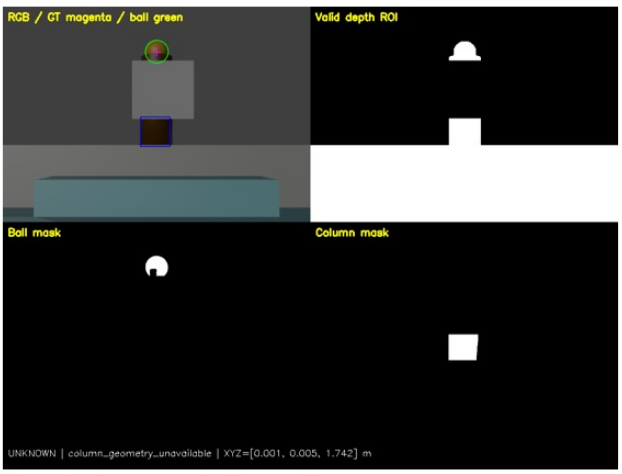
012_occlusion_upward_diagnostic.png



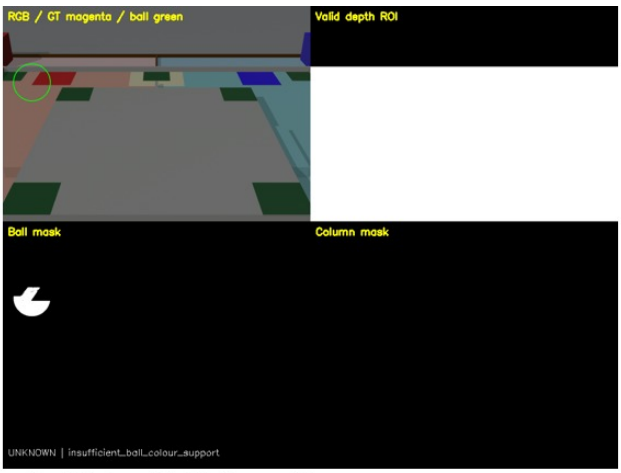
013_occlusion_upward_diagnostic.png



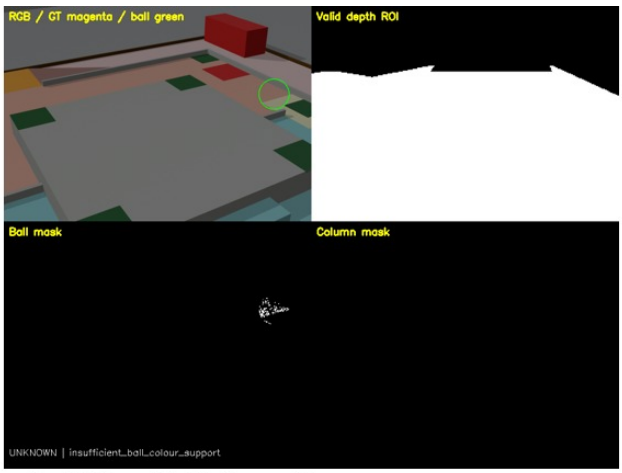
014_occlusion_upward_diagnostic.png



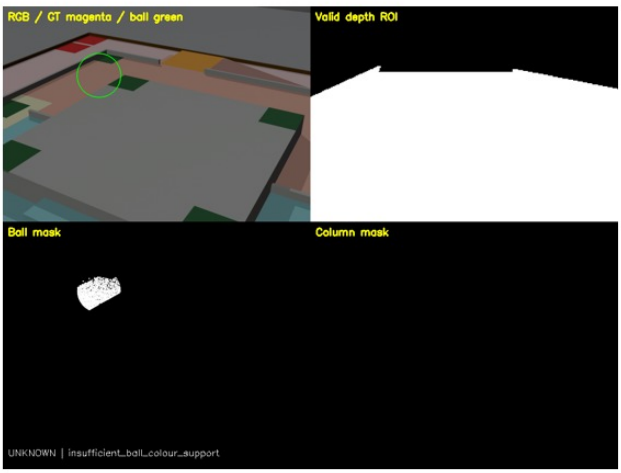
015_occlusion_upward_diagnostic.png



empty_001_top_h3.50_diagnostic.png



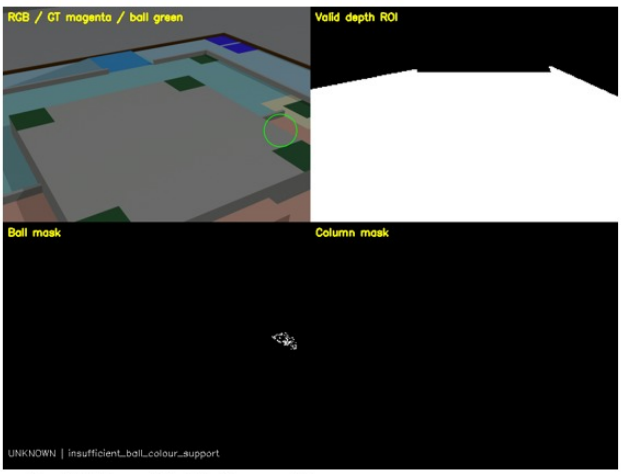
empty_002_top_h3.50_diagnostic.png



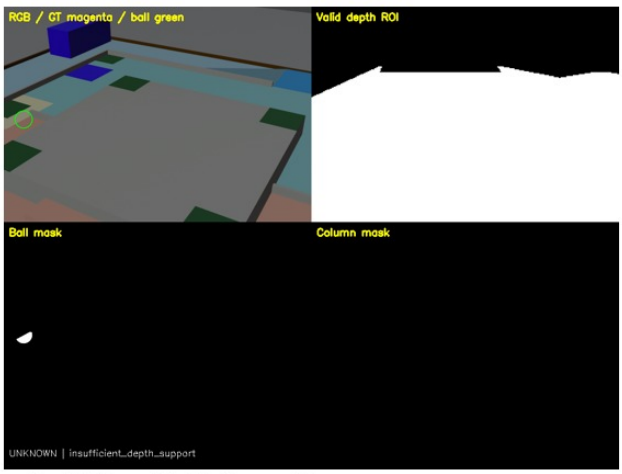
empty_003_top_h3.50_diagnostic.png



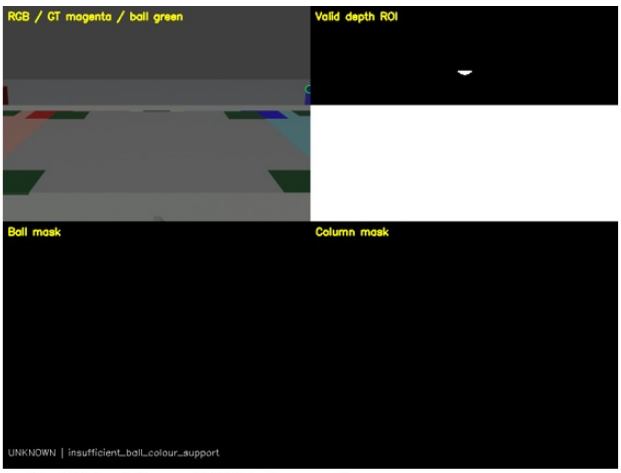
empty_004_top_h3.50_diagnostic.png



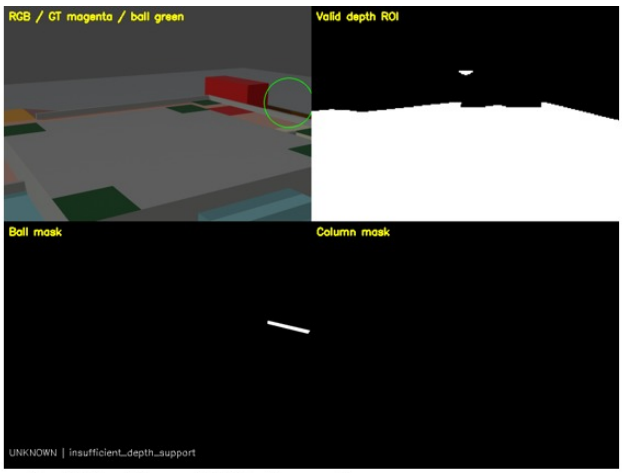
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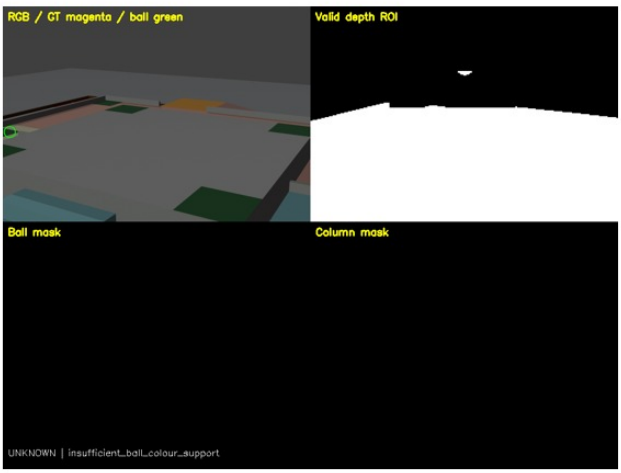
empty_006_top_h3.50_diagnostic.png



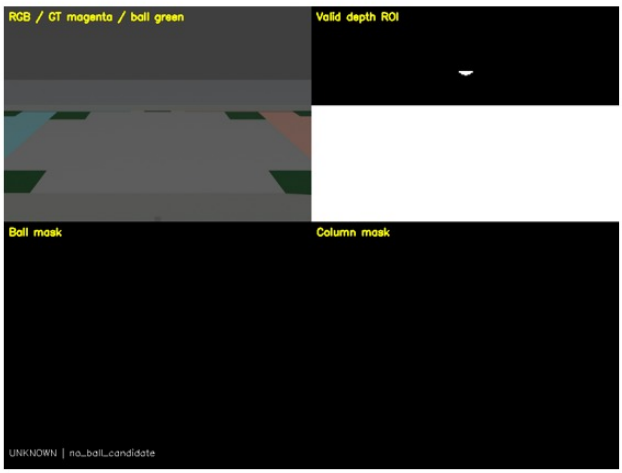
empty_007_oblique_h2.20_diagnostic.png



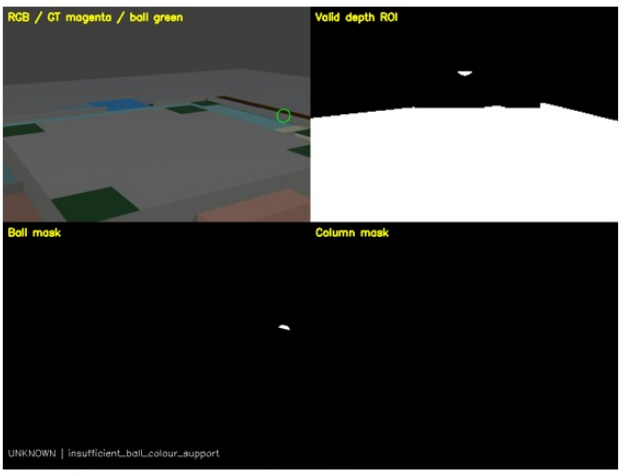
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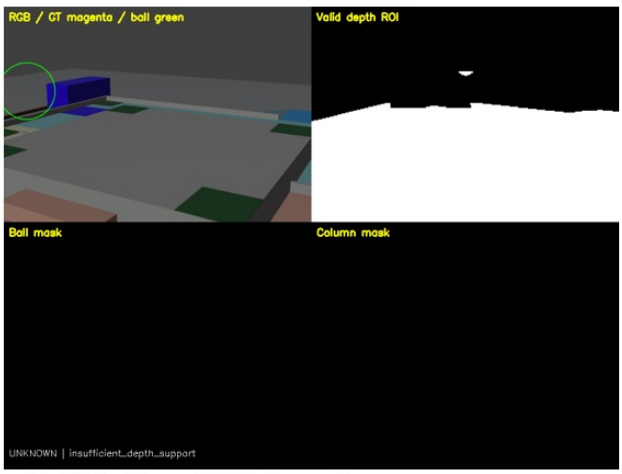
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empty_010_oblique_h2.20_diagnostic.png

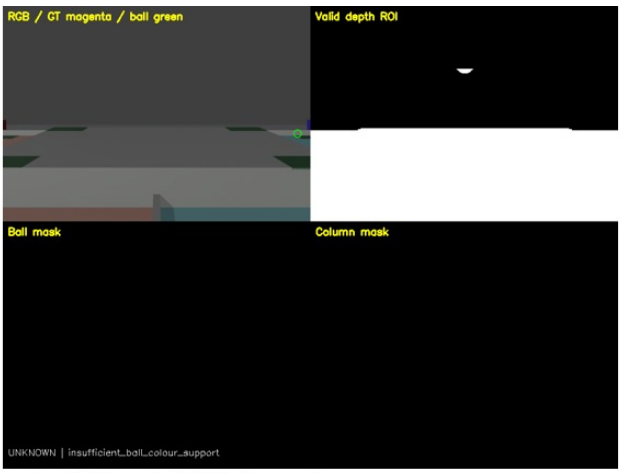


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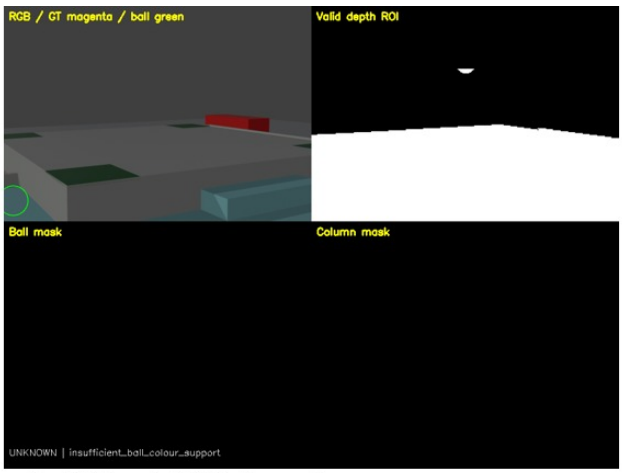


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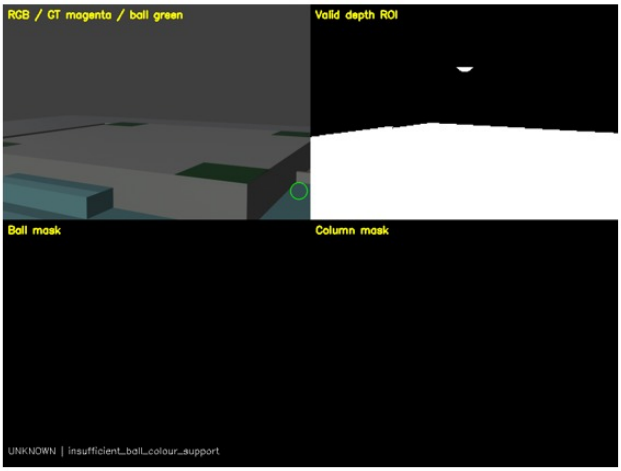
几何不确定：第 6 页



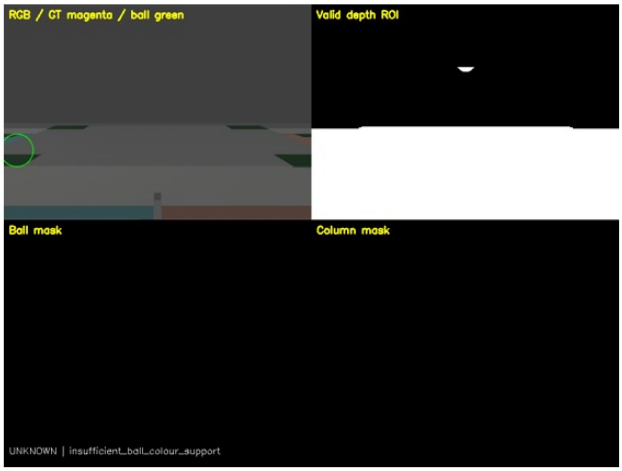
empty_013_level_h1.50_diagnostic.png



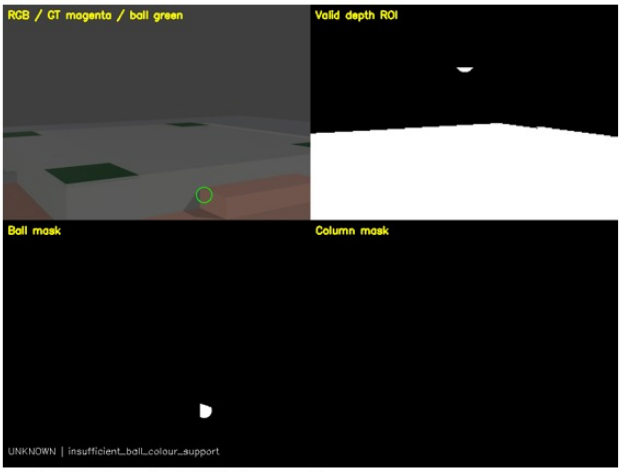
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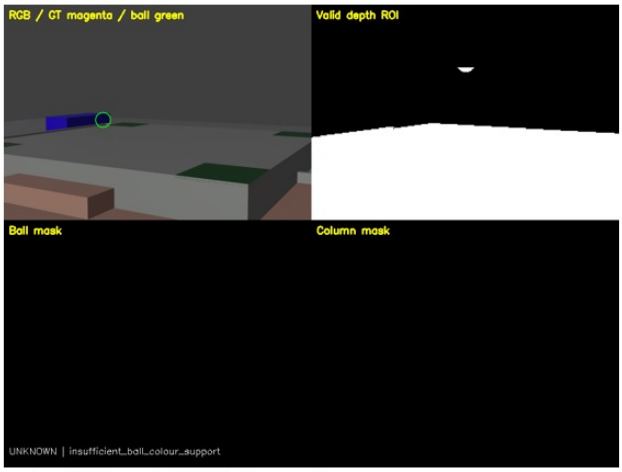
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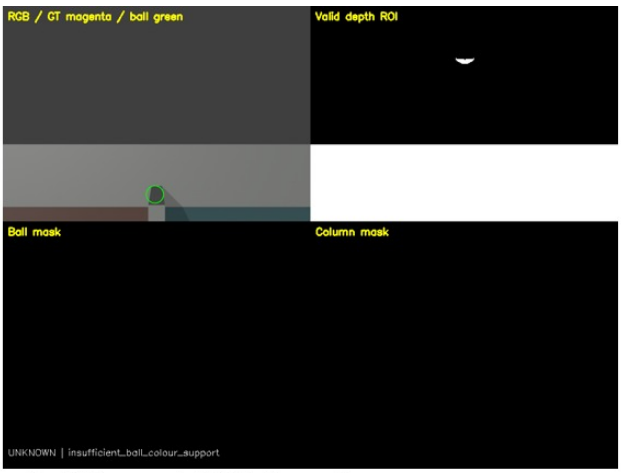
empty_016_level_h1.50_diagnostic.png



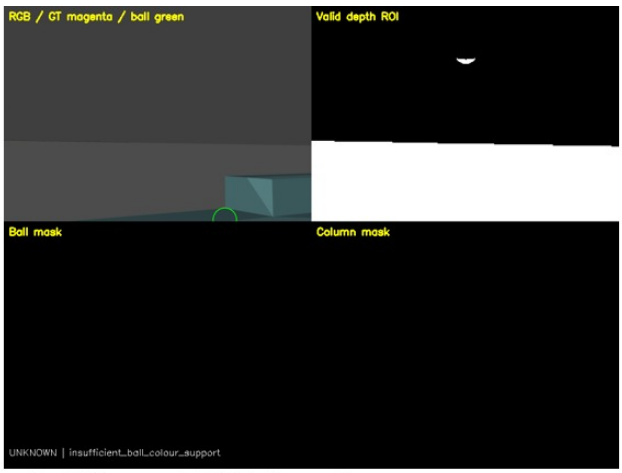
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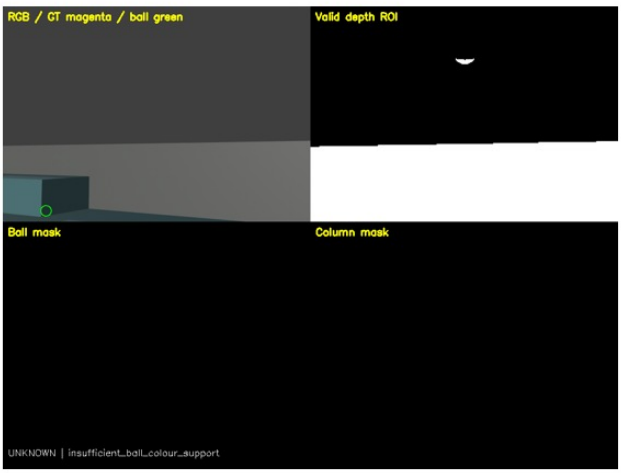
empty_018_level_h1.50_diagnostic.png



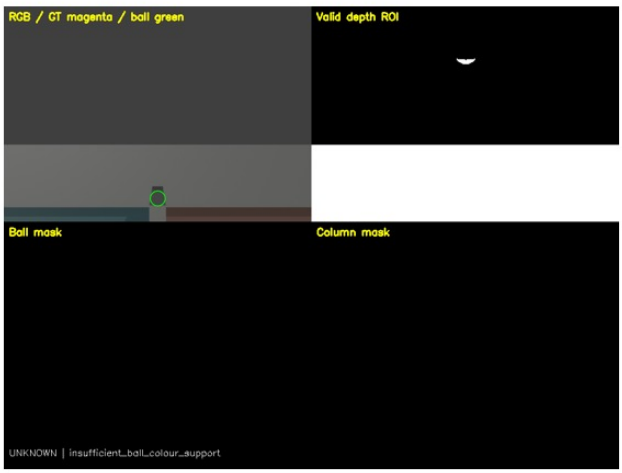
empty_019_upward_h0.85_diagnostic.png



empty_020_upward_h0.85_diagnostic.png



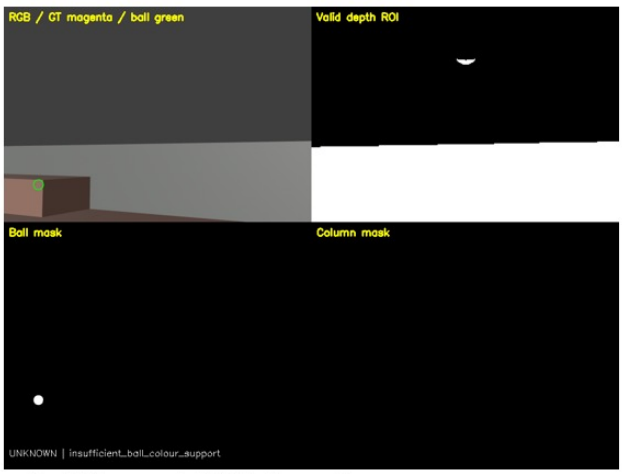
empty_021_upward_h0.85_diagnostic.png



empty_022_upward_h0.85_diagnostic.png



empty_023_upward_h0.85_diagnostic.png

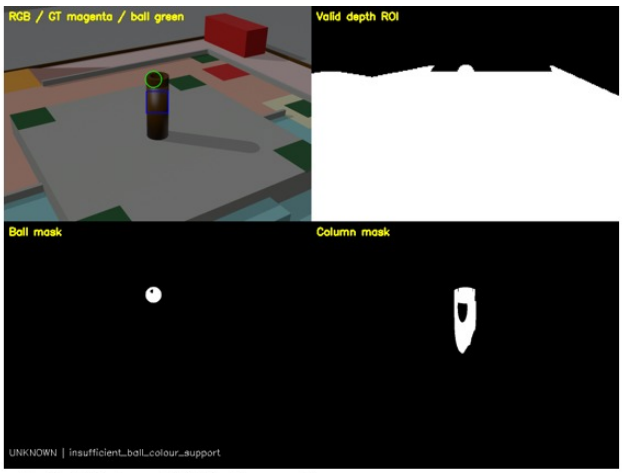


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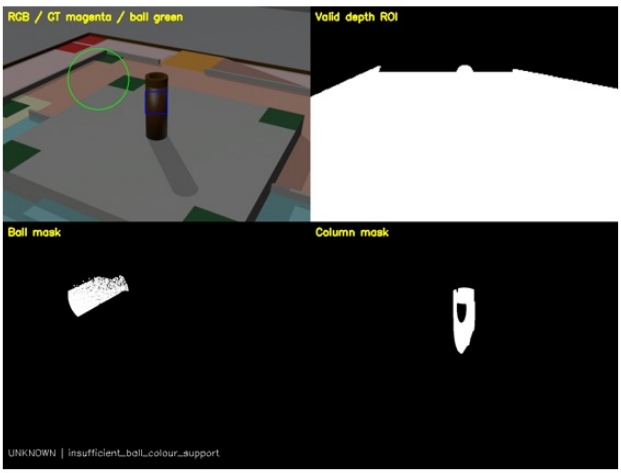
几何不确定：第 8 页



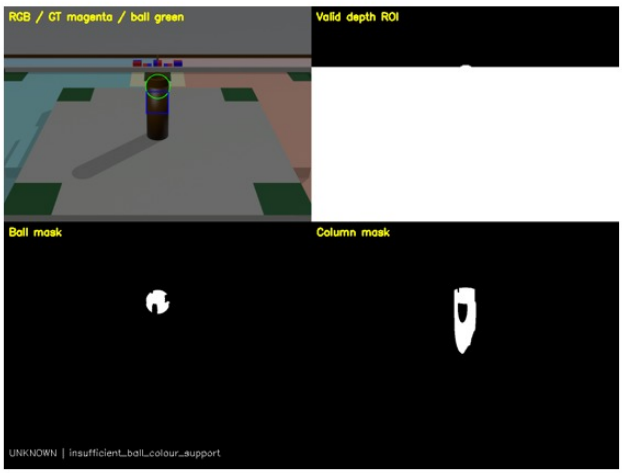
no_ball_001_top_h3.50_diagnostic.png



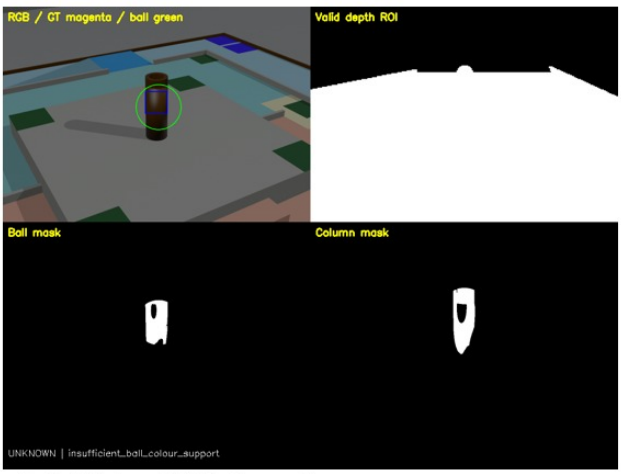
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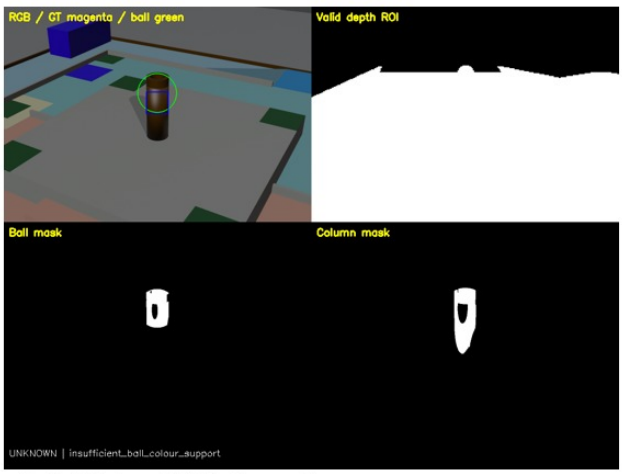
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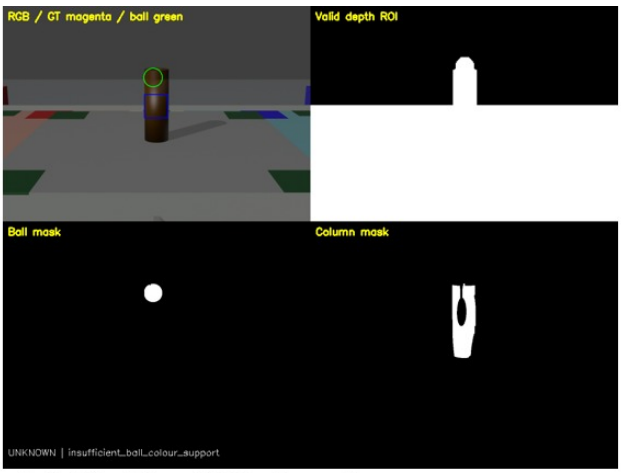
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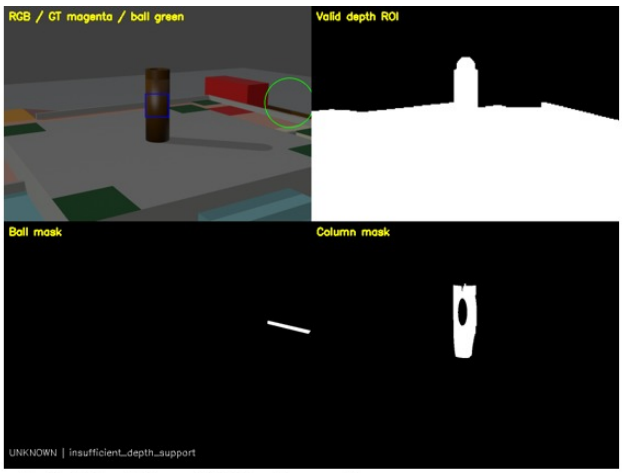
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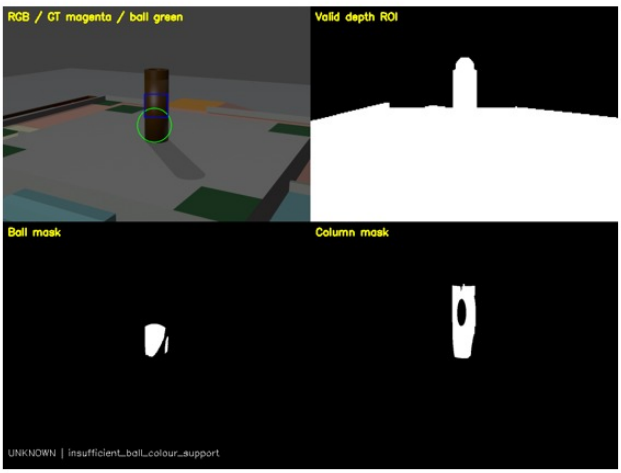
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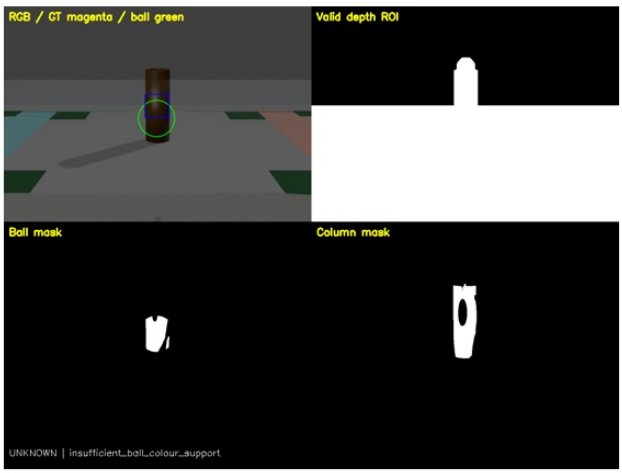
no_ball_007_oblique_h2.20_diagnostic.png



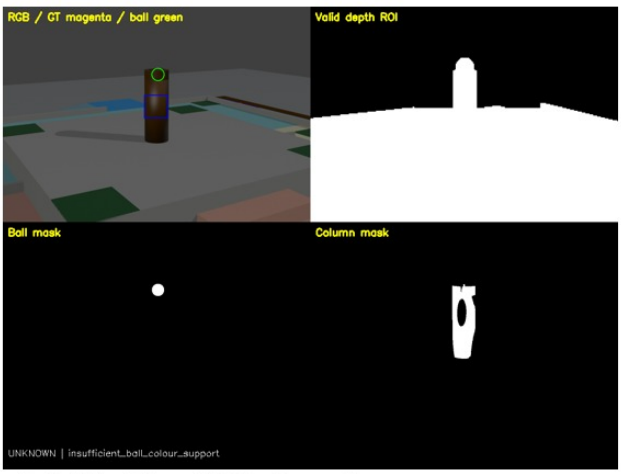
no_ball_008_oblique_h2.20_diagnostic.png



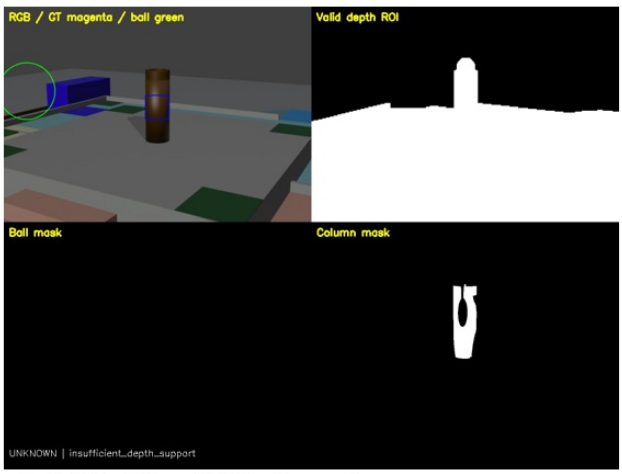
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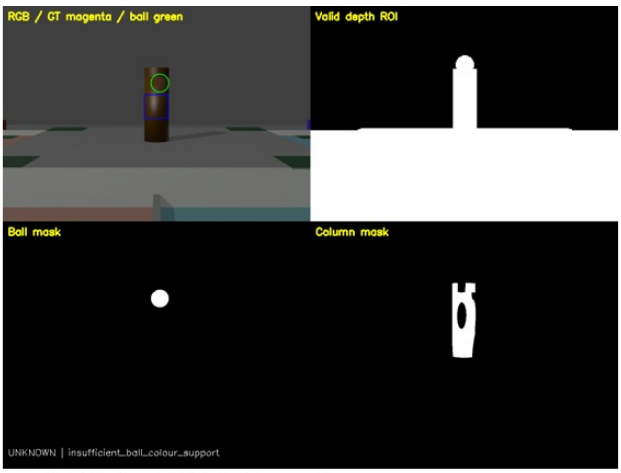
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no_ball_011_oblique_h2.20_diagnostic.png



no_ball_012_oblique_h2.20_diagnostic.png



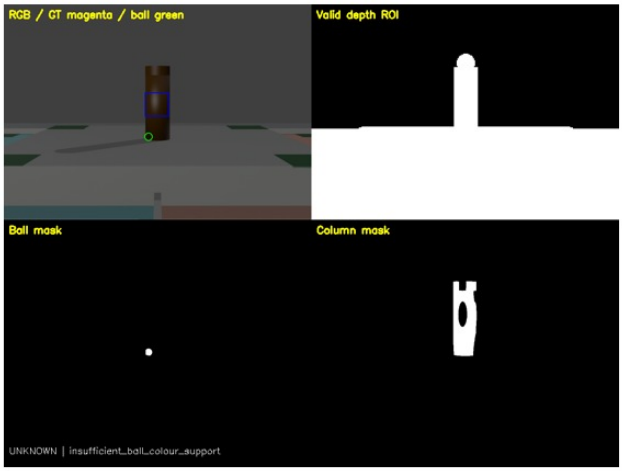
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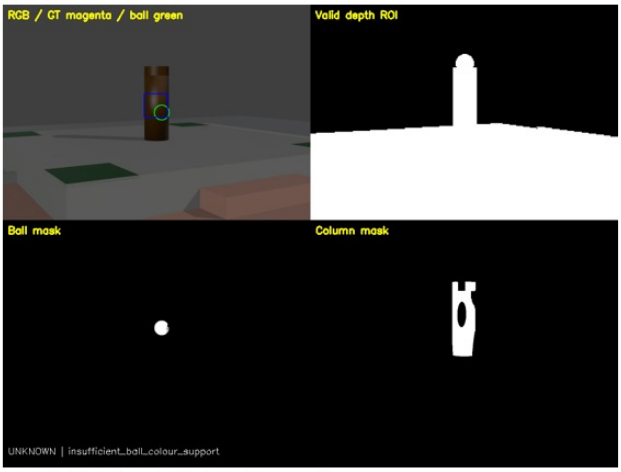
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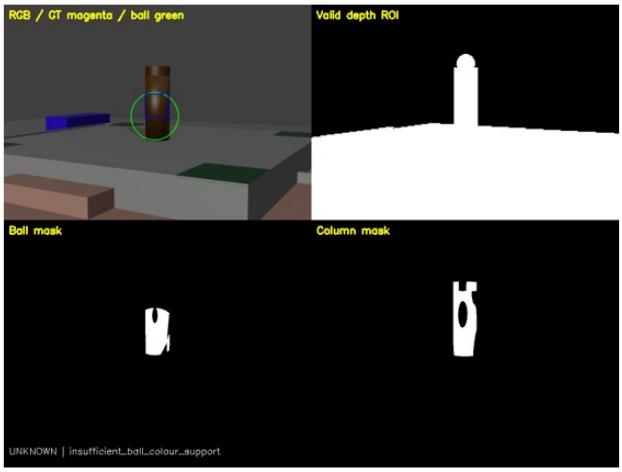
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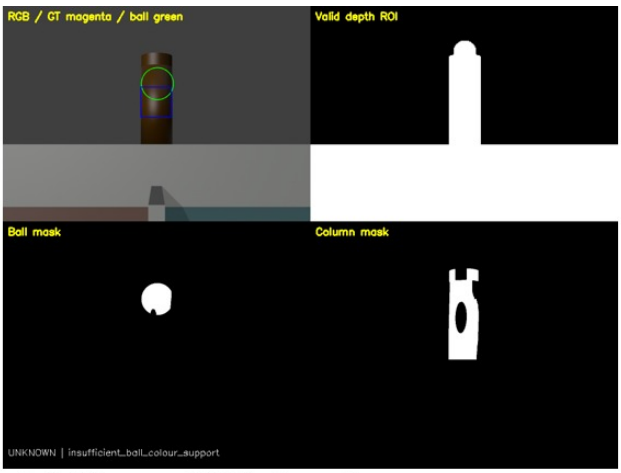
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no_ball_017_level_h1.50_diagnostic.png



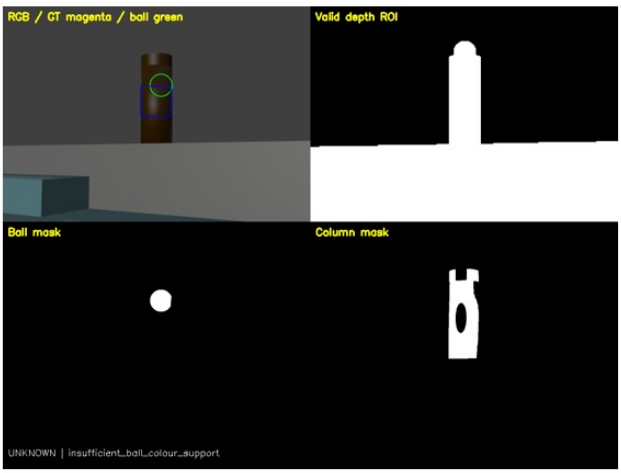
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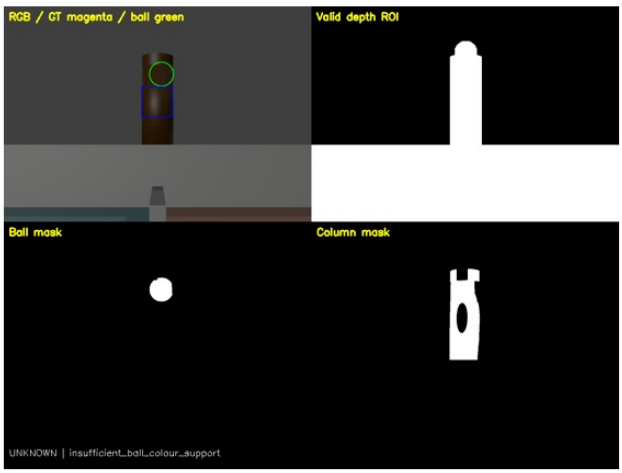
no_ball_019_upward_h0.85_diagnostic.png



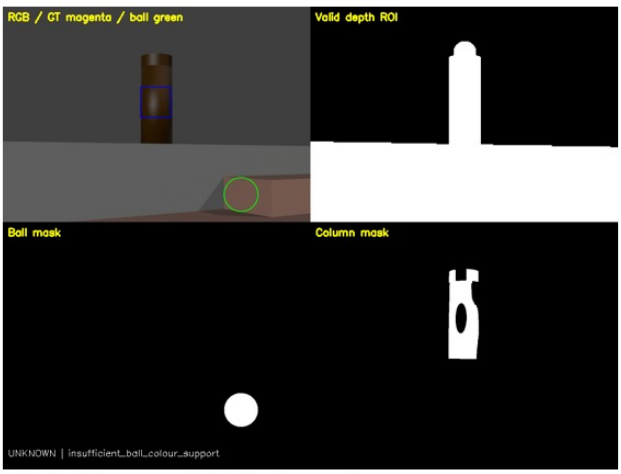
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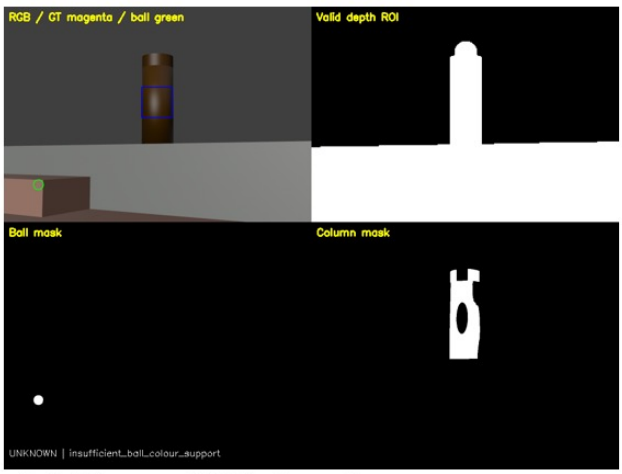
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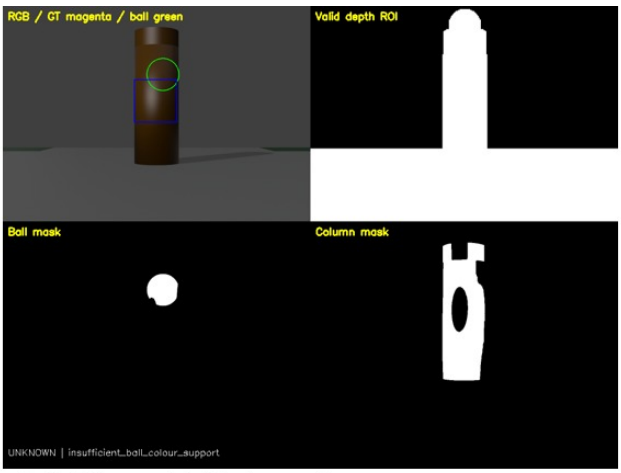
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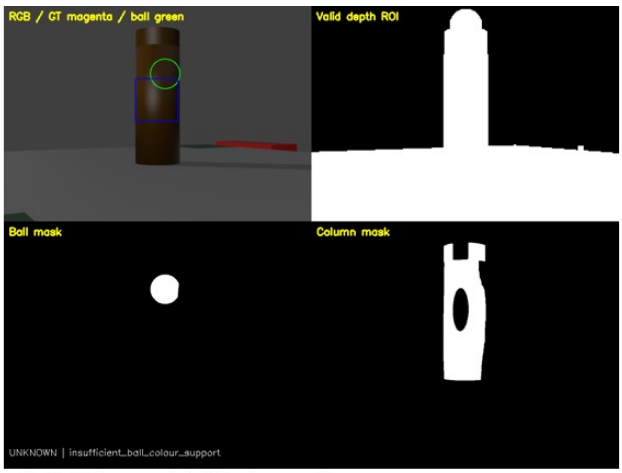
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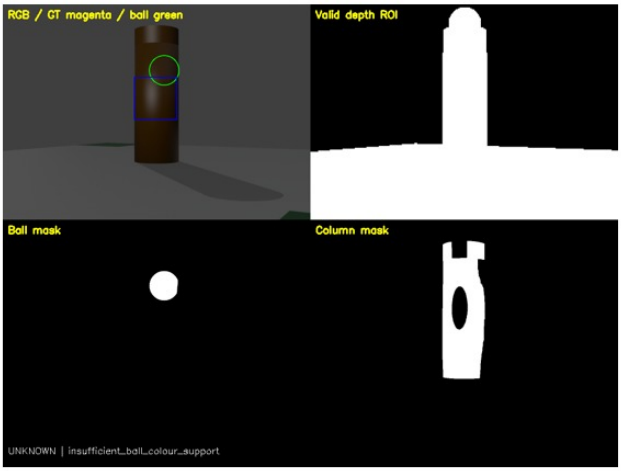
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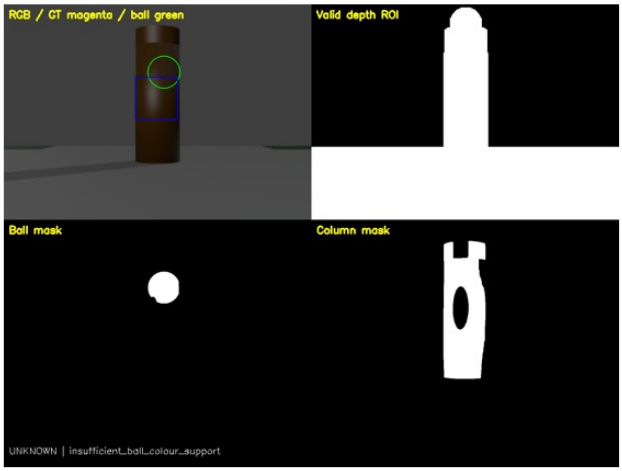
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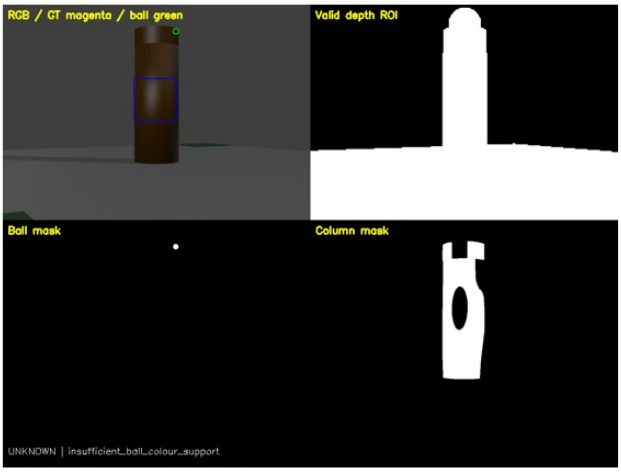
no_ball_026_near_h1.15_diagnostic.png



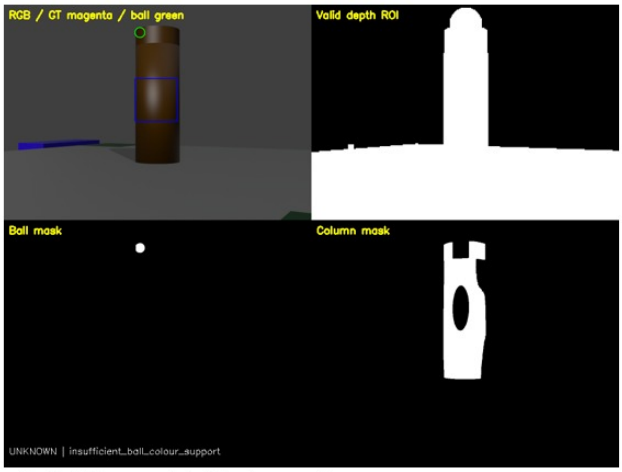
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no_ball_028_near_h1.15_diagnostic.png

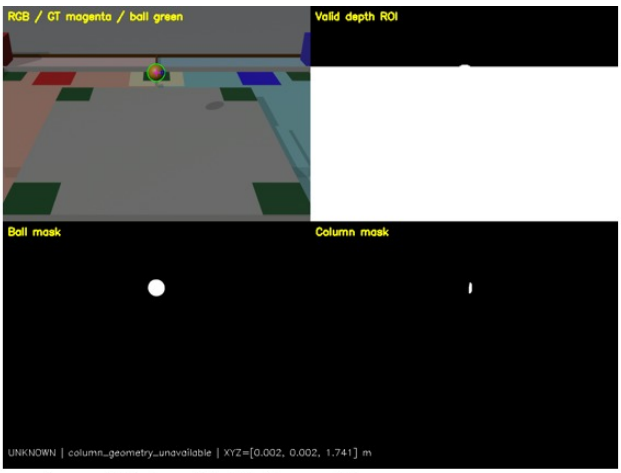


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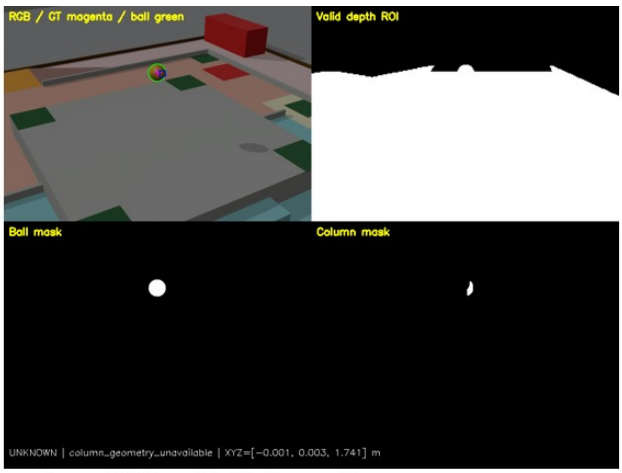


no_ball_030_near_h1.15_diagnostic.png

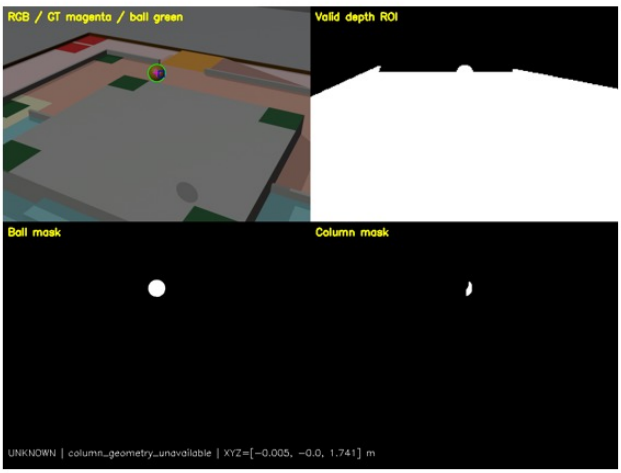
几何不确定：第 13 页



no_column_001_top_h3.50_diagnostic.png



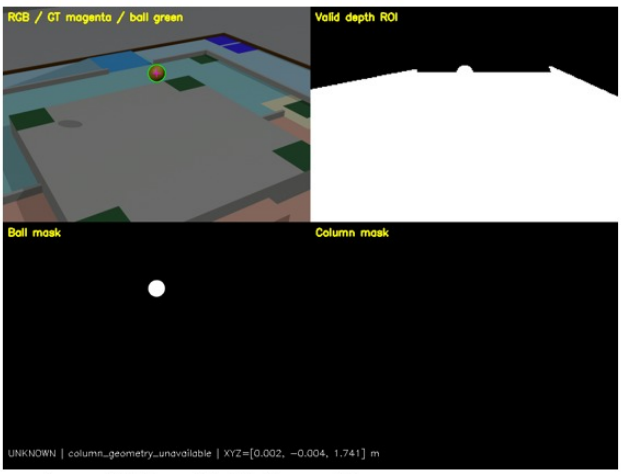
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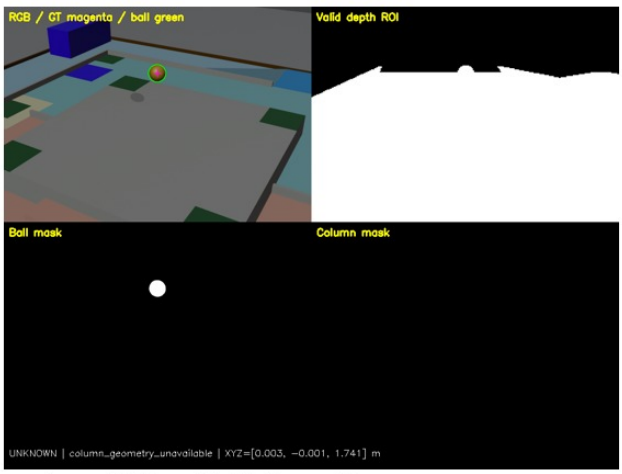
no_column_003_top_h3.50_diagnostic.png



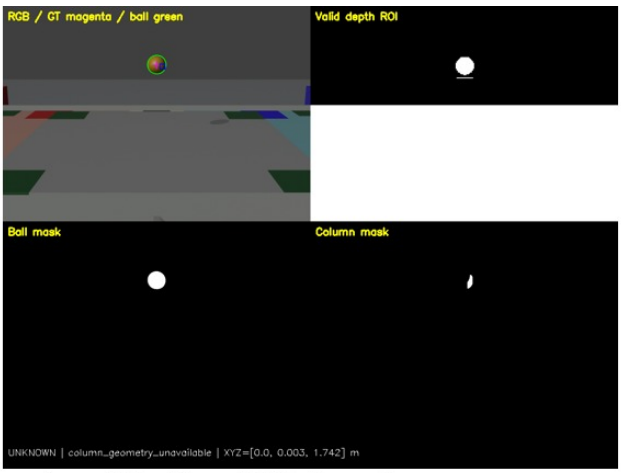
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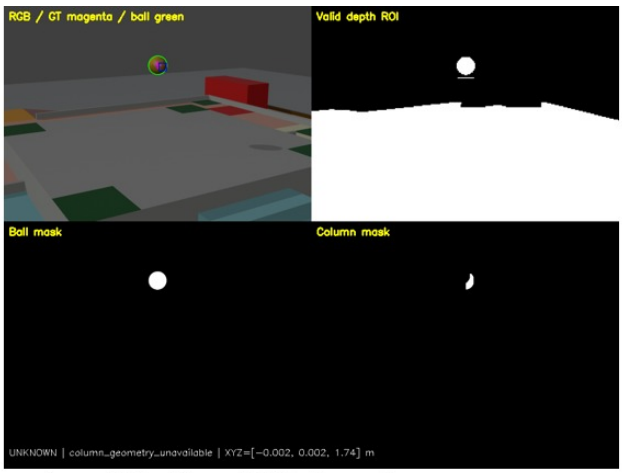
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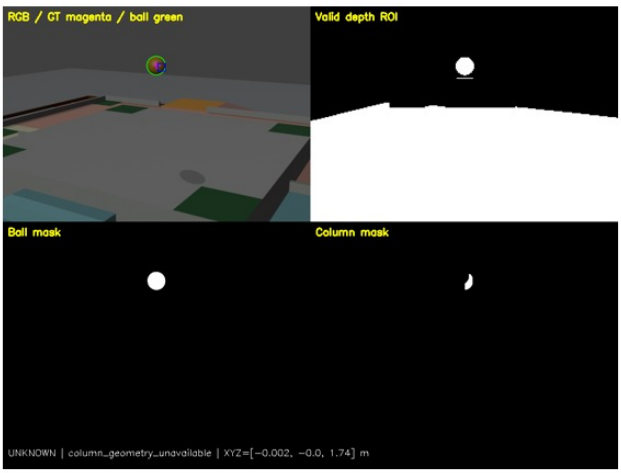
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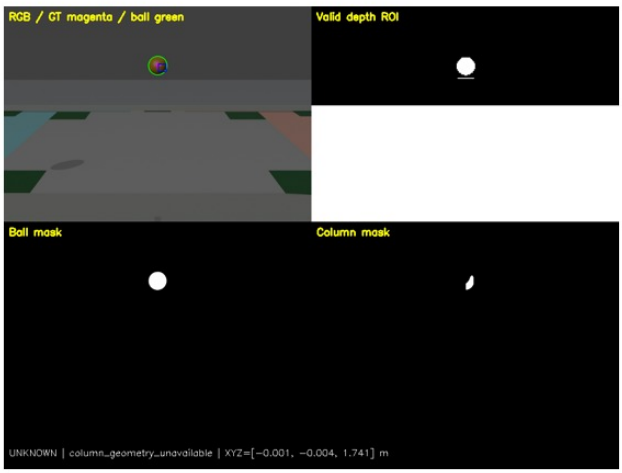
no_column_007_oblique_h2.20_diagnostic.png



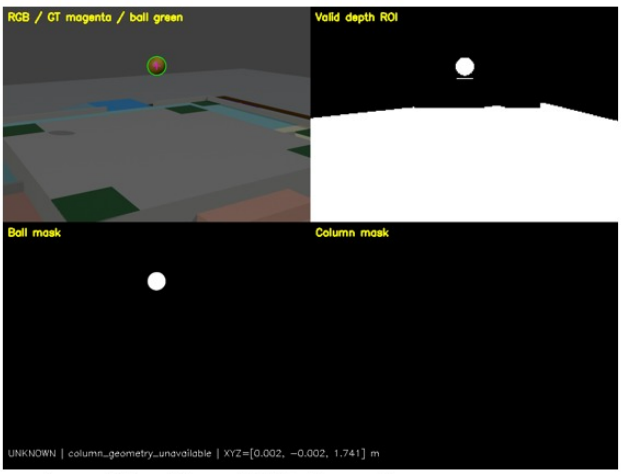
no_column_008_oblique_h2.20_diagnostic.png



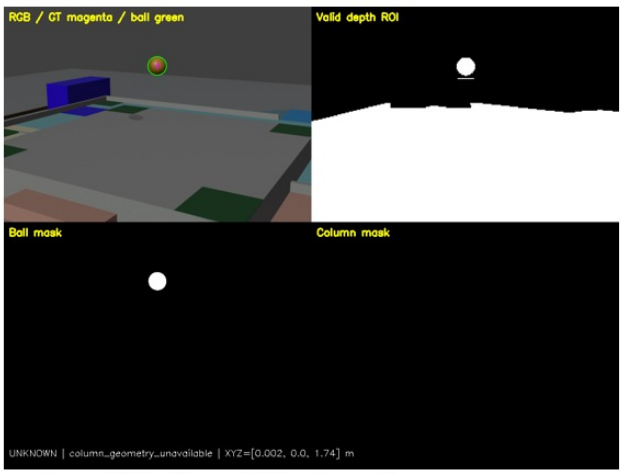
no_column_009_oblique_h2.20_diagnostic.png



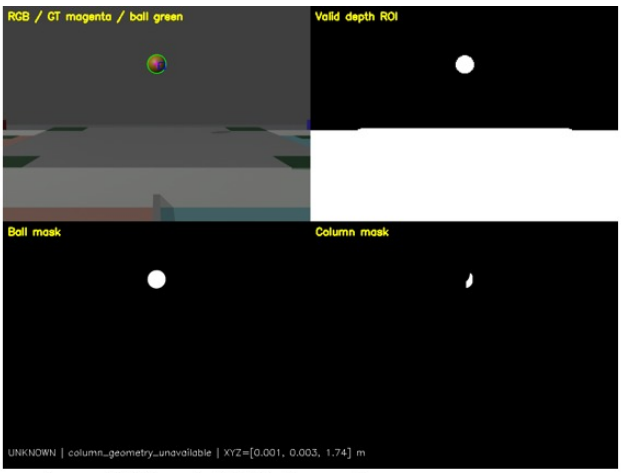
no_column_010_oblique_h2.20_diagnostic.png



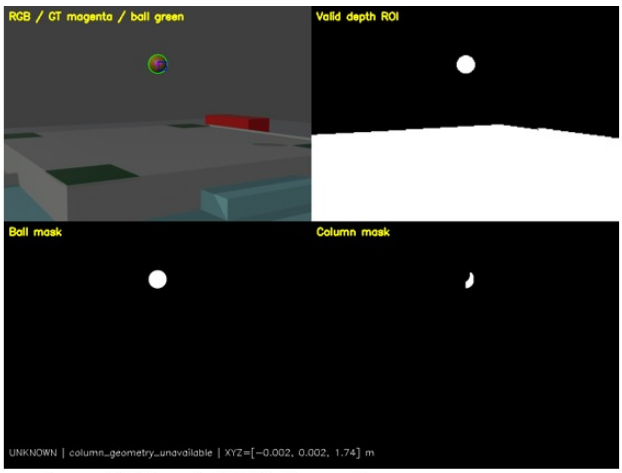
no_column_011_oblique_h2.20_diagnostic.png



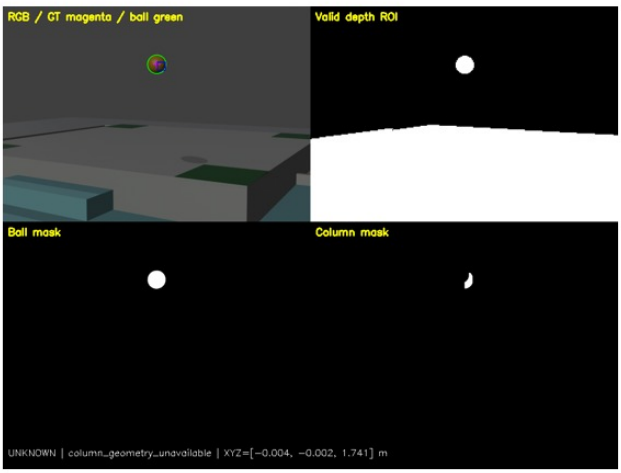
no_column_012_oblique_h2.20_diagnostic.png



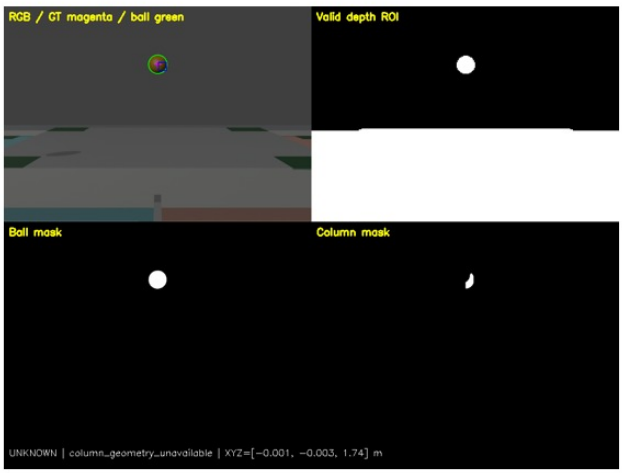
no_column_013_level_h1.50_diagnostic.png



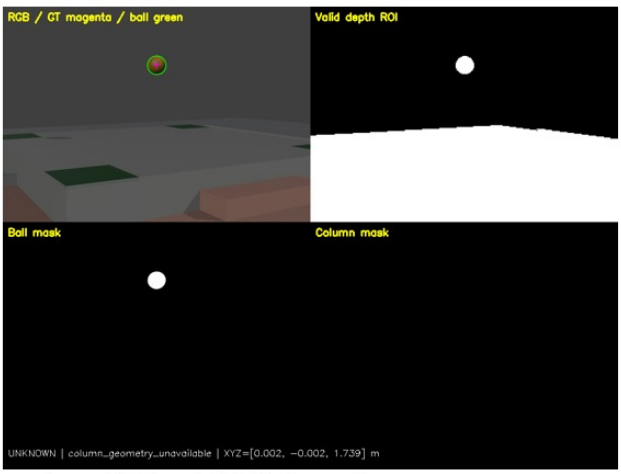
no_column_014_level_h1.50_diagnostic.png



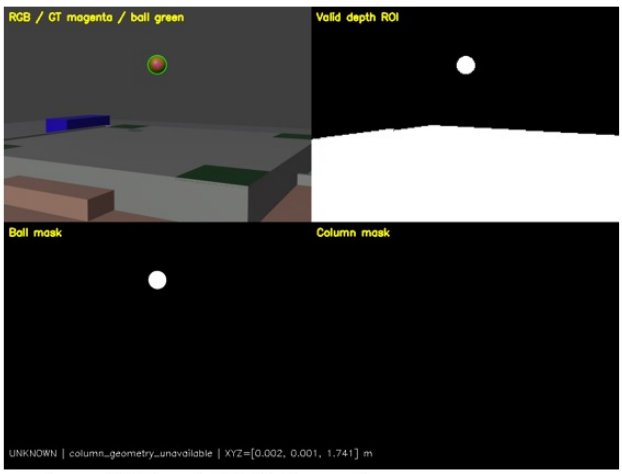
no_column_015_level_h1.50_diagnostic.png



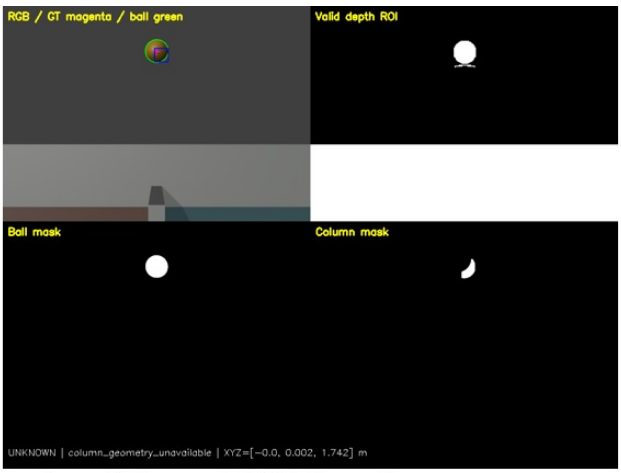
no_column_016_level_h1.50_diagnostic.png



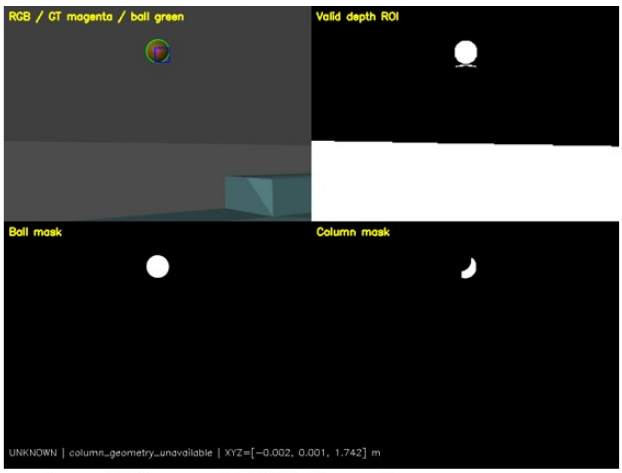
no_column_017_level_h1.50_diagnostic.png



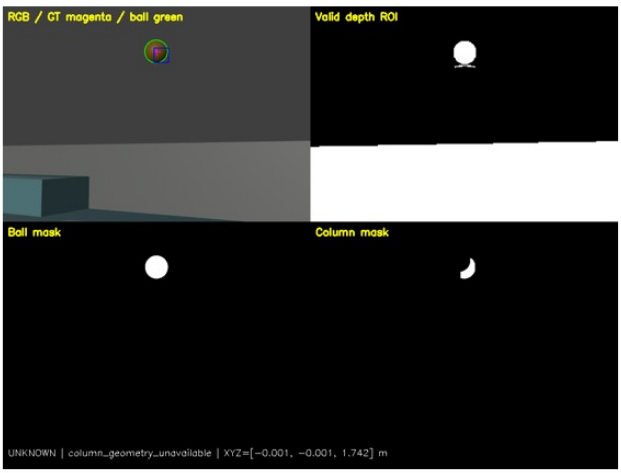
no_column_018_level_h1.50_diagnostic.png



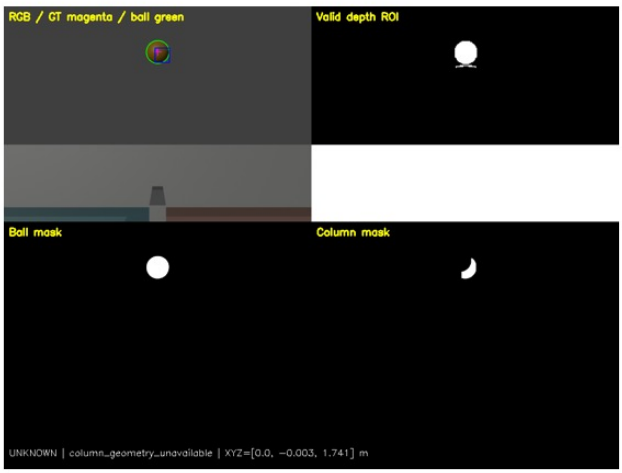
no_column_019_upward_h0.85_diagnostic.png



no_column_020_upward_h0.85_diagnostic.png



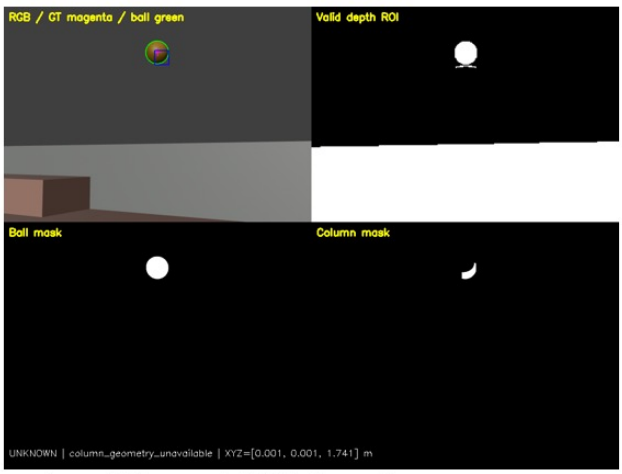
no_column_021_upward_h0.85_diagnostic.png



no_column_022_upward_h0.85_diagnostic.png



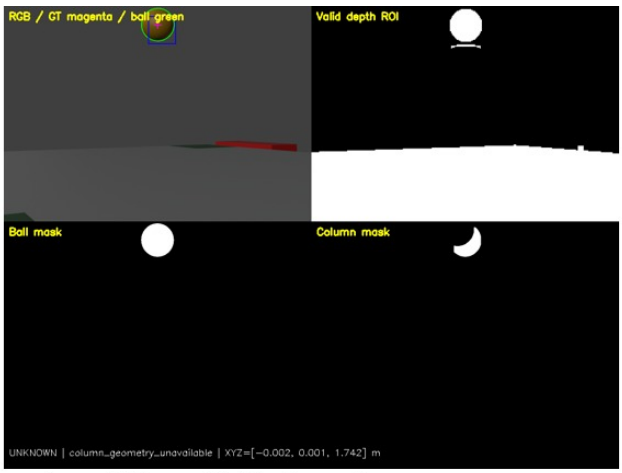
no_column_023_upward_h0.85_diagnostic.png



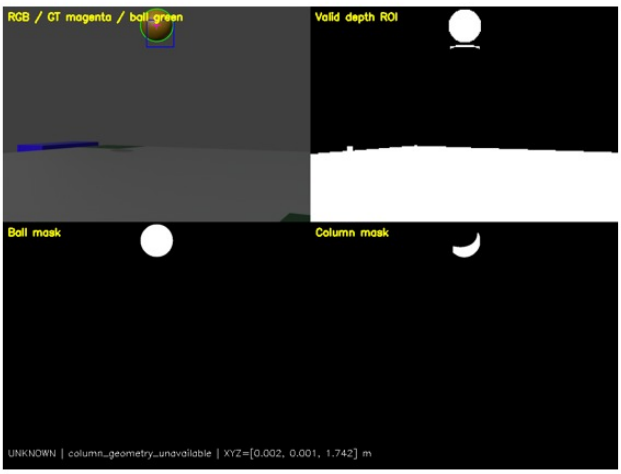
no_column_024_upward_h0.85_diagnostic.png



no_column_025_near_h1.15_diagnostic.png



no_column_026_near_h1.15_diagnostic.png



no_column_030_near_h1.15_diagnostic.png

7 逐图 RGB-D 评测记录

333 张样本逐张计分。球像素误差为候选中心与投影真值的距离；球 3D 误差仅在输出坐标存在时计算。

集合	图像	视角/高度	状态	球	柱	px误差	3D误差cm	原因
positive	001_top_h3.50.png	top / 3.50	ON_COLUMN	是	是	9.2	0.7	rgbd_geometry_ok
positive	002_top_h3.50.png	top / 3.50	ON_COLUMN	是	是	16.1	1.9	rgbd_geometry_ok
positive	003_top_h3.50.png	top / 3.50	ON_COLUMN	是	是	1.3	0.2	rgbd_geometry_ok
positive	004_top_h3.50.png	top / 3.50	ON_COLUMN	是	是	18.3	1.7	rgbd_geometry_ok
positive	005_top_h3.50.png	top / 3.50	ON_COLUMN	是	是	7.1	0.3	rgbd_geometry_ok
positive	006_top_h3.50.png	top / 3.50	ON_COLUMN	是	是	16.1	1.9	rgbd_geometry_ok
positive	007_top_h3.50.png	top / 3.50	ON_COLUMN	是	是	21.6	2.2	rgbd_geometry_ok
positive	008_top_h3.50.png	top / 3.50	ON_COLUMN	是	是	1.9	0.5	rgbd_geometry_ok
positive	009_top_h3.50.png	top / 3.50	ON_COLUMN	是	是	1.3	0.2	rgbd_geometry_ok
positive	010_top_h3.50.png	top / 3.50	ON_COLUMN	是	是	9.4	0.4	rgbd_geometry_ok
positive	011_top_h3.50.png	top / 3.50	ON_COLUMN	是	是	3.3	0.9	rgbd_geometry_ok
positive	012_top_h3.50.png	top / 3.50	ON_COLUMN	是	是	13.3	1.7	rgbd_geometry_ok
positive	013_oblique_h2.20.png	oblique / 2.20	ON_COLUMN	是	是	5.3	1.4	rgbd_geometry_ok
positive	014_oblique_h2.20.png	oblique / 2.20	ON_COLUMN	是	是	2.0	0.5	rgbd_geometry_ok
positive	015_oblique_h2.20.png	oblique / 2.20	ON_COLUMN	是	是	2.0	0.5	rgbd_geometry_ok
positive	016_oblique_h2.20.png	oblique / 2.20	ON_COLUMN	是	是	4.6	1.2	rgbd_geometry_ok
positive	017_oblique_h2.20.png	oblique / 2.20	ON_COLUMN	是	是	2.0	0.5	rgbd_geometry_ok
positive	018_oblique_h2.20.png	oblique / 2.20	ON_COLUMN	是	是	4.7	1.2	rgbd_geometry_ok
positive	019_oblique_h2.20.png	oblique / 2.20	ON_COLUMN	是	是	17.3	1.6	rgbd_geometry_ok
positive	020_oblique_h2.20.png	oblique / 2.20	ON_COLUMN	是	是	4.3	1.0	rgbd_geometry_ok
positive	021_oblique_h2.20.png	oblique / 2.20	ON_COLUMN	是	是	2.5	0.6	rgbd_geometry_ok
positive	022_oblique_h2.20.png	oblique / 2.20	OFF_COLUMN	是	是	18.2	6.4	rgbd_geometry_ok
positive	023_oblique_h2.20.png	oblique / 2.20	ON_COLUMN	是	是	3.3	0.8	rgbd_geometry_ok
positive	024_oblique_h2.20.png	oblique / 2.20	ON_COLUMN	是	是	3.9	1.0	rgbd_geometry_ok
positive	025_level_h1.50.png	level / 1.50	ON_COLUMN	是	是	1.9	0.5	rgbd_geometry_ok
positive	026_level_h1.50.png	level / 1.50	ON_COLUMN	是	是	1.9	0.5	rgbd_geometry_ok
positive	027_level_h1.50.png	level / 1.50	ON_COLUMN	是	是	4.5	1.2	rgbd_geometry_ok
positive	028_level_h1.50.png	level / 1.50	ON_COLUMN	是	是	2.9	0.7	rgbd_geometry_ok
positive	029_level_h1.50.png	level / 1.50	ON_COLUMN	是	是	5.1	1.3	rgbd_geometry_ok
positive	030_level_h1.50.png	level / 1.50	ON_COLUMN	是	是	4.6	1.1	rgbd_geometry_ok
positive	031_level_h1.50.png	level / 1.50	ON_COLUMN	是	是	1.0	0.5	rgbd_geometry_ok
positive	032_level_h1.50.png	level / 1.50	ON_COLUMN	是	是	1.0	0.5	rgbd_geometry_ok
positive	033_level_h1.50.png	level / 1.50	ON_COLUMN	是	是	5.9	1.5	rgbd_geometry_ok
positive	034_level_h1.50.png	level / 1.50	ON_COLUMN	是	是	5.1	1.3	rgbd_geometry_ok
positive	035_level_h1.50.png	level / 1.50	ON_COLUMN	是	是	4.1	1.0	rgbd_geometry_ok
positive	036_level_h1.50.png	level / 1.50	ON_COLUMN	是	是	1.7	0.7	rgbd_geometry_ok
positive	037_upward_h0.85.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
positive	038_upward_h0.85.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
positive	039_upward_h0.85.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
positive	040_upward_h0.85.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
positive	041_upward_h0.85.png	upward / 0.85	ON_COLUMN	是	是	4.4	0.8	rgbd_geometry_ok
positive	042_upward_h0.85.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
positive	043_upward_h0.85.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
positive	044_upward_h0.85.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
positive	045_upward_h0.85.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
positive	046_upward_h0.85.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
positive	047_upward_h0.85.png	upward / 0.85	ON_COLUMN	是	是	3.9	0.6	rgbd_geometry_ok
positive	048_upward_h0.85.png	upward / 0.85	ON_COLUMN	是	是	3.0	0.6	rgbd_geometry_ok
positive	049_near_h1.15.png	near / 1.15	ON_COLUMN	是	是	3.0	0.3	rgbd_geometry_ok
positive	050_near_h1.15.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
positive	051_near_h1.15.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
positive	052_near_h1.15.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
positive	053_near_h1.15.png	near / 1.15	ON_COLUMN	是	是	1.1	0.6	rgbd_geometry_ok
positive	054_near_h1.15.png	near / 1.15	ON_COLUMN	是	是	0.5	0.7	rgbd_geometry_ok
positive	055_near_h1.15.png	near / 1.15	ON_COLUMN	是	是	3.0	0.3	rgbd_geometry_ok
positive	056_near_h1.15.png	near / 1.15	ON_COLUMN	是	是	3.0	0.3	rgbd_geometry_ok
positive	057_near_h1.15.png	near / 1.15	ON_COLUMN	是	是	3.0	0.3	rgbd_geometry_ok
positive	058_near_h1.15.png	near / 1.15	ON_COLUMN	是	是	7.1	0.8	rgbd_geometry_ok
positive	059_near_h1.15.png	near / 1.15	ON_COLUMN	是	是	1.1	0.6	rgbd_geometry_ok
positive	060_near_h1.15.png	near / 1.15	ON_COLUMN	是	是	3.0	0.3	rgbd_geometry_ok
negative	no_ball_001_top_h3.50.png	top / 3.50	UNKNOWN	否	是	—	—	insufficient_ball_colour_support

集合	图像	视角/高度	状态	球	柱	px误差	3D误差cm	原因
negative	no_ball_002_top_h3.50.png	top / 3.50	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_003_top_h3.50.png	top / 3.50	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_004_top_h3.50.png	top / 3.50	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_005_top_h3.50.png	top / 3.50	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_006_top_h3.50.png	top / 3.50	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_007_oblique_h2.20.png	oblique / 2.20	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_008_oblique_h2.20.png	oblique / 2.20	UNKNOWN	否	是	—	—	insufficient_depth_support
negative	no_ball_009_oblique_h2.20.png	oblique / 2.20	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_010_oblique_h2.20.png	oblique / 2.20	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_011_oblique_h2.20.png	oblique / 2.20	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_012_oblique_h2.20.png	oblique / 2.20	UNKNOWN	否	是	—	—	insufficient_depth_support
negative	no_ball_013_level_h1.50.png	level / 1.50	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_014_level_h1.50.png	level / 1.50	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_015_level_h1.50.png	level / 1.50	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_016_level_h1.50.png	level / 1.50	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_017_level_h1.50.png	level / 1.50	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_018_level_h1.50.png	level / 1.50	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_019_upward_h0.85.png	upward / 0.85	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_020_upward_h0.85.png	upward / 0.85	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_021_upward_h0.85.png	upward / 0.85	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_022_upward_h0.85.png	upward / 0.85	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_023_upward_h0.85.png	upward / 0.85	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_024_upward_h0.85.png	upward / 0.85	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_025_near_h1.15.png	near / 1.15	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_026_near_h1.15.png	near / 1.15	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_027_near_h1.15.png	near / 1.15	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_028_near_h1.15.png	near / 1.15	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_029_near_h1.15.png	near / 1.15	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_ball_030_near_h1.15.png	near / 1.15	UNKNOWN	否	是	—	—	insufficient_ball_colour_support
negative	no_column_001_top_h3.50.png	top / 3.50	UNKNOWN	是	否	1.5	0.4	column_geometry_unavailable
negative	no_column_002_top_h3.50.png	top / 3.50	UNKNOWN	是	否	1.5	0.4	column_geometry_unavailable
negative	no_column_003_top_h3.50.png	top / 3.50	UNKNOWN	是	否	0.5	0.6	column_geometry_unavailable
negative	no_column_004_top_h3.50.png	top / 3.50	UNKNOWN	是	否	17.5	1.6	column_geometry_unavailable
negative	no_column_005_top_h3.50.png	top / 3.50	UNKNOWN	是	否	0.5	0.5	column_geometry_unavailable
negative	no_column_006_top_h3.50.png	top / 3.50	UNKNOWN	是	否	1.5	0.4	column_geometry_unavailable
negative	no_column_007_oblique_h2.20.png	oblique / 2.20	UNKNOWN	是	否	2.0	0.3	column_geometry_unavailable
negative	no_column_008_oblique_h2.20.png	oblique / 2.20	UNKNOWN	是	否	0.5	0.5	column_geometry_unavailable
negative	no_column_009_oblique_h2.20.png	oblique / 2.20	UNKNOWN	是	否	1.5	0.4	column_geometry_unavailable
negative	no_column_010_oblique_h2.20.png	oblique / 2.20	UNKNOWN	是	否	1.1	0.5	column_geometry_unavailable
negative	no_column_011_oblique_h2.20.png	oblique / 2.20	UNKNOWN	是	否	1.1	0.4	column_geometry_unavailable

集合	图像	视角/高度	状态	球	柱	px误差	3D误差cm	原因
negative	no_column_012_oblique_h2.20.png	oblique / 2.20	UNKNOWN	是	否	1.5	0.4	column_geometry_unavailable
negative	no_column_013_level_h1.50.png	level / 1.50	UNKNOWN	是	否	0.5	0.5	column_geometry_unavailable
negative	no_column_014_level_h1.50.png	level / 1.50	UNKNOWN	是	否	0.5	0.5	column_geometry_unavailable
negative	no_column_015_level_h1.50.png	level / 1.50	UNKNOWN	是	否	1.0	0.5	column_geometry_unavailable
negative	no_column_016_level_h1.50.png	level / 1.50	UNKNOWN	是	否	0.5	0.5	column_geometry_unavailable
negative	no_column_017_level_h1.50.png	level / 1.50	UNKNOWN	是	否	1.3	0.6	column_geometry_unavailable
negative	no_column_018_level_h1.50.png	level / 1.50	UNKNOWN	是	否	1.7	0.3	column_geometry_unavailable
negative	no_column_019_upward_h0.85.png	upward / 0.85	UNKNOWN	是	否	1.7	0.3	column_geometry_unavailable
negative	no_column_020_upward_h0.85.png	upward / 0.85	UNKNOWN	是	否	1.7	0.3	column_geometry_unavailable
negative	no_column_021_upward_h0.85.png	upward / 0.85	UNKNOWN	是	否	2.2	0.2	column_geometry_unavailable
negative	no_column_022_upward_h0.85.png	upward / 0.85	UNKNOWN	是	否	0.8	0.4	column_geometry_unavailable
negative	no_column_023_upward_h0.85.png	upward / 0.85	UNKNOWN	是	否	0.8	0.4	column_geometry_unavailable
negative	no_column_024_upward_h0.85.png	upward / 0.85	UNKNOWN	是	否	1.6	0.3	column_geometry_unavailable
negative	no_column_025_near_h1.15.png	near / 1.15	UNKNOWN	是	否	1.1	0.2	column_geometry_unavailable
negative	no_column_026_near_h1.15.png	near / 1.15	UNKNOWN	是	否	1.1	0.3	column_geometry_unavailable
negative	no_column_030_near_h1.15.png	near / 1.15	UNKNOWN	是	否	1.1	0.2	column_geometry_unavailable
negative	empty_001_top_h3.50.png	top / 3.50	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_002_top_h3.50.png	top / 3.50	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_003_top_h3.50.png	top / 3.50	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_004_top_h3.50.png	top / 3.50	UNKNOWN	否	否	—	—	insufficient_depth_support
negative	empty_005_top_h3.50.png	top / 3.50	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_006_top_h3.50.png	top / 3.50	UNKNOWN	否	否	—	—	insufficient_depth_support
negative	empty_007_oblique_h2.20.png	oblique / 2.20	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_008_oblique_h2.20.png	oblique / 2.20	UNKNOWN	否	否	—	—	insufficient_depth_support
negative	empty_009_oblique_h2.20.png	oblique / 2.20	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_010_oblique_h2.20.png	oblique / 2.20	UNKNOWN	否	否	—	—	no_ball_candidate
negative	empty_011_oblique_h2.20.png	oblique / 2.20	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_012_oblique_h2.20.png	oblique / 2.20	UNKNOWN	否	否	—	—	insufficient_depth_support
negative	empty_013_level_h1.50.png	level / 1.50	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_014_level_h1.50.png	level / 1.50	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_015_level_h1.50.png	level / 1.50	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_016_level_h1.50.png	level / 1.50	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_017_level_h1.50.png	level / 1.50	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_018_level_h1.50.png	level / 1.50	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_019_upward_h0.85.png	upward / 0.85	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_020_upward_h0.85.png	upward / 0.85	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_021_upward_h0.85.png	upward / 0.85	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_022_upward_h0.85.png	upward / 0.85	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_023_upward_h0.85.png	upward / 0.85	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
negative	empty_024_upward_h0.85.png	upward / 0.85	UNKNOWN	否	否	—	—	insufficient_ball_colour_support
robust	000_normal_top.png	top / 3.50	ON_COLUMN	是	是	9.2	0.7	rgbd_geometry_ok

集合	图像	视角/高度	状态	球	柱	px误差	3D误差cm	原因
robust	000_dark_top.png	top / 3.50	ON_COLUMN	是	是	3.6	1.0	rgbd_geometry_ok
robust	000_bright_top.png	top / 3.50	ON_COLUMN	是	是	1.3	0.2	rgbd_geometry_ok
robust	000_glare_top.png	top / 3.50	ON_COLUMN	是	是	8.8	3.2	rgbd_geometry_ok
robust	000_blur_top.png	top / 3.50	ON_COLUMN	是	是	7.6	2.4	rgbd_geometry_ok
robust	000_noise_top.png	top / 3.50	ON_COLUMN	是	是	14.6	1.3	rgbd_geometry_ok
robust	000_tilt_top.png	top / 3.50	ON_COLUMN	是	是	19.7	2.1	rgbd_geometry_ok
robust	000_side_crop_top.png	top / 3.50	ON_COLUMN	是	是	11.3	0.9	rgbd_geometry_ok
robust	000_occlusion_top.png	top / 3.50	UNKNOWN	否	是	407.8	—	insufficient_ball_colour_support
robust	000_gamma_top.png	top / 3.50	ON_COLUMN	是	是	21.3	2.3	rgbd_geometry_ok
robust	001_normal_top.png	top / 3.50	ON_COLUMN	是	是	16.1	1.9	rgbd_geometry_ok
robust	001_dark_top.png	top / 3.50	ON_COLUMN	是	是	1.3	0.2	rgbd_geometry_ok
robust	001_bright_top.png	top / 3.50	ON_COLUMN	是	是	4.1	1.2	rgbd_geometry_ok
robust	001_glare_top.png	top / 3.50	ON_COLUMN	是	是	1.3	0.2	rgbd_geometry_ok
robust	001_blur_top.png	top / 3.50	ON_COLUMN	是	是	2.8	1.2	rgbd_geometry_ok
robust	001_noise_top.png	top / 3.50	ON_COLUMN	是	是	0.5	0.3	rgbd_geometry_ok
robust	001_tilt_top.png	top / 3.50	ON_COLUMN	是	是	2.2	0.5	rgbd_geometry_ok
robust	001_side_crop_top.png	top / 3.50	ON_COLUMN	是	是	6.4	1.8	rgbd_geometry_ok
robust	001_occlusion_top.png	top / 3.50	UNKNOWN	否	是	211.4	—	insufficient_ball_colour_support
robust	001_gamma_top.png	top / 3.50	ON_COLUMN	是	是	14.7	1.7	rgbd_geometry_ok
robust	002_normal_top.png	top / 3.50	ON_COLUMN	是	是	1.3	0.2	rgbd_geometry_ok
robust	002_dark_top.png	top / 3.50	ON_COLUMN	是	是	1.3	0.2	rgbd_geometry_ok
robust	002_bright_top.png	top / 3.50	ON_COLUMN	是	是	16.2	1.5	rgbd_geometry_ok
robust	002_glare_top.png	top / 3.50	ON_COLUMN	是	是	15.3	1.4	rgbd_geometry_ok
robust	002_blur_top.png	top / 3.50	ON_COLUMN	是	是	2.6	1.0	rgbd_geometry_ok
robust	002_noise_top.png	top / 3.50	ON_COLUMN	是	是	0.5	0.3	rgbd_geometry_ok
robust	002_tilt_top.png	top / 3.50	ON_COLUMN	是	是	7.7	0.3	rgbd_geometry_ok
robust	002_side_crop_top.png	top / 3.50	ON_COLUMN	是	是	7.7	2.2	rgbd_geometry_ok
robust	002_occlusion_top.png	top / 3.50	UNKNOWN	否	是	141.9	—	insufficient_ball_colour_support
robust	002_gamma_top.png	top / 3.50	ON_COLUMN	是	是	15.3	1.4	rgbd_geometry_ok
robust	003_normal_top.png	top / 3.50	ON_COLUMN	是	是	18.3	1.7	rgbd_geometry_ok
robust	003_dark_top.png	top / 3.50	UNKNOWN	是	否	5.5	1.7	column_geometry_unavailable
robust	003_bright_top.png	top / 3.50	ON_COLUMN	是	是	1.3	0.2	rgbd_geometry_ok
robust	003_glare_top.png	top / 3.50	ON_COLUMN	是	是	14.3	1.6	rgbd_geometry_ok
robust	003_blur_top.png	top / 3.50	ON_COLUMN	是	是	5.0	1.6	rgbd_geometry_ok
robust	003_noise_top.png	top / 3.50	ON_COLUMN	是	是	5.5	1.7	rgbd_geometry_ok
robust	003_tilt_top.png	top / 3.50	ON_COLUMN	是	是	1.0	0.5	rgbd_geometry_ok
robust	003_side_crop_top.png	top / 3.50	ON_COLUMN	是	是	19.1	1.8	rgbd_geometry_ok
robust	003_occlusion_top.png	top / 3.50	UNKNOWN	否	是	504.8	—	insufficient_ball_colour_support
robust	003_gamma_top.png	top / 3.50	ON_COLUMN	是	是	15.5	1.8	rgbd_geometry_ok
robust	004_normal_oblique.png	oblique / 2.20	ON_COLUMN	是	是	5.3	1.4	rgbd_geometry_ok
robust	004_dark_oblique.png	oblique / 2.20	ON_COLUMN	是	是	2.5	0.6	rgbd_geometry_ok
robust	004_bright_oblique.png	oblique / 2.20	ON_COLUMN	是	是	0.5	0.6	rgbd_geometry_ok
robust	004_glare_oblique.png	oblique / 2.20	ON_COLUMN	是	是	20.3	1.8	rgbd_geometry_ok
robust	004_blur_oblique.png	oblique / 2.20	ON_COLUMN	是	是	5.3	1.4	rgbd_geometry_ok
robust	004_noise_oblique.png	oblique / 2.20	ON_COLUMN	是	是	17.7	1.2	rgbd_geometry_ok
robust	004_tilt_oblique.png	oblique / 2.20	ON_COLUMN	是	是	11.8	3.9	rgbd_geometry_ok
robust	004_side_crop_oblique.png	oblique / 2.20	ON_COLUMN	是	是	11.3	3.0	rgbd_geometry_ok
robust	004_occlusion_oblique.png	oblique / 2.20	UNKNOWN	否	是	202.4	—	insufficient_ball_colour_support
robust	004_gamma_oblique.png	oblique / 2.20	ON_COLUMN	是	是	2.0	0.5	rgbd_geometry_ok
robust	005_normal_oblique.png	oblique / 2.20	ON_COLUMN	是	是	2.0	0.5	rgbd_geometry_ok
robust	005_dark_oblique.png	oblique / 2.20	ON_COLUMN	是	是	2.0	0.5	rgbd_geometry_ok
robust	005_bright_oblique.png	oblique / 2.20	ON_COLUMN	是	是	2.0	0.5	rgbd_geometry_ok
robust	005_glare_oblique.png	oblique / 2.20	ON_COLUMN	是	是	2.1	1.1	rgbd_geometry_ok
robust	005_blur_oblique.png	oblique / 2.20	ON_COLUMN	是	是	18.2	1.6	rgbd_geometry_ok
robust	005_noise_oblique.png	oblique / 2.20	ON_COLUMN	是	是	26.9	3.0	rgbd_geometry_ok
robust	005_tilt_oblique.png	oblique / 2.20	ON_COLUMN	是	是	1.6	0.4	rgbd_geometry_ok
robust	005_side_crop_oblique.png	oblique / 2.20	ON_COLUMN	是	是	9.9	2.6	rgbd_geometry_ok
robust	005_occlusion_oblique.png	oblique / 2.20	UNKNOWN	否	是	7.8	—	insufficient_ball_colour_support
robust	005_gamma_oblique.png	oblique / 2.20	ON_COLUMN	是	是	3.9	1.0	rgbd_geometry_ok
robust	006_normal_oblique.png	oblique / 2.20	ON_COLUMN	是	是	2.0	0.5	rgbd_geometry_ok
robust	006_dark_oblique.png	oblique / 2.20	ON_COLUMN	是	是	1.5	0.7	rgbd_geometry_ok

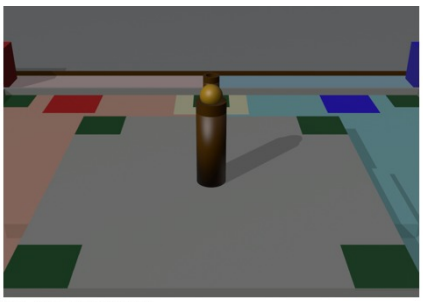
集合	图像	视角/高度	状态	球	柱	px误差	3D误差cm	原因
robust	006_bright_oblique.png	oblique / 2.20	ON_COLUMN	是	是	2.0	0.5	rgbd_geometry_ok
robust	006_glare_oblique.png	oblique / 2.20	ON_COLUMN	是	是	1.1	0.4	rgbd_geometry_ok
robust	006_blur_oblique.png	oblique / 2.20	ON_COLUMN	是	是	3.2	0.9	rgbd_geometry_ok
robust	006_noise_oblique.png	oblique / 2.20	ON_COLUMN	是	是	1.1	0.4	rgbd_geometry_ok
robust	006_tilt_oblique.png	oblique / 2.20	ON_COLUMN	是	是	1.7	0.9	rgbd_geometry_ok
robust	006_side_crop_oblique.png	oblique / 2.20	ON_COLUMN	是	是	7.8	2.0	rgbd_geometry_ok
robust	006_occlusion_oblique.png	oblique / 2.20	UNKNOWN	否	是	442.6	—	insufficient_depth_support
robust	006_gamma_oblique.png	oblique / 2.20	ON_COLUMN	是	是	2.0	0.5	rgbd_geometry_ok
robust	007_normal_oblique.png	oblique / 2.20	ON_COLUMN	是	是	4.6	1.2	rgbd_geometry_ok
robust	007_bright_oblique.png	oblique / 2.20	ON_COLUMN	是	是	19.0	1.8	rgbd_geometry_ok
robust	007_glare_oblique.png	oblique / 2.20	ON_COLUMN	是	是	23.0	2.3	rgbd_geometry_ok
robust	007_blur_oblique.png	oblique / 2.20	ON_COLUMN	是	是	1.5	0.7	rgbd_geometry_ok
robust	007_noise_oblique.png	oblique / 2.20	ON_COLUMN	是	是	18.6	1.4	rgbd_geometry_ok
robust	007_tilt_oblique.png	oblique / 2.20	ON_COLUMN	是	是	19.8	1.8	rgbd_geometry_ok
robust	007_side_crop_oblique.png	oblique / 2.20	ON_COLUMN	是	是	11.3	3.0	rgbd_geometry_ok
robust	007_occlusion_oblique.png	oblique / 2.20	UNKNOWN	否	是	5.3	—	insufficient_ball_colour_support
robust	007_gamma_oblique.png	oblique / 2.20	ON_COLUMN	是	是	5.6	1.5	rgbd_geometry_ok
robust	008_normal_level.png	level / 1.50	ON_COLUMN	是	是	1.9	0.5	rgbd_geometry_ok
robust	008_dark_level.png	level / 1.50	ON_COLUMN	是	是	5.3	1.5	rgbd_geometry_ok
robust	008_bright_level.png	level / 1.50	ON_COLUMN	是	是	1.9	0.5	rgbd_geometry_ok
robust	008_glare_level.png	level / 1.50	ON_COLUMN	是	是	3.8	1.0	rgbd_geometry_ok
robust	008_blur_level.png	level / 1.50	ON_COLUMN	是	是	6.0	1.6	rgbd_geometry_ok
robust	008_noise_level.png	level / 1.50	ON_COLUMN	是	是	2.4	0.6	rgbd_geometry_ok
robust	008_tilt_level.png	level / 1.50	OFF_COLUMN	是	是	23.7	8.4	rgbd_geometry_ok
robust	008_side_crop_level.png	level / 1.50	ON_COLUMN	是	是	5.6	2.3	rgbd_geometry_ok
robust	008_occlusion_level.png	level / 1.50	UNKNOWN	否	是	6.4	—	insufficient_ball_colour_support
robust	008_gamma_level.png	level / 1.50	ON_COLUMN	是	是	1.9	0.5	rgbd_geometry_ok
robust	009_normal_level.png	level / 1.50	ON_COLUMN	是	是	1.9	0.5	rgbd_geometry_ok
robust	009_dark_level.png	level / 1.50	ON_COLUMN	是	是	1.0	0.5	rgbd_geometry_ok
robust	009_bright_level.png	level / 1.50	ON_COLUMN	是	是	0.5	0.7	rgbd_geometry_ok
robust	009_glare_level.png	level / 1.50	ON_COLUMN	是	是	1.0	0.5	rgbd_geometry_ok
robust	009_blur_level.png	level / 1.50	ON_COLUMN	是	是	0.5	0.7	rgbd_geometry_ok
robust	009_noise_level.png	level / 1.50	ON_COLUMN	是	是	2.9	0.7	rgbd_geometry_ok
robust	009_tilt_level.png	level / 1.50	ON_COLUMN	是	是	1.6	0.5	rgbd_geometry_ok
robust	009_side_crop_level.png	level / 1.50	ON_COLUMN	是	是	8.0	2.0	rgbd_geometry_ok
robust	009_occlusion_level.png	level / 1.50	UNKNOWN	否	是	5.5	—	insufficient_ball_colour_support
robust	009_gamma_level.png	level / 1.50	ON_COLUMN	是	是	1.9	0.5	rgbd_geometry_ok
robust	010_normal_level.png	level / 1.50	ON_COLUMN	是	是	4.5	1.2	rgbd_geometry_ok
robust	010_bright_level.png	level / 1.50	ON_COLUMN	是	是	1.9	0.5	rgbd_geometry_ok
robust	010_glare_level.png	level / 1.50	ON_COLUMN	是	是	0.5	0.7	rgbd_geometry_ok
robust	010_blur_level.png	level / 1.50	ON_COLUMN	是	是	0.5	0.7	rgbd_geometry_ok
robust	010_noise_level.png	level / 1.50	ON_COLUMN	是	是	4.6	1.1	rgbd_geometry_ok
robust	010_tilt_level.png	level / 1.50	ON_COLUMN	是	是	1.6	0.5	rgbd_geometry_ok
robust	010_side_crop_level.png	level / 1.50	ON_COLUMN	是	是	5.0	1.9	rgbd_geometry_ok
robust	010_occlusion_level.png	level / 1.50	UNKNOWN	否	是	3.2	—	insufficient_ball_colour_support
robust	010_gamma_level.png	level / 1.50	ON_COLUMN	是	是	1.9	0.5	rgbd_geometry_ok
robust	011_normal_level.png	level / 1.50	ON_COLUMN	是	是	2.9	0.7	rgbd_geometry_ok
robust	011_bright_level.png	level / 1.50	ON_COLUMN	是	是	0.5	0.7	rgbd_geometry_ok
robust	011_glare_level.png	level / 1.50	ON_COLUMN	是	是	7.5	2.4	rgbd_geometry_ok
robust	011_blur_level.png	level / 1.50	UNKNOWN	否	是	109.7	—	insufficient_ball_colour_support
robust	011_noise_level.png	level / 1.50	ON_COLUMN	是	是	1.0	0.5	rgbd_geometry_ok
robust	011_tilt_level.png	level / 1.50	ON_COLUMN	是	是	1.6	0.5	rgbd_geometry_ok
robust	011_side_crop_level.png	level / 1.50	ON_COLUMN	是	是	5.0	1.9	rgbd_geometry_ok
robust	011_occlusion_level.png	level / 1.50	UNKNOWN	否	是	3.8	—	insufficient_ball_colour_support
robust	011_gamma_level.png	level / 1.50	ON_COLUMN	是	是	1.7	0.7	rgbd_geometry_ok
robust	012_normal_upward.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
robust	012_dark_upward.png	upward / 0.85	ON_COLUMN	是	是	2.1	1.1	rgbd_geometry_ok
robust	012_bright_upward.png	upward / 0.85	ON_COLUMN	是	是	2.2	0.6	rgbd_geometry_ok
robust	012_glare_upward.png	upward / 0.85	ON_COLUMN	是	是	1.6	0.7	rgbd_geometry_ok
robust	012_blur_upward.png	upward / 0.85	ON_COLUMN	是	是	3.6	0.7	rgbd_geometry_ok
robust	012_noise_upward.png	upward / 0.85	ON_COLUMN	是	是	0.8	0.7	rgbd_geometry_ok

集合	图像	视角/高度	状态	球	柱	px误差	3D误差cm	原因
robust	012_tilt_upward.png	upward / 0.85	ON_COLUMN	是	是	3.2	0.6	rgbd_geometry_ok
robust	012_side_crop_upward.png	upward / 0.85	ON_COLUMN	是	是	11.8	2.3	rgbd_geometry_ok
robust	012_occlusion_upward.png	upward / 0.85	UNKNOWN	是	否	1.7	0.5	column_geometry_unavailable
robust	012_gamma_upward.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
robust	013_normal_upward.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
robust	013_dark_upward.png	upward / 0.85	ON_COLUMN	是	是	2.6	0.5	rgbd_geometry_ok
robust	013_bright_upward.png	upward / 0.85	ON_COLUMN	是	是	0.8	0.7	rgbd_geometry_ok
robust	013_glare_upward.png	upward / 0.85	ON_COLUMN	是	是	0.8	0.7	rgbd_geometry_ok
robust	013_blur_upward.png	upward / 0.85	ON_COLUMN	是	是	0.8	0.7	rgbd_geometry_ok
robust	013_noise_upward.png	upward / 0.85	ON_COLUMN	是	是	0.8	0.7	rgbd_geometry_ok
robust	013_tilt_upward.png	upward / 0.85	ON_COLUMN	是	是	1.1	0.6	rgbd_geometry_ok
robust	013_side_crop_upward.png	upward / 0.85	ON_COLUMN	是	是	3.1	1.4	rgbd_geometry_ok
robust	013_occlusion_upward.png	upward / 0.85	UNKNOWN	是	否	1.7	0.5	column_geometry_unavailable
robust	013_gamma_upward.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
robust	014_normal_upward.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
robust	014_bright_upward.png	upward / 0.85	ON_COLUMN	是	是	0.8	0.7	rgbd_geometry_ok
robust	014_glare_upward.png	upward / 0.85	ON_COLUMN	是	是	0.8	0.7	rgbd_geometry_ok
robust	014_blur_upward.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
robust	014_noise_upward.png	upward / 0.85	ON_COLUMN	是	是	2.2	0.6	rgbd_geometry_ok
robust	014_tilt_upward.png	upward / 0.85	ON_COLUMN	是	是	2.5	0.5	rgbd_geometry_ok
robust	014_side_crop_upward.png	upward / 0.85	ON_COLUMN	是	是	6.1	1.6	rgbd_geometry_ok
robust	014_occlusion_upward.png	upward / 0.85	UNKNOWN	是	否	1.7	0.5	column_geometry_unavailable
robust	014_gamma_upward.png	upward / 0.85	ON_COLUMN	是	是	0.8	0.7	rgbd_geometry_ok
robust	015_normal_upward.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
robust	015_dark_upward.png	upward / 0.85	ON_COLUMN	是	是	11.3	2.4	rgbd_geometry_ok
robust	015_bright_upward.png	upward / 0.85	ON_COLUMN	是	是	0.8	0.7	rgbd_geometry_ok
robust	015_glare_upward.png	upward / 0.85	ON_COLUMN	是	是	0.8	0.7	rgbd_geometry_ok
robust	015_blur_upward.png	upward / 0.85	OFF_COLUMN	是	是	13.6	3.5	rgbd_geometry_ok
robust	015_noise_upward.png	upward / 0.85	ON_COLUMN	是	是	5.7	1.2	rgbd_geometry_ok
robust	015_tilt_upward.png	upward / 0.85	ON_COLUMN	是	是	5.3	0.9	rgbd_geometry_ok
robust	015_side_crop_upward.png	upward / 0.85	ON_COLUMN	是	是	6.4	1.6	rgbd_geometry_ok
robust	015_occlusion_upward.png	upward / 0.85	UNKNOWN	是	否	1.7	0.5	column_geometry_unavailable
robust	015_gamma_upward.png	upward / 0.85	ON_COLUMN	是	是	1.7	0.5	rgbd_geometry_ok
robust	016_normal_near.png	near / 1.15	ON_COLUMN	是	是	3.0	0.3	rgbd_geometry_ok
robust	016_bright_near.png	near / 1.15	ON_COLUMN	是	是	1.8	0.6	rgbd_geometry_ok
robust	016_glare_near.png	near / 1.15	ON_COLUMN	是	是	3.0	0.3	rgbd_geometry_ok
robust	016_blur_near.png	near / 1.15	ON_COLUMN	是	是	3.3	0.4	rgbd_geometry_ok
robust	016_noise_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	016_tilt_near.png	near / 1.15	ON_COLUMN	是	是	0.6	0.6	rgbd_geometry_ok
robust	016_side_crop_near.png	near / 1.15	ON_COLUMN	是	是	13.0	1.8	rgbd_geometry_ok
robust	016_occlusion_near.png	near / 1.15	ON_COLUMN	是	是	3.0	0.3	rgbd_geometry_ok
robust	016_gamma_near.png	near / 1.15	ON_COLUMN	是	是	3.0	0.3	rgbd_geometry_ok
robust	017_normal_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	017_bright_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	017_glare_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	017_blur_near.png	near / 1.15	ON_COLUMN	是	是	1.1	0.6	rgbd_geometry_ok
robust	017_noise_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	017_tilt_near.png	near / 1.15	ON_COLUMN	是	是	2.3	0.4	rgbd_geometry_ok
robust	017_side_crop_near.png	near / 1.15	ON_COLUMN	是	是	1.2	0.7	rgbd_geometry_ok
robust	017_occlusion_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	017_gamma_near.png	near / 1.15	ON_COLUMN	是	是	2.5	0.8	rgbd_geometry_ok
robust	018_normal_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	018_bright_near.png	near / 1.15	ON_COLUMN	是	是	1.1	0.6	rgbd_geometry_ok
robust	018_glare_near.png	near / 1.15	ON_COLUMN	是	是	4.2	0.4	rgbd_geometry_ok
robust	018_blur_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	018_noise_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	018_tilt_near.png	near / 1.15	ON_COLUMN	是	是	2.3	0.4	rgbd_geometry_ok
robust	018_side_crop_near.png	near / 1.15	ON_COLUMN	是	是	17.5	2.4	rgbd_geometry_ok
robust	018_occlusion_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	018_gamma_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	019_normal_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	019_bright_near.png	near / 1.15	ON_COLUMN	是	是	4.2	0.4	rgbd_geometry_ok
robust	019_glare_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok

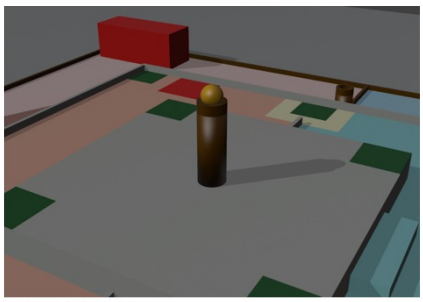
集合	图像	视角/高度	状态	球	柱	px误差	3D误差cm	原因
robust	019_blur_near.png	near / 1.15	ON_COLUMN	是	是	1.8	0.6	rgbd_geometry_ok
robust	019_noise_near.png	near / 1.15	ON_COLUMN	是	是	16.6	2.8	rgbd_geometry_ok
robust	019_tilt_near.png	near / 1.15	ON_COLUMN	是	是	2.7	0.3	rgbd_geometry_ok
robust	019_side_crop_near.png	near / 1.15	ON_COLUMN	是	是	19.6	2.7	rgbd_geometry_ok
robust	019_occlusion_near.png	near / 1.15	ON_COLUMN	是	是	2.0	0.4	rgbd_geometry_ok
robust	019_gamma_near.png	near / 1.15	ON_COLUMN	是	是	3.3	0.4	rgbd_geometry_ok

8 图集和工程接口

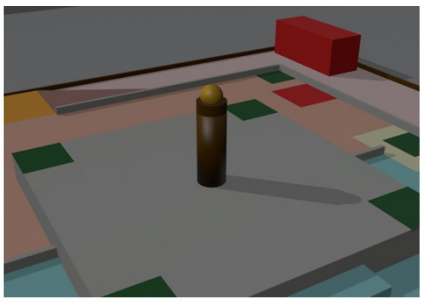
Blender 有目标视角集：60 张，第 1 页



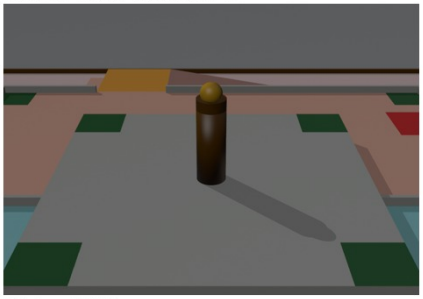
001_top_h3.50.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



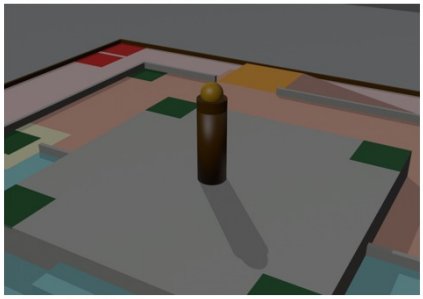
002_top_h3.50.png
top h=3.50m
cam XYZ=(3.90, 2.25, 3.50) m



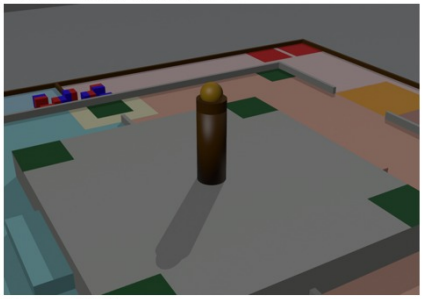
003_top_h3.50.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m



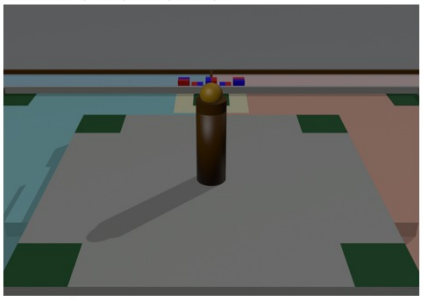
004_top_h3.50.png
top h=3.50m
cam XYZ=(0.00, 4.50, 3.50) m



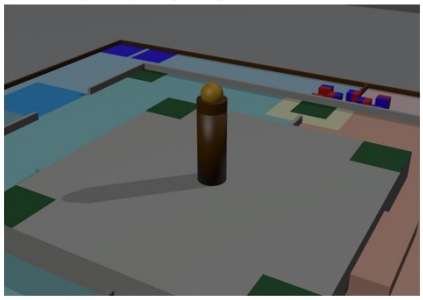
005_top_h3.50.png
top h=3.50m
cam XYZ=(-2.25, 3.90, 3.50) m



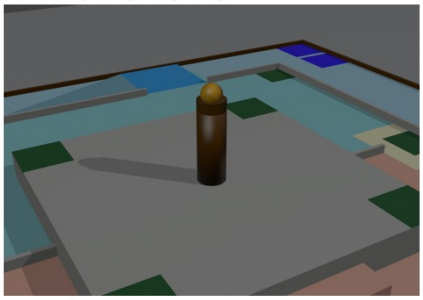
006_top_h3.50.png
top h=3.50m
cam XYZ=(-3.90, 2.25, 3.50) m



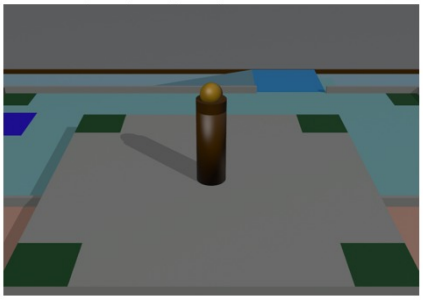
007_top_h3.50.png
top h=3.50m
cam XYZ=(-4.50, 0.00, 3.50) m



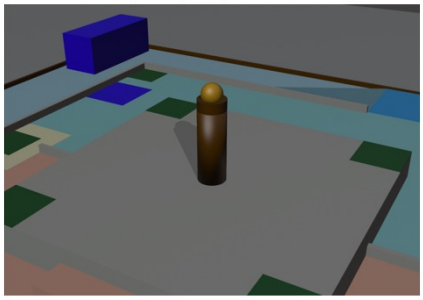
008_top_h3.50.png
top h=3.50m
cam XYZ=(-3.90, -2.25, 3.50) m



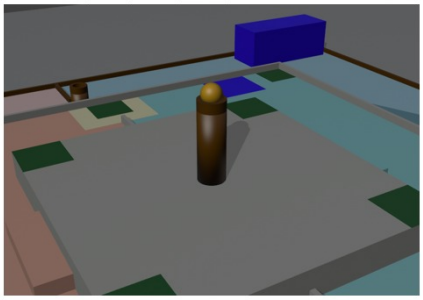
009_top_h3.50.png
top h=3.50m
cam XYZ=(-2.25, -3.90, 3.50) m



010_top_h3.50.png
top h=3.50m
cam XYZ=(-0.00, -4.50, 3.50) m

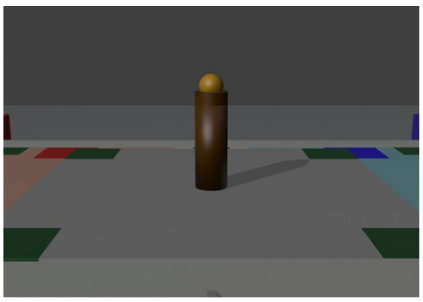


011_top_h3.50.png
top h=3.50m
cam XYZ=(2.25, -3.90, 3.50) m

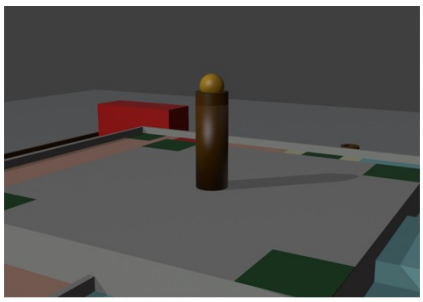


012_top_h3.50.png
top h=3.50m
cam XYZ=(3.90, -2.25, 3.50) m

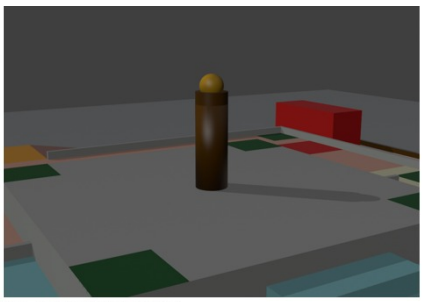
Blender 有目标视角集：60 张，第 2 页



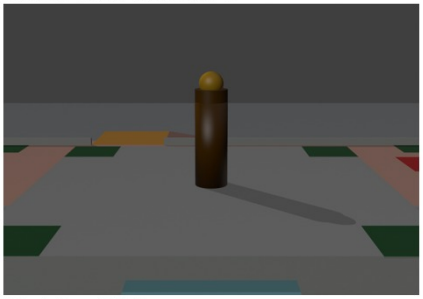
013_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m



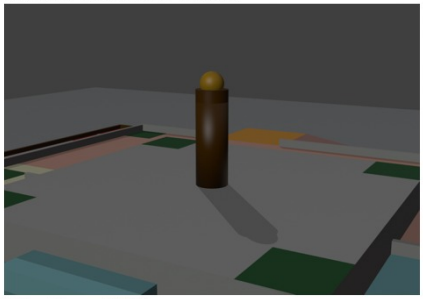
014_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(3.90, 2.25, 2.20) m



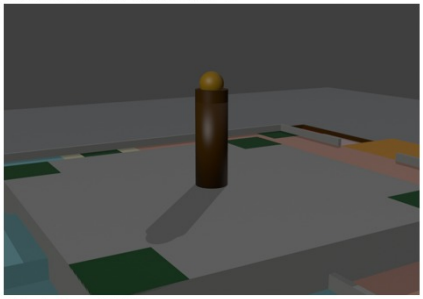
015_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



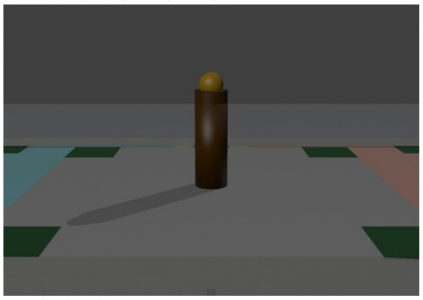
016_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(0.00, 4.50, 2.20) m



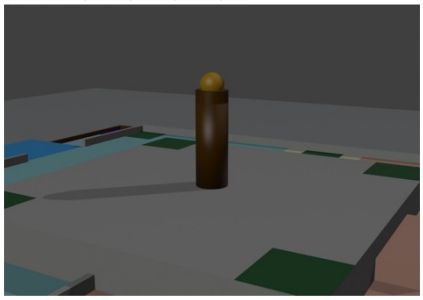
017_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-2.25, 3.90, 2.20) m



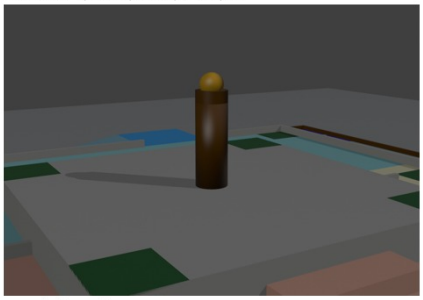
018_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-3.90, 2.25, 2.20) m



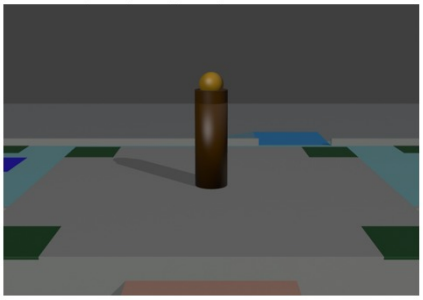
019_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-4.50, 0.00, 2.20) m



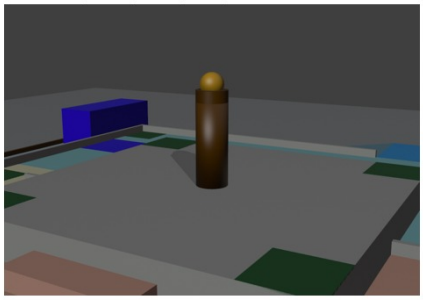
020_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-3.90, -2.25, 2.20) m



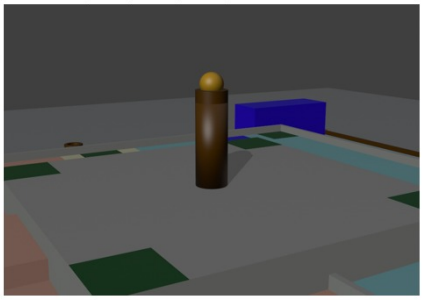
021_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-2.25, -3.90, 2.20) m



022_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-0.00, -4.50, 2.20) m

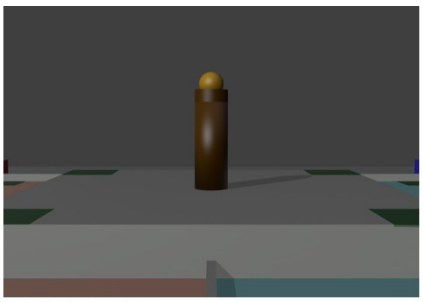


023_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(2.25, -3.90, 2.20) m

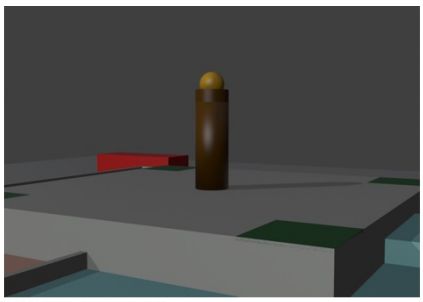


024_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(3.90, -2.25, 2.20) m

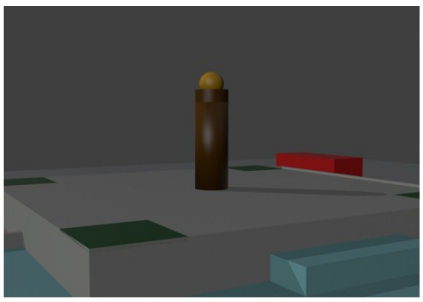
Blender 有目标视角集：60 张，第 3 页



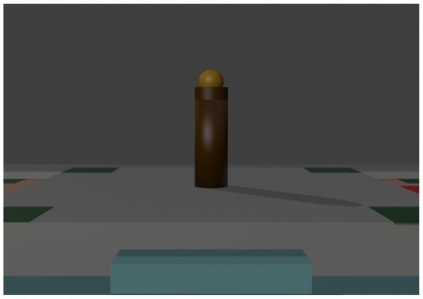
025_level_h1.50.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



026_level_h1.50.png
level h=1.50m
cam XYZ=(3.90, 2.25, 1.50) m



027_level_h1.50.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



028_level_h1.50.png
level h=1.50m
cam XYZ=(0.00, 4.50, 1.50) m



029_level_h1.50.png
level h=1.50m
cam XYZ=(-2.25, 3.90, 1.50) m



030_level_h1.50.png
level h=1.50m
cam XYZ=(-3.90, 2.25, 1.50) m



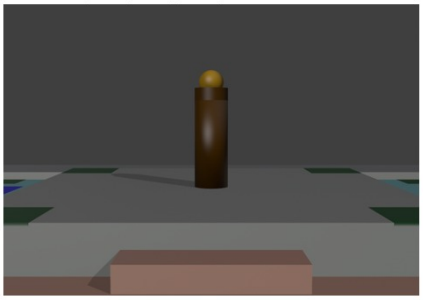
031_level_h1.50.png
level h=1.50m
cam XYZ=(-4.50, 0.00, 1.50) m



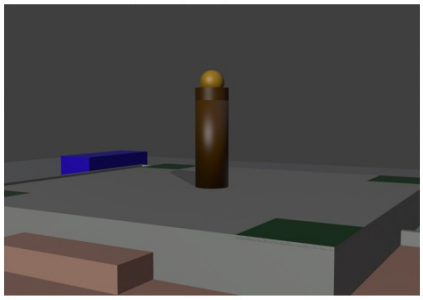
032_level_h1.50.png
level h=1.50m
cam XYZ=(-3.90, -2.25, 1.50) m



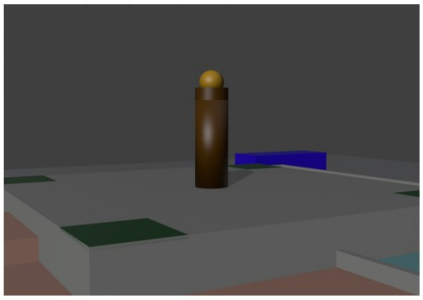
033_level_h1.50.png
level h=1.50m
cam XYZ=(-2.25, -3.90, 1.50) m



034_level_h1.50.png
level h=1.50m
cam XYZ=(-0.00, -4.50, 1.50) m

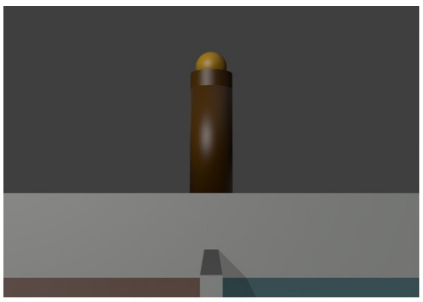


035_level_h1.50.png
level h=1.50m
cam XYZ=(2.25, -3.90, 1.50) m

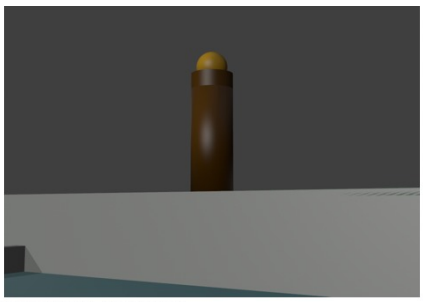


036_level_h1.50.png
level h=1.50m
cam XYZ=(3.90, -2.25, 1.50) m

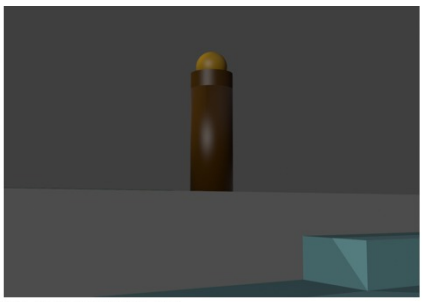
Blender 有目标视角集：60 张，第 4 页



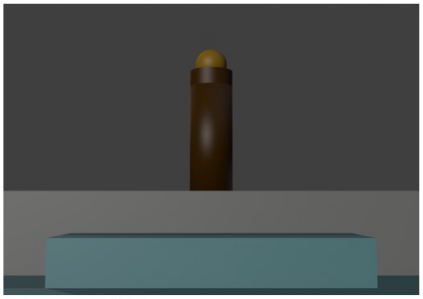
037_upward_h0.85.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m



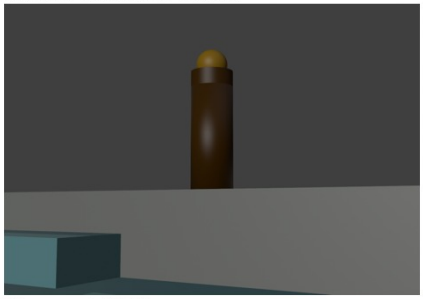
038_upward_h0.85.png
upward h=0.85m
cam XYZ=(3.03, 1.75, 0.85) m



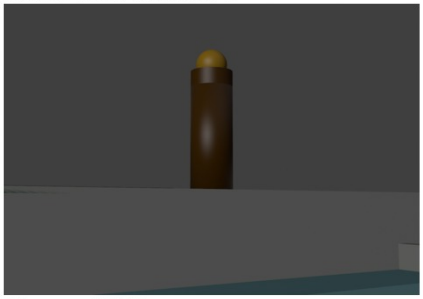
039_upward_h0.85.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



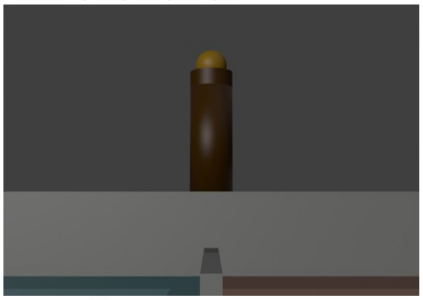
040_upward_h0.85.png
upward h=0.85m
cam XYZ=(0.00, 3.50, 0.85) m



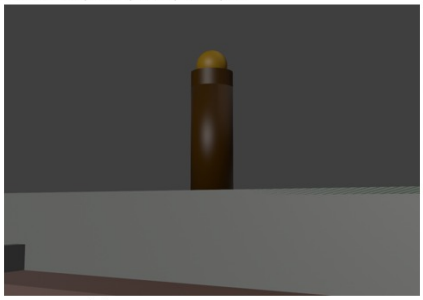
041_upward_h0.85.png
upward h=0.85m
cam XYZ=(-1.75, 3.03, 0.85) m



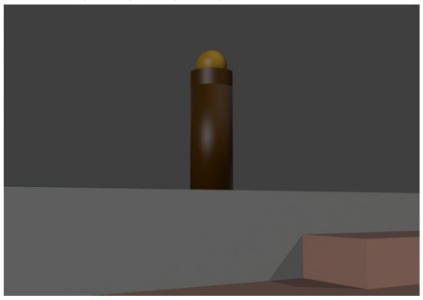
042_upward_h0.85.png
upward h=0.85m
cam XYZ=(-3.03, 1.75, 0.85) m



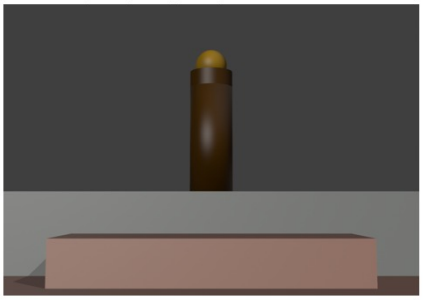
043_upward_h0.85.png
upward h=0.85m
cam XYZ=(-3.50, 0.00, 0.85) m



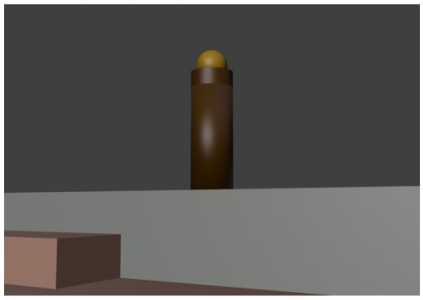
044_upward_h0.85.png
upward h=0.85m
cam XYZ=(-3.03, -1.75, 0.85) m



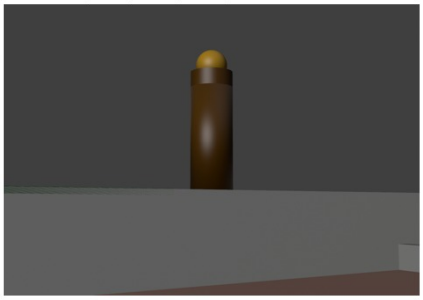
045_upward_h0.85.png
upward h=0.85m
cam XYZ=(-1.75, -3.03, 0.85) m



046_upward_h0.85.png
upward h=0.85m
cam XYZ=(-0.00, -3.50, 0.85) m

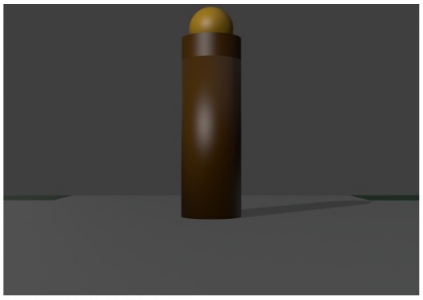


047_upward_h0.85.png
upward h=0.85m
cam XYZ=(1.75, -3.03, 0.85) m

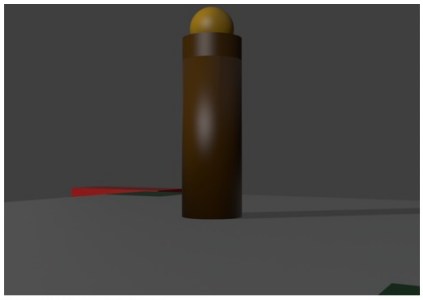


048_upward_h0.85.png
upward h=0.85m
cam XYZ=(3.03, -1.75, 0.85) m

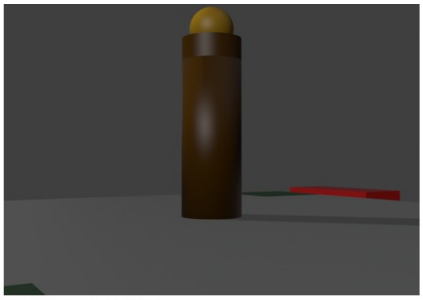
Blender 有目标视角集：60 张，第 5 页



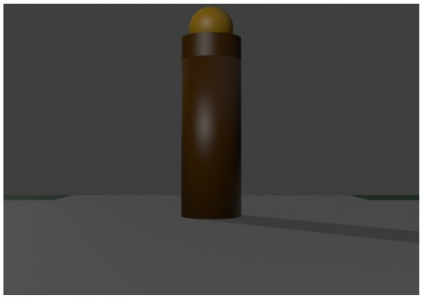
049_near_h1.15.png
near h=1.15m
cam XYZ=(2.50, 0.00, 1.15) m



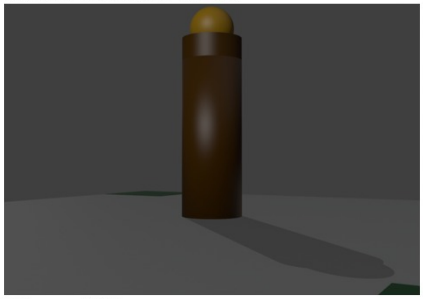
050_near_h1.15.png
near h=1.15m
cam XYZ=(2.17, 1.25, 1.15) m



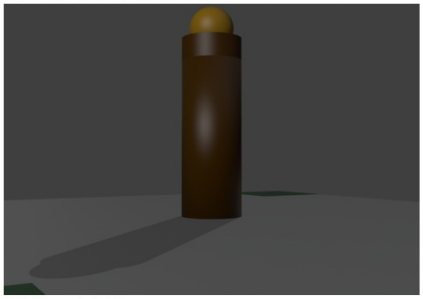
051_near_h1.15.png
near h=1.15m
cam XYZ=(1.25, 2.17, 1.15) m



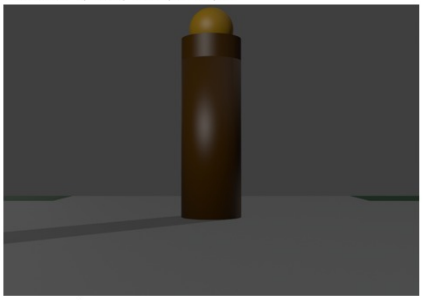
052_near_h1.15.png
near h=1.15m
cam XYZ=(0.00, 2.50, 1.15) m



053_near_h1.15.png
near h=1.15m
cam XYZ=(-1.25, 2.17, 1.15) m



054_near_h1.15.png
near h=1.15m
cam XYZ=(-2.17, 1.25, 1.15) m



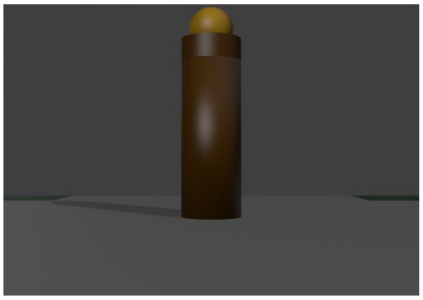
055_near_h1.15.png
near h=1.15m
cam XYZ=(-2.50, 0.00, 1.15) m



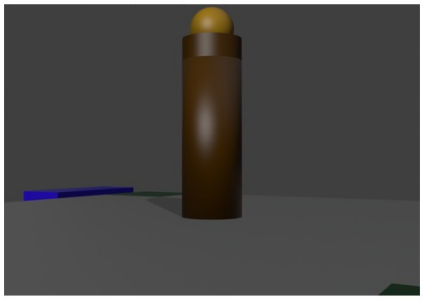
056_near_h1.15.png
near h=1.15m
cam XYZ=(-2.17, -1.25, 1.15) m



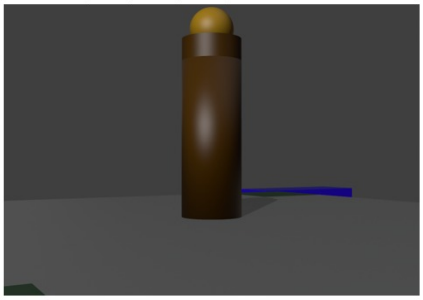
057_near_h1.15.png
near h=1.15m
cam XYZ=(-1.25, -2.17, 1.15) m



058_near_h1.15.png
near h=1.15m
cam XYZ=(-0.00, -2.50, 1.15) m

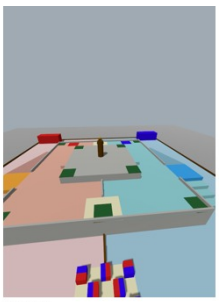


059_near_h1.15.png
near h=1.15m
cam XYZ=(1.25, -2.17, 1.15) m

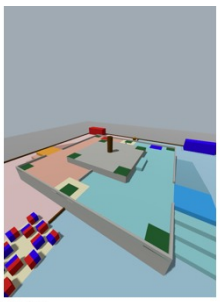


060_near_h1.15.png
near h=1.15m
cam XYZ=(2.17, -1.25, 1.15) m

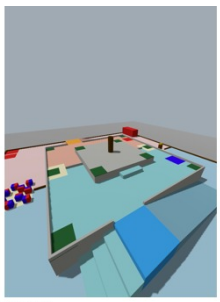
Gazebo RGB 视角集：60 张，第 1 页



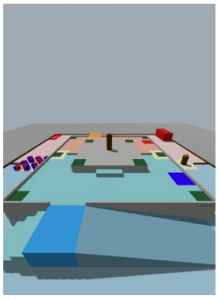
001_top_h3.20.png
top h=3.20m
cam XYZ=(5.00, 0.00, 3.20) m



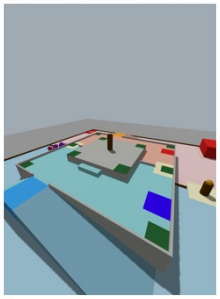
002_top_h3.20.png
top h=3.20m
cam XYZ=(4.33, 2.50, 3.20) m



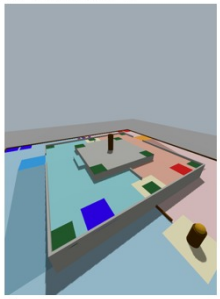
003_top_h3.20.png
top h=3.20m
cam XYZ=(2.50, 4.33, 3.20) m



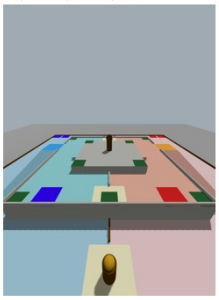
004_top_h3.20.png
top h=3.20m
cam XYZ=(0.00, 5.00, 3.20) m



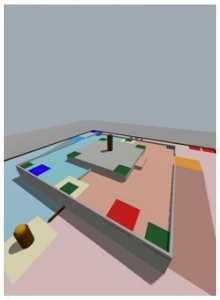
005_top_h3.20.png
top h=3.20m
cam XYZ=(-2.50, 4.33, 3.20) m



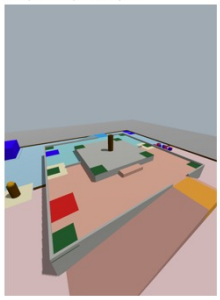
006_top_h3.20.png
top h=3.20m
cam XYZ=(-4.33, 2.50, 3.20) m



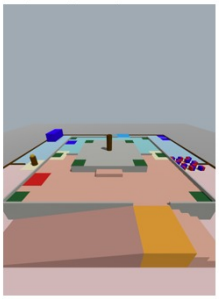
007_top_h3.20.png
top h=3.20m
cam XYZ=(-5.00, 0.00, 3.20) m



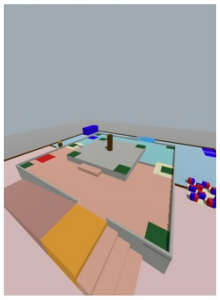
008_top_h3.20.png
top h=3.20m
cam XYZ=(-4.33, -2.50, 3.20) m



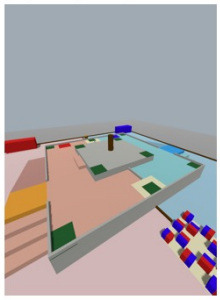
009_top_h3.20.png
top h=3.20m
cam XYZ=(-2.50, -4.33, 3.20) m



010_top_h3.20.png
top h=3.20m
cam XYZ=(-0.00, -5.00, 3.20) m

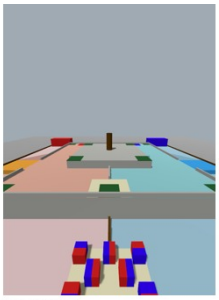


011_top_h3.20.png
top h=3.20m
cam XYZ=(2.50, -4.33, 3.20) m

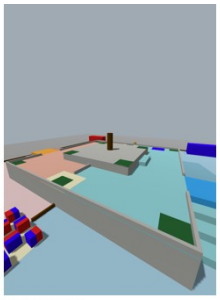


012_top_h3.20.png
top h=3.20m
cam XYZ=(4.33, -2.50, 3.20) m

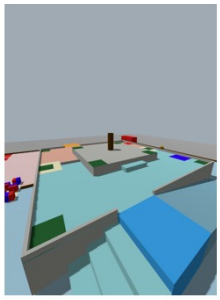
Gazebo RGB 视角集：60 张，第 2 页



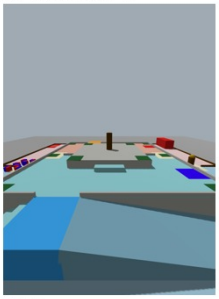
013_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(5.00, 0.00, 2.20) m



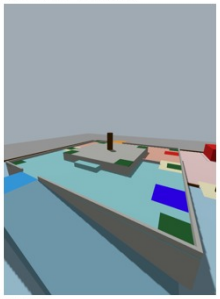
014_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(4.33, 2.50, 2.20) m



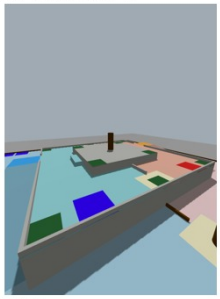
015_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(2.50, 4.33, 2.20) m



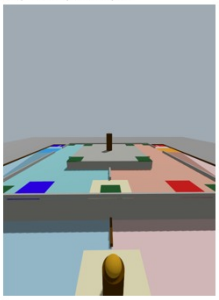
016_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(0.00, 5.00, 2.20) m



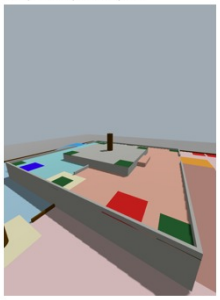
017_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-2.50, 4.33, 2.20) m



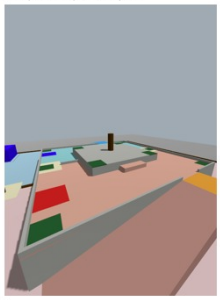
018_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-4.33, 2.50, 2.20) m



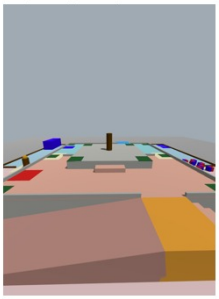
019_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-5.00, 0.00, 2.20) m



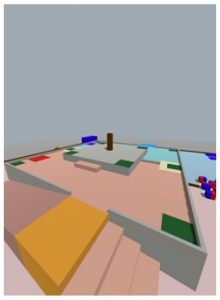
020_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-4.33, -2.50, 2.20) m



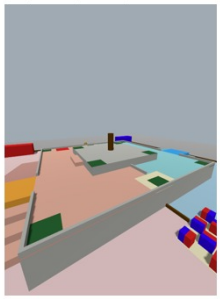
021_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-2.50, -4.33, 2.20) m



022_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-0.00, -5.00, 2.20) m

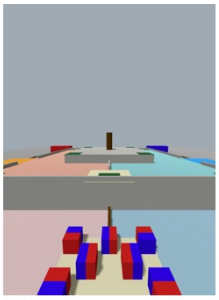


023_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(2.50, -4.33, 2.20) m

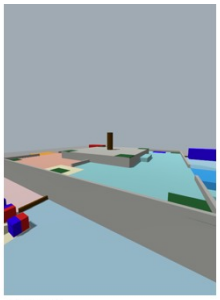


024_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(4.33, -2.50, 2.20) m

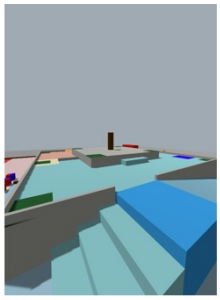
Gazebo RGB 视角集：60 张，第 3 页



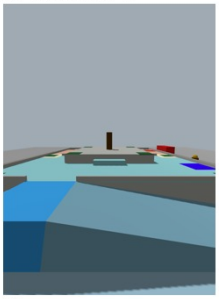
025_level_h1.45.png
level h=1.45m
cam XYZ=(5.00, 0.00, 1.45) m



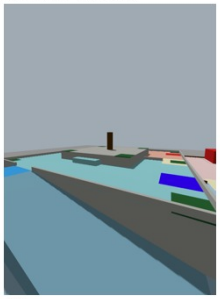
026_level_h1.45.png
level h=1.45m
cam XYZ=(4.33, 2.50, 1.45) m



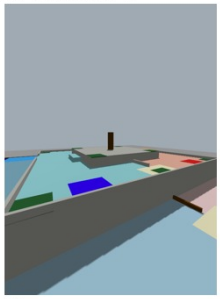
027_level_h1.45.png
level h=1.45m
cam XYZ=(2.50, 4.33, 1.45) m



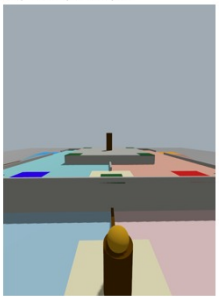
028_level_h1.45.png
level h=1.45m
cam XYZ=(0.00, 5.00, 1.45) m



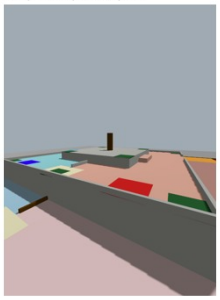
029_level_h1.45.png
level h=1.45m
cam XYZ=(-2.50, 4.33, 1.45) m



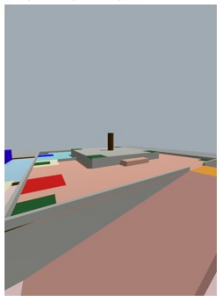
030_level_h1.45.png
level h=1.45m
cam XYZ=(-4.33, 2.50, 1.45) m



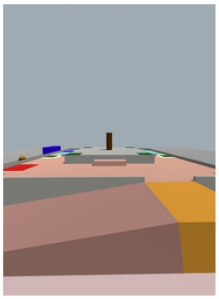
031_level_h1.45.png
level h=1.45m
cam XYZ=(-5.00, 0.00, 1.45) m



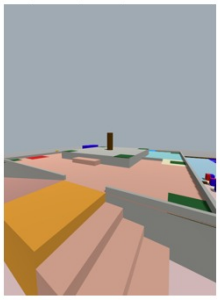
032_level_h1.45.png
level h=1.45m
cam XYZ=(-4.33, -2.50, 1.45) m



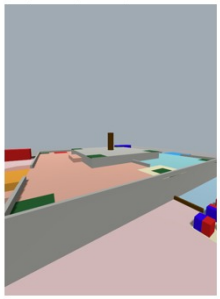
033_level_h1.45.png
level h=1.45m
cam XYZ=(-2.50, -4.33, 1.45) m



034_level_h1.45.png
level h=1.45m
cam XYZ=(-0.00, -5.00, 1.45) m

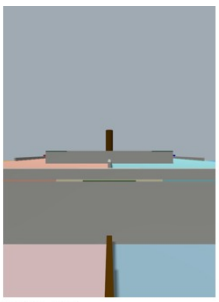


035_level_h1.45.png
level h=1.45m
cam XYZ=(2.50, -4.33, 1.45) m

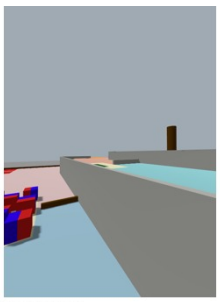


036_level_h1.45.png
level h=1.45m
cam XYZ=(4.33, -2.50, 1.45) m

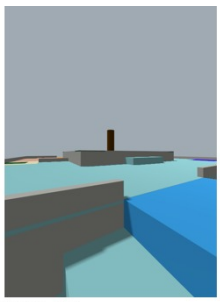
Gazebo RGB 视角集：60 张，第 4 页



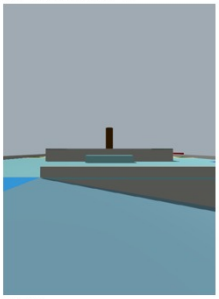
037_upward_h0.85.png
upward h=0.85m
cam XYZ=(4.00, 0.00, 0.85) m



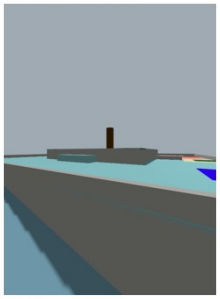
038_upward_h0.85.png
upward h=0.85m
cam XYZ=(3.46, 2.00, 0.85) m



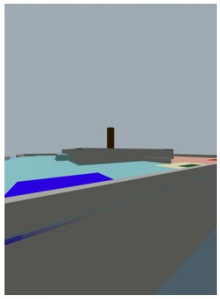
039_upward_h0.85.png
upward h=0.85m
cam XYZ=(2.00, 3.46, 0.85) m



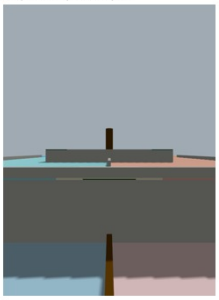
040_upward_h0.85.png
upward h=0.85m
cam XYZ=(0.00, 4.00, 0.85) m



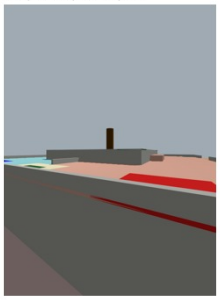
041_upward_h0.85.png
upward h=0.85m
cam XYZ=(-2.00, 3.46, 0.85) m



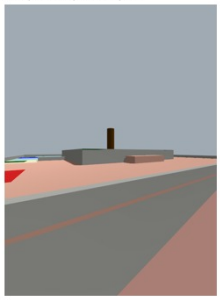
042_upward_h0.85.png
upward h=0.85m
cam XYZ=(-3.46, 2.00, 0.85) m



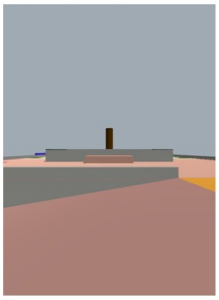
043_upward_h0.85.png
upward h=0.85m
cam XYZ=(-4.00, 0.00, 0.85) m



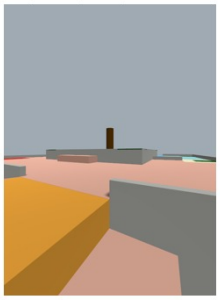
044_upward_h0.85.png
upward h=0.85m
cam XYZ=(-3.46, -2.00, 0.85) m



045_upward_h0.85.png
upward h=0.85m
cam XYZ=(-2.00, -3.46, 0.85) m



046_upward_h0.85.png
upward h=0.85m
cam XYZ=(-0.00, -4.00, 0.85) m

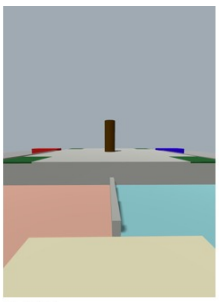


047_upward_h0.85.png
upward h=0.85m
cam XYZ=(2.00, -3.46, 0.85) m

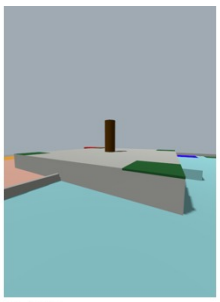


048_upward_h0.85.png
upward h=0.85m
cam XYZ=(3.46, -2.00, 0.85) m

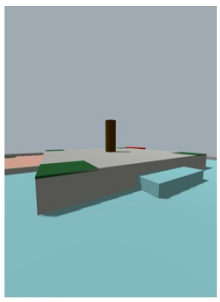
Gazebo RGB 视角集：60 张，第 5 页



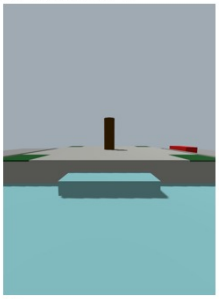
049_near_h1.15.png
near h=1.15m
cam XYZ=(2.80, 0.00, 1.15) m



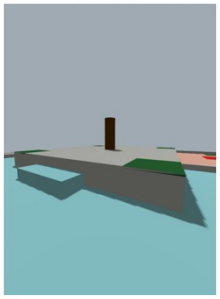
050_near_h1.15.png
near h=1.15m
cam XYZ=(2.42, 1.40, 1.15) m



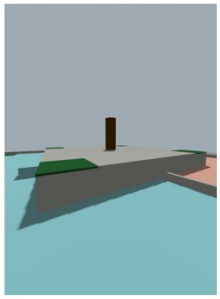
051_near_h1.15.png
near h=1.15m
cam XYZ=(1.40, 2.42, 1.15) m



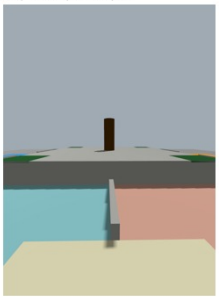
052_near_h1.15.png
near h=1.15m
cam XYZ=(0.00, 2.80, 1.15) m



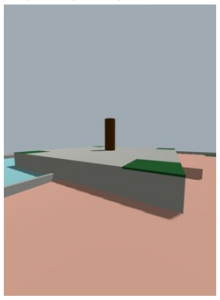
053_near_h1.15.png
near h=1.15m
cam XYZ=(-1.40, 2.42, 1.15) m



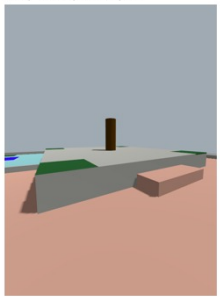
054_near_h1.15.png
near h=1.15m
cam XYZ=(-2.42, 1.40, 1.15) m



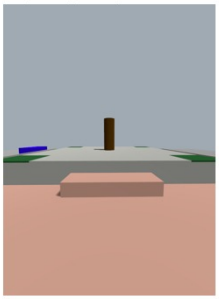
055_near_h1.15.png
near h=1.15m
cam XYZ=(-2.80, 0.00, 1.15) m



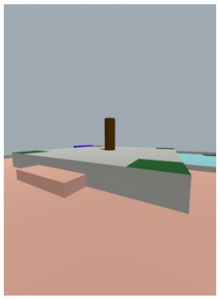
056_near_h1.15.png
near h=1.15m
cam XYZ=(-2.42, -1.40, 1.15) m



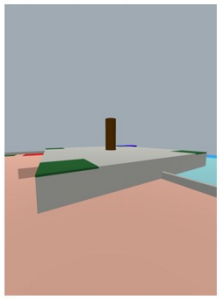
057_near_h1.15.png
near h=1.15m
cam XYZ=(-1.40, -2.42, 1.15) m



058_near_h1.15.png
near h=1.15m
cam XYZ=(-0.00, -2.80, 1.15) m

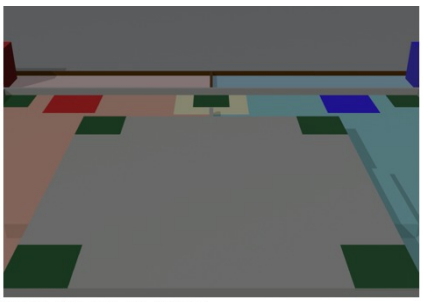


059_near_h1.15.png
near h=1.15m
cam XYZ=(1.40, -2.42, 1.15) m

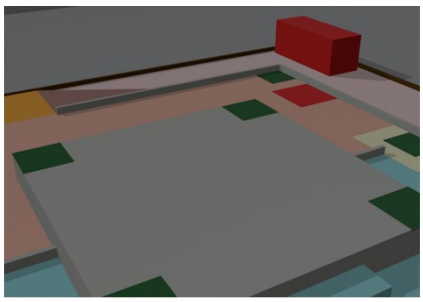


060_near_h1.15.png
near h=1.15m
cam XYZ=(2.42, -1.40, 1.15) m

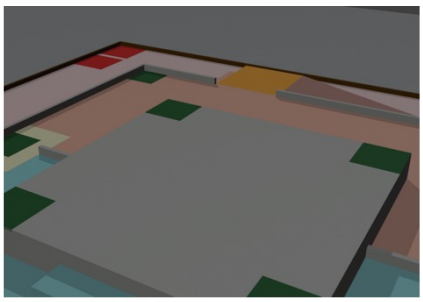
无球、无柱和空场：81 张，第 1 页



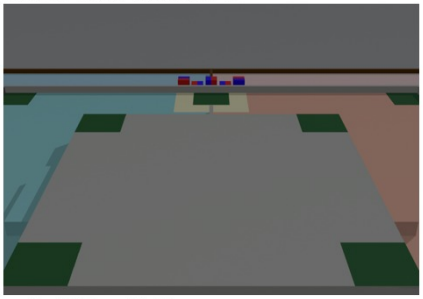
empty_001_top_h3.50.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



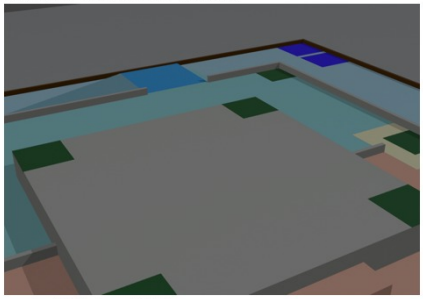
empty_002_top_h3.50.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m



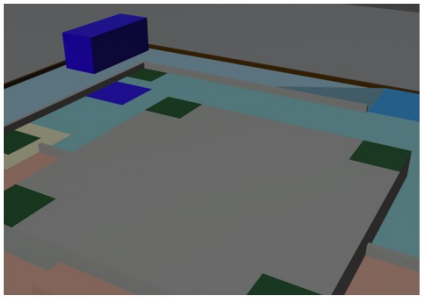
empty_003_top_h3.50.png
top h=3.50m
cam XYZ=(-2.25, 3.90, 3.50) m



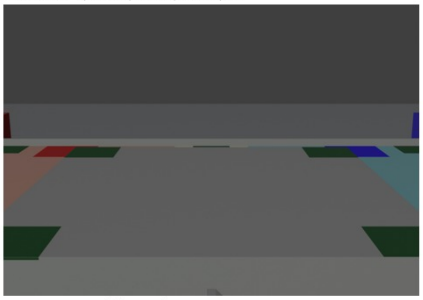
empty_004_top_h3.50.png
top h=3.50m
cam XYZ=(-4.50, 0.00, 3.50) m



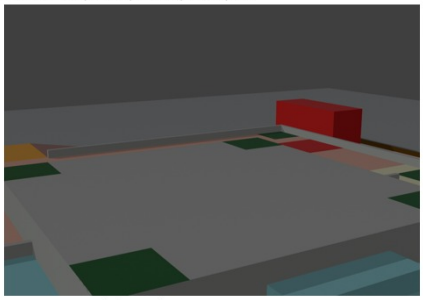
empty_005_top_h3.50.png
top h=3.50m
cam XYZ=(-2.25, -3.90, 3.50) m



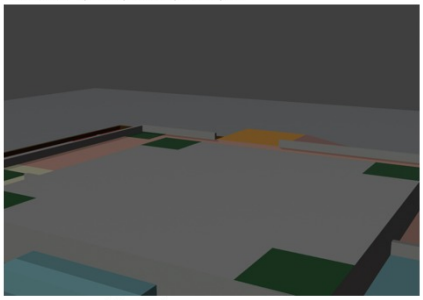
empty_006_top_h3.50.png
top h=3.50m
cam XYZ=(2.25, -3.90, 3.50) m



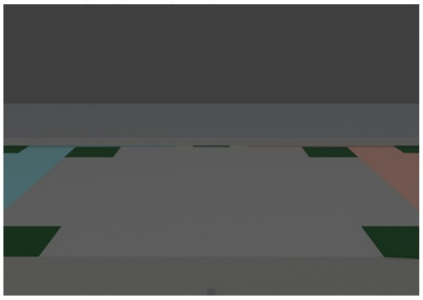
empty_007_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m



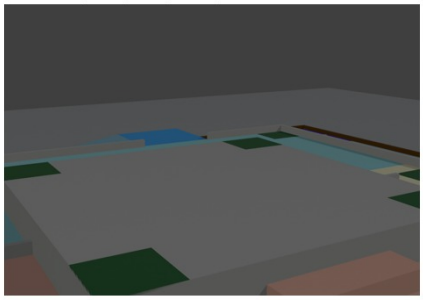
empty_008_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



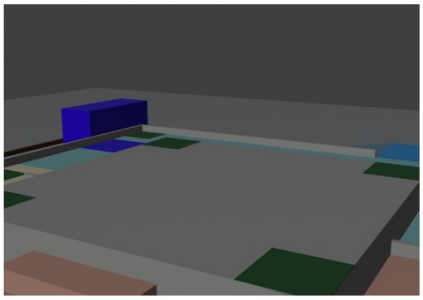
empty_009_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-2.25, 3.90, 2.20) m



empty_010_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-4.50, 0.00, 2.20) m



empty_011_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-2.25, -3.90, 2.20) m

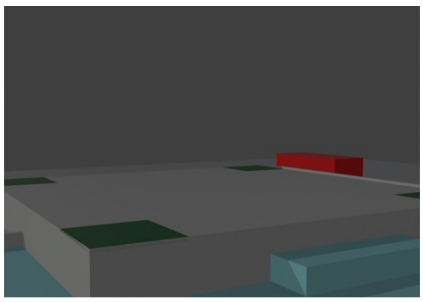


empty_012_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(2.25, -3.90, 2.20) m

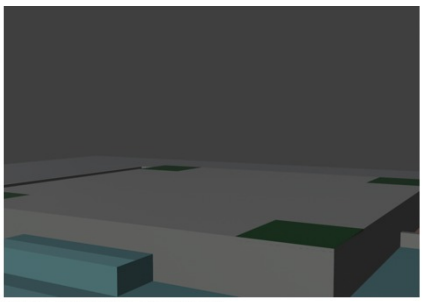
无球、无柱和空场：81 张，第 2 页



empty_013_level_h1.50.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



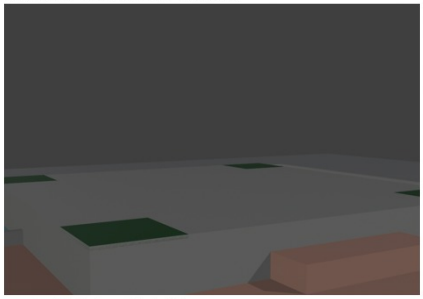
empty_014_level_h1.50.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



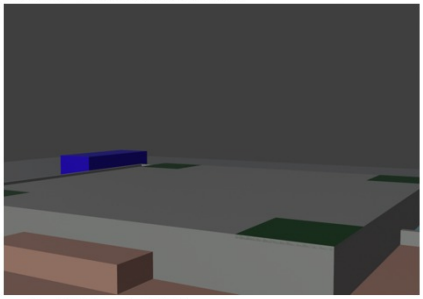
empty_015_level_h1.50.png
level h=1.50m
cam XYZ=(-2.25, 3.90, 1.50) m



empty_016_level_h1.50.png
level h=1.50m
cam XYZ=(-4.50, 0.00, 1.50) m



empty_017_level_h1.50.png
level h=1.50m
cam XYZ=(-2.25, -3.90, 1.50) m



empty_018_level_h1.50.png
level h=1.50m
cam XYZ=(2.25, -3.90, 1.50) m



empty_019_upward_h0.85.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m



empty_020_upward_h0.85.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



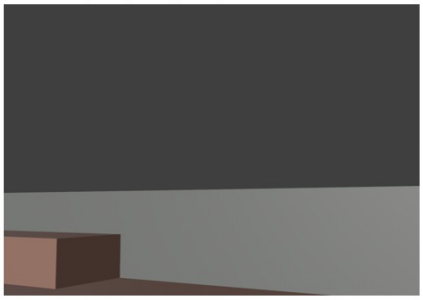
empty_021_upward_h0.85.png
upward h=0.85m
cam XYZ=(-1.75, 3.03, 0.85) m



empty_022_upward_h0.85.png
upward h=0.85m
cam XYZ=(-3.50, 0.00, 0.85) m

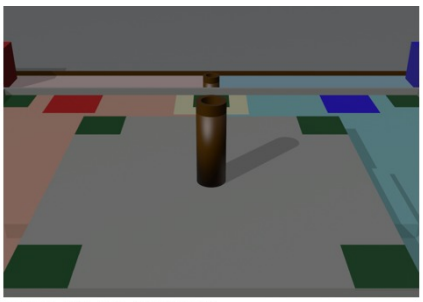


empty_023_upward_h0.85.png
upward h=0.85m
cam XYZ=(-1.75, -3.03, 0.85) m

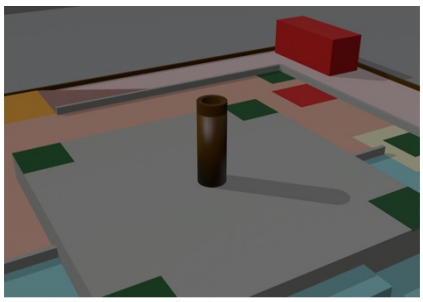


empty_024_upward_h0.85.png
upward h=0.85m
cam XYZ=(1.75, -3.03, 0.85) m

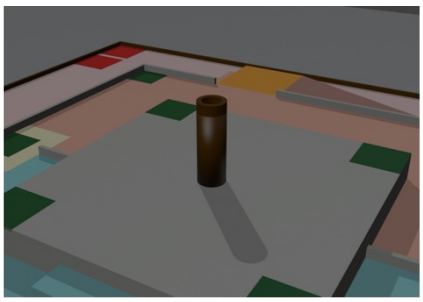
无球、无柱和空场：81 张，第 3 页



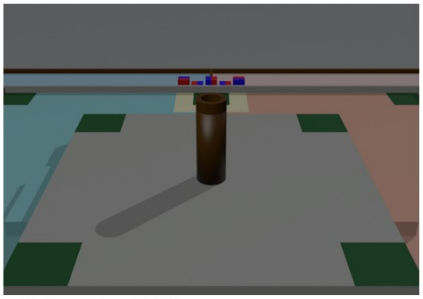
no_ball_001_top_h3.50.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



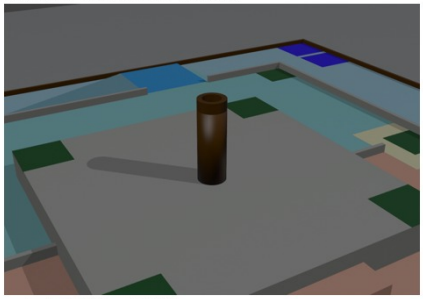
no_ball_002_top_h3.50.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m



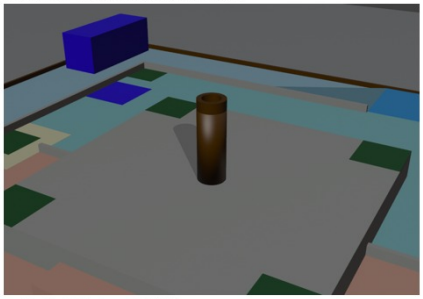
no_ball_003_top_h3.50.png
top h=3.50m
cam XYZ=(-2.25, 3.90, 3.50) m



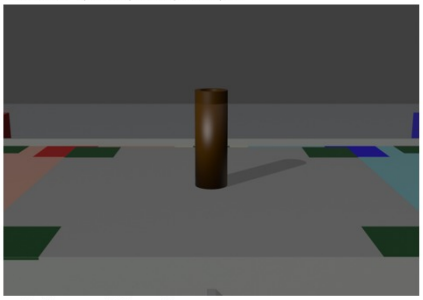
no_ball_004_top_h3.50.png
top h=3.50m
cam XYZ=(-4.50, 0.00, 3.50) m



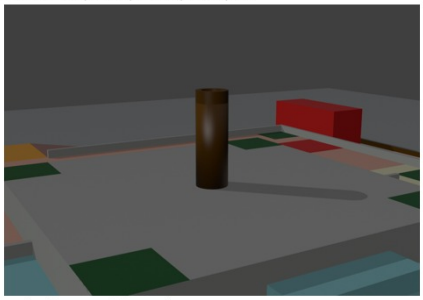
no_ball_005_top_h3.50.png
top h=3.50m
cam XYZ=(-2.25, -3.90, 3.50) m



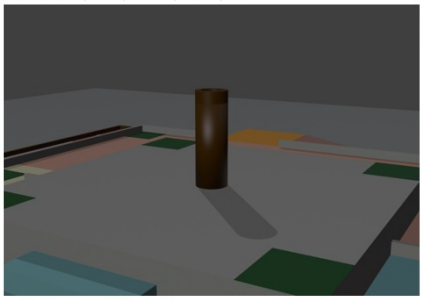
no_ball_006_top_h3.50.png
top h=3.50m
cam XYZ=(2.25, -3.90, 3.50) m



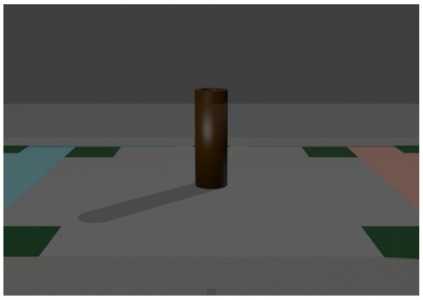
no_ball_007_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m



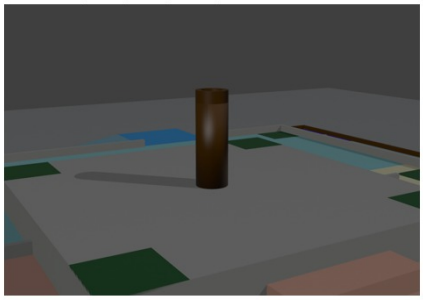
no_ball_008_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



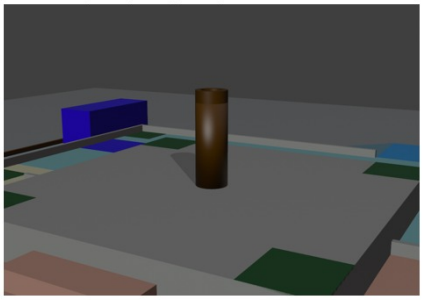
no_ball_009_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-2.25, 3.90, 2.20) m



no_ball_010_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-4.50, 0.00, 2.20) m

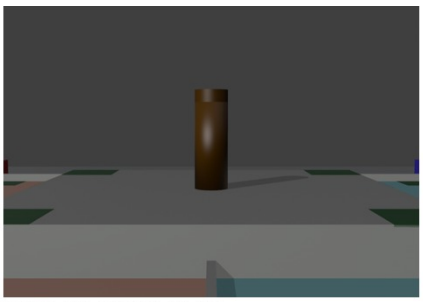


no_ball_011_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-2.25, -3.90, 2.20) m

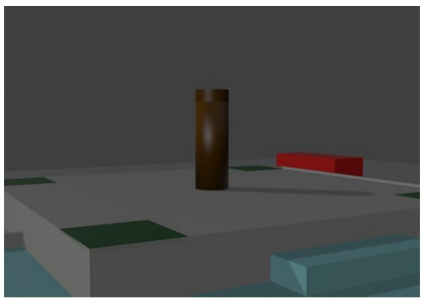


no_ball_012_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(2.25, -3.90, 2.20) m

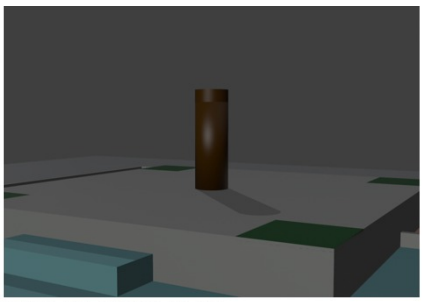
无球、无柱和空场：81 张，第 4 页



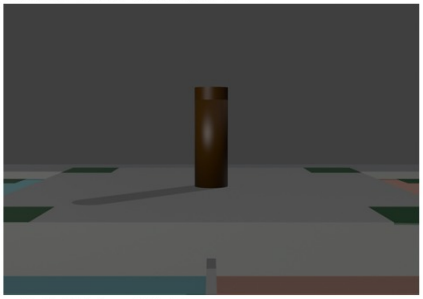
no_ball_013_level_h1.50.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



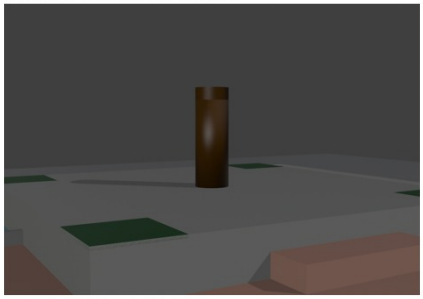
no_ball_014_level_h1.50.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



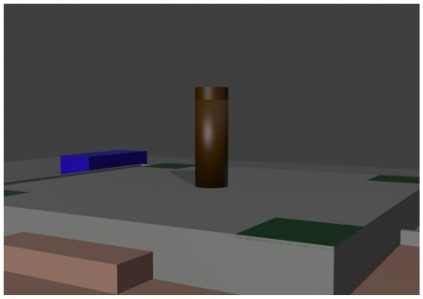
no_ball_015_level_h1.50.png
level h=1.50m
cam XYZ=(-2.25, 3.90, 1.50) m



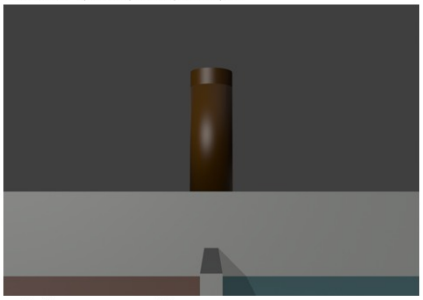
no_ball_016_level_h1.50.png
level h=1.50m
cam XYZ=(-4.50, 0.00, 1.50) m



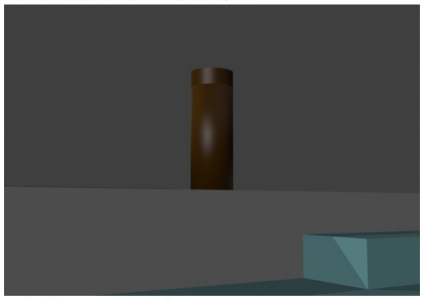
no_ball_017_level_h1.50.png
level h=1.50m
cam XYZ=(-2.25, -3.90, 1.50) m



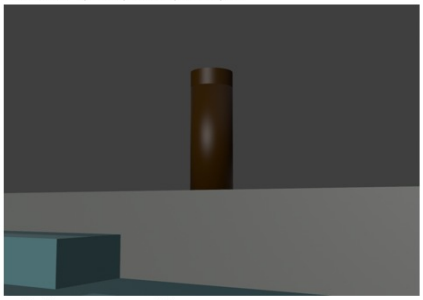
no_ball_018_level_h1.50.png
level h=1.50m
cam XYZ=(2.25, -3.90, 1.50) m



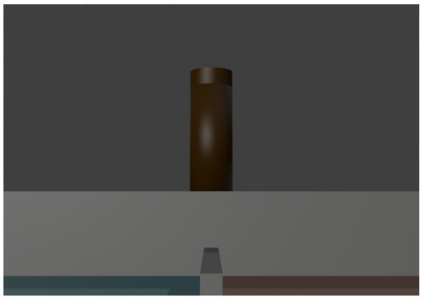
no_ball_019_upward_h0.85.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m



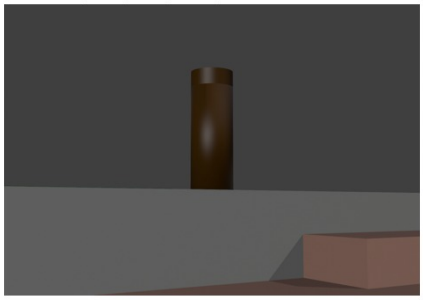
no_ball_020_upward_h0.85.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



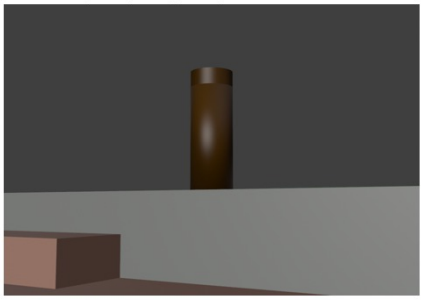
no_ball_021_upward_h0.85.png
upward h=0.85m
cam XYZ=(-1.75, 3.03, 0.85) m



no_ball_022_upward_h0.85.png
upward h=0.85m
cam XYZ=(-3.50, 0.00, 0.85) m

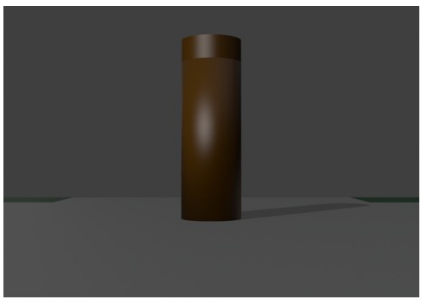


no_ball_023_upward_h0.85.png
upward h=0.85m
cam XYZ=(-1.75, -3.03, 0.85) m

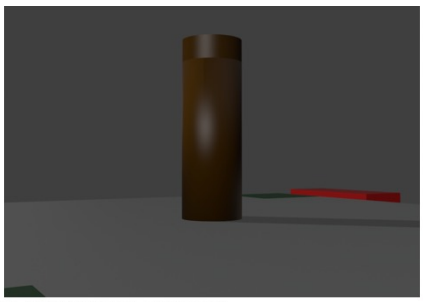


no_ball_024_upward_h0.85.png
upward h=0.85m
cam XYZ=(1.75, -3.03, 0.85) m

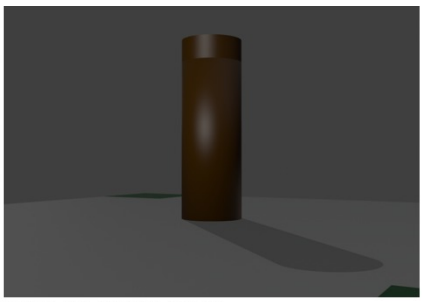
无球、无柱和空场：81 张，第 5 页



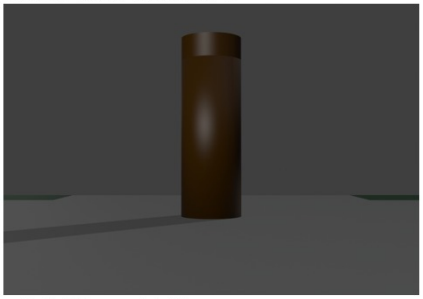
no_ball_025_near_h1.15.png
near h=1.15m
cam XYZ=(2.50, 0.00, 1.15) m



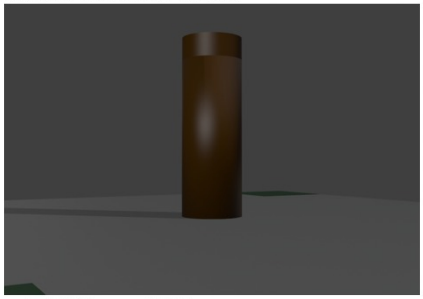
no_ball_026_near_h1.15.png
near h=1.15m
cam XYZ=(1.25, 2.17, 1.15) m



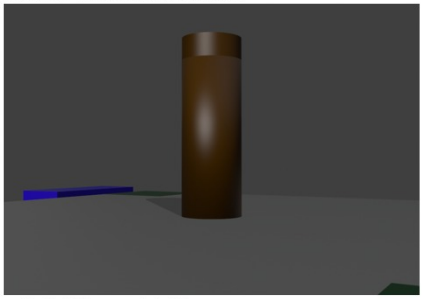
no_ball_027_near_h1.15.png
near h=1.15m
cam XYZ=(-1.25, 2.17, 1.15) m



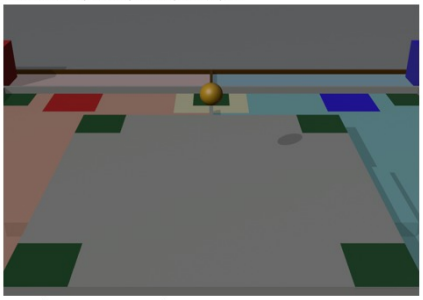
no_ball_028_near_h1.15.png
near h=1.15m
cam XYZ=(-2.50, 0.00, 1.15) m



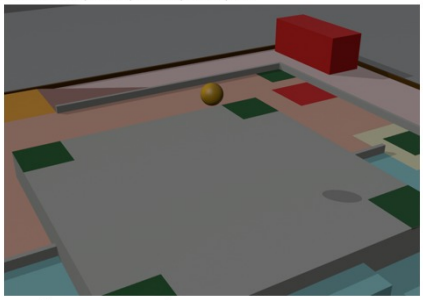
no_ball_029_near_h1.15.png
near h=1.15m
cam XYZ=(-1.25, -2.17, 1.15) m



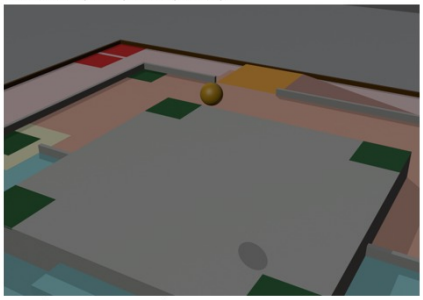
no_ball_030_near_h1.15.png
near h=1.15m
cam XYZ=(1.25, -2.17, 1.15) m



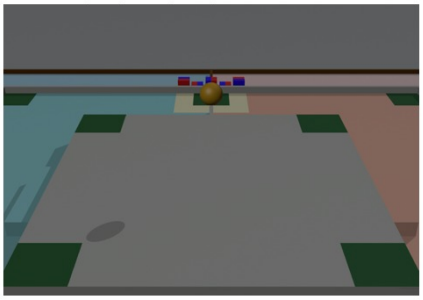
no_column_001_top_h3.50.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



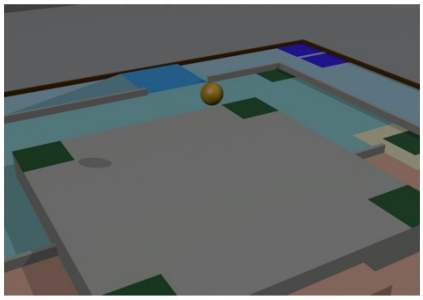
no_column_002_top_h3.50.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m



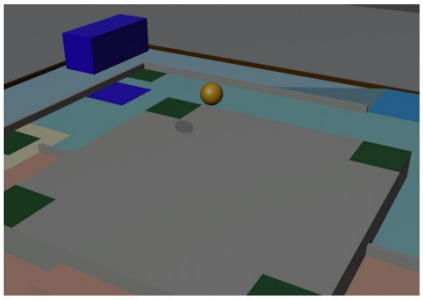
no_column_003_top_h3.50.png
top h=3.50m
cam XYZ=(-2.25, 3.90, 3.50) m



no_column_004_top_h3.50.png
top h=3.50m
cam XYZ=(-4.50, 0.00, 3.50) m

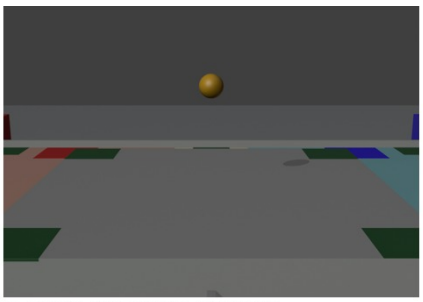


no_column_005_top_h3.50.png
top h=3.50m
cam XYZ=(-2.25, -3.90, 3.50) m

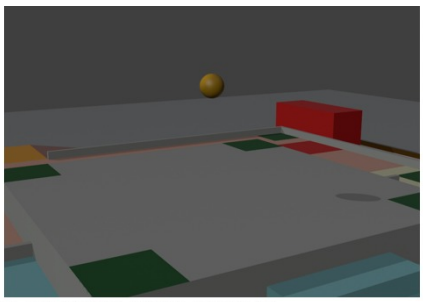


no_column_006_top_h3.50.png
top h=3.50m
cam XYZ=(2.25, -3.90, 3.50) m

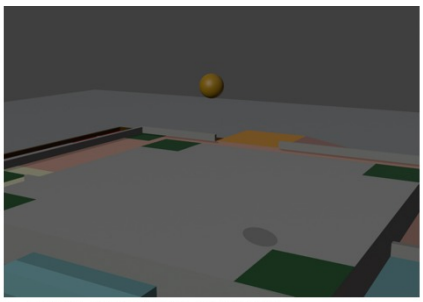
无球、无柱和空场：81 张，第 6 页



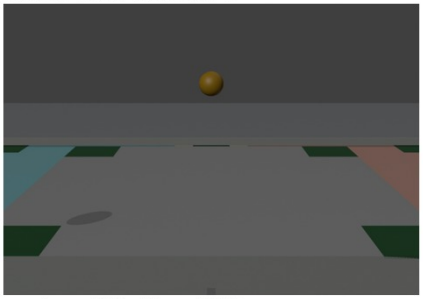
no_column_007_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m



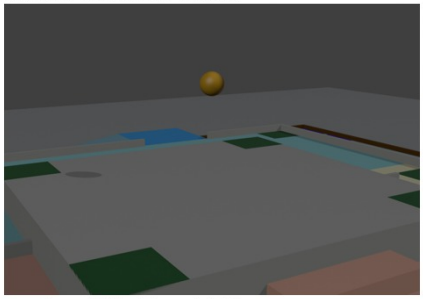
no_column_008_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



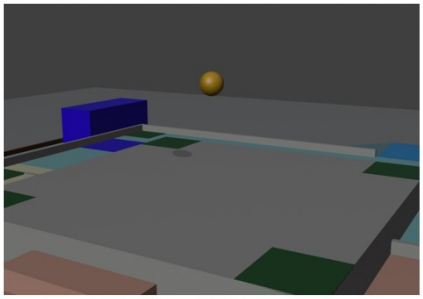
no_column_009_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-2.25, 3.90, 2.20) m



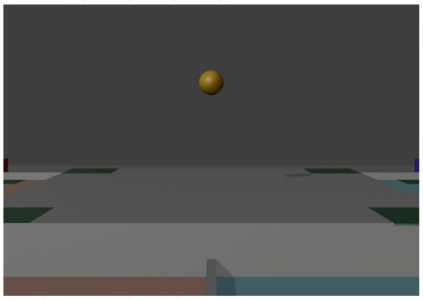
no_column_010_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-4.50, 0.00, 2.20) m



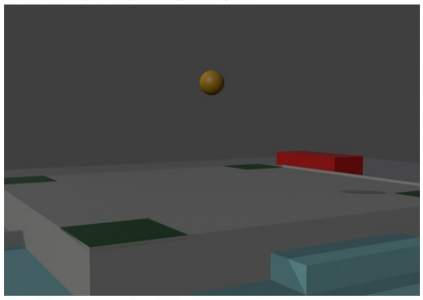
no_column_011_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(-2.25, -3.90, 2.20) m



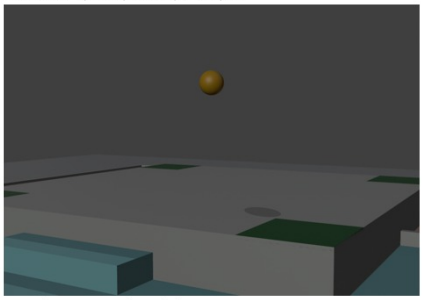
no_column_012_oblique_h2.20.png
oblique h=2.20m
cam XYZ=(2.25, -3.90, 2.20) m



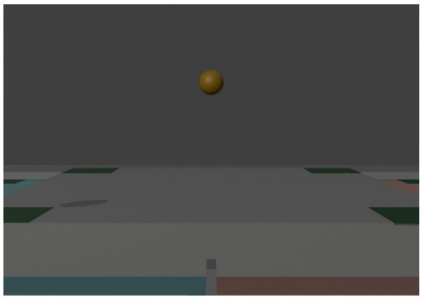
no_column_013_level_h1.50.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



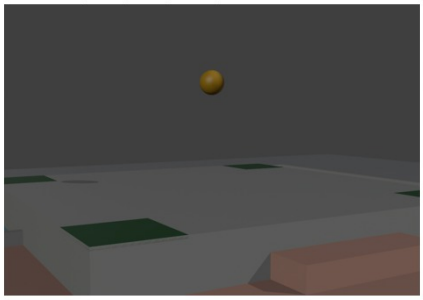
no_column_014_level_h1.50.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



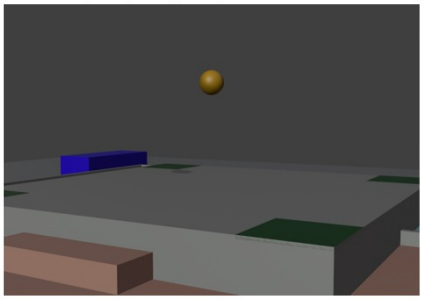
no_column_015_level_h1.50.png
level h=1.50m
cam XYZ=(-2.25, 3.90, 1.50) m



no_column_016_level_h1.50.png
level h=1.50m
cam XYZ=(-4.50, 0.00, 1.50) m

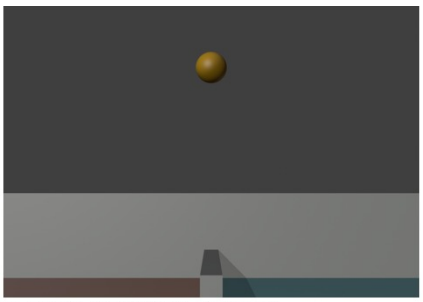


no_column_017_level_h1.50.png
level h=1.50m
cam XYZ=(-2.25, -3.90, 1.50) m



no_column_018_level_h1.50.png
level h=1.50m
cam XYZ=(2.25, -3.90, 1.50) m

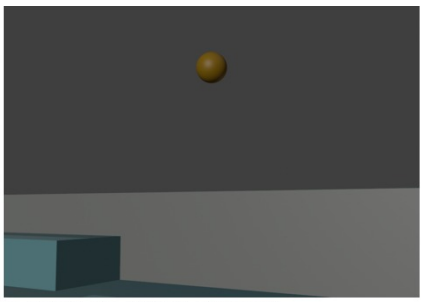
无球、无柱和空场：81 张，第 7 页



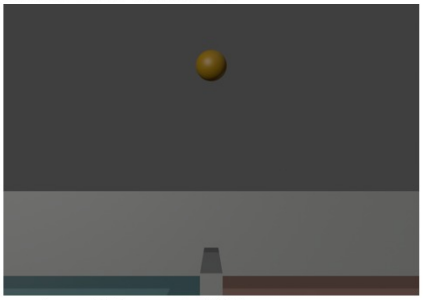
no_column_019_upward_h0.85.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m



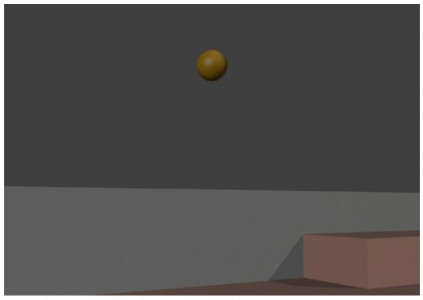
no_column_020_upward_h0.85.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



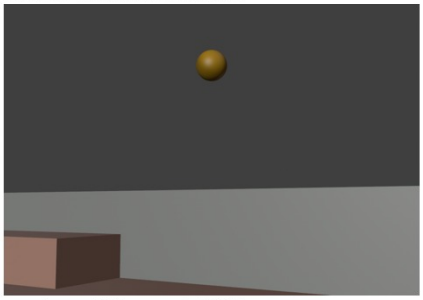
no_column_021_upward_h0.85.png
upward h=0.85m
cam XYZ=(-1.75, 3.03, 0.85) m



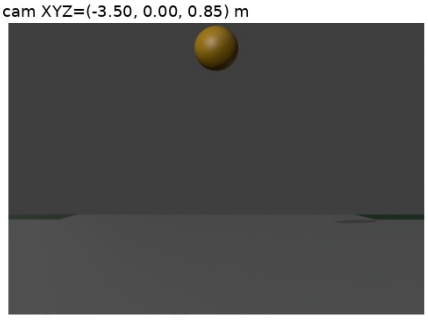
no_column_022_upward_h0.85.png
upward h=0.85m
cam XYZ=(-3.50, 0.00, 0.85) m



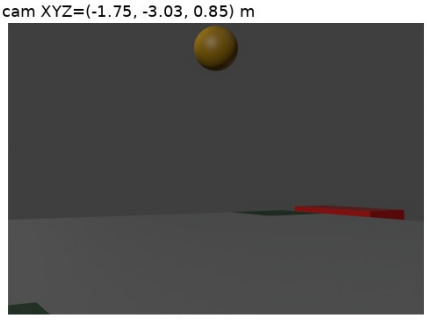
no_column_023_upward_h0.85.png
upward h=0.85m
cam XYZ=(-1.75, -3.03, 0.85) m



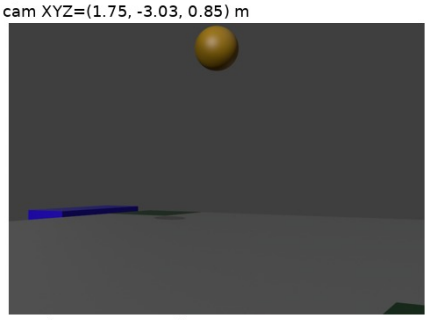
no_column_024_upward_h0.85.png
upward h=0.85m
cam XYZ=(1.75, -3.03, 0.85) m



no_column_025_near_h1.15.png
near h=1.15m
cam XYZ=(2.50, 0.00, 1.15) m

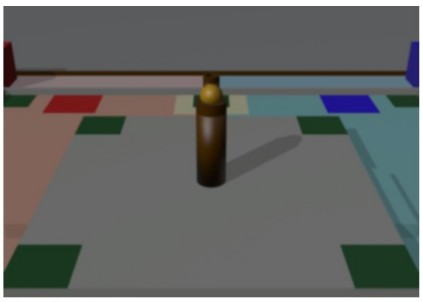


no_column_026_near_h1.15.png
near h=1.15m
cam XYZ=(1.25, 2.17, 1.15) m

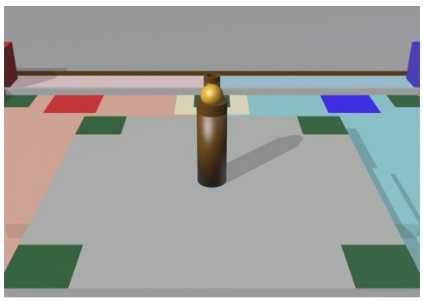


no_column_030_near_h1.15.png
near h=1.15m
cam XYZ=(1.25, -2.17, 1.15) m

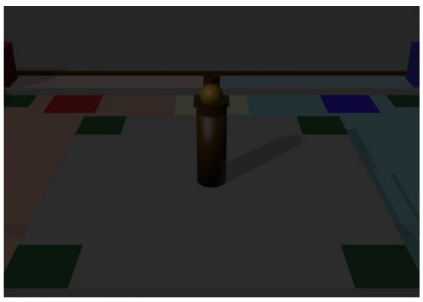
鲁棒集：192 张，第 1 页



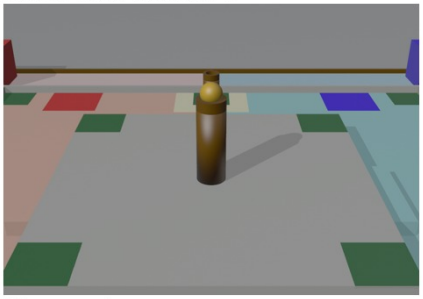
000_blur_top.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



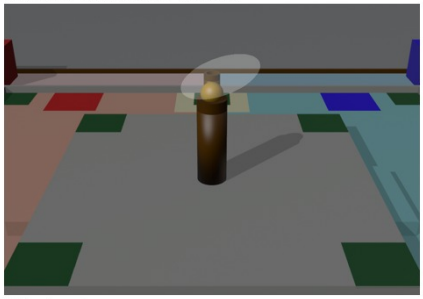
000_bright_top.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



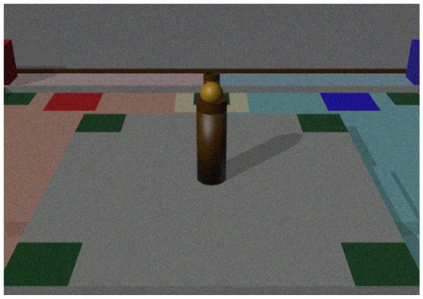
000_dark_top.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



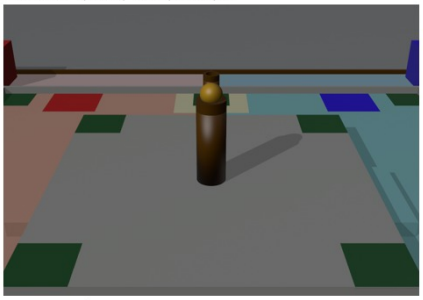
000_gamma_top.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



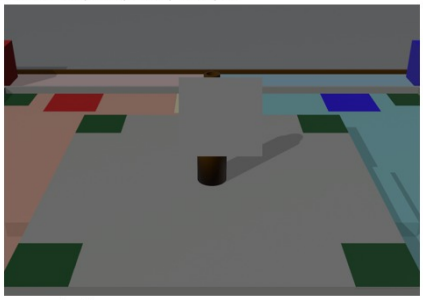
000_glare_top.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



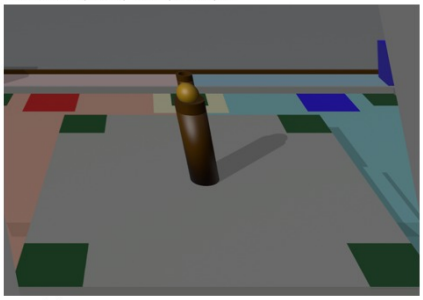
000_noise_top.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



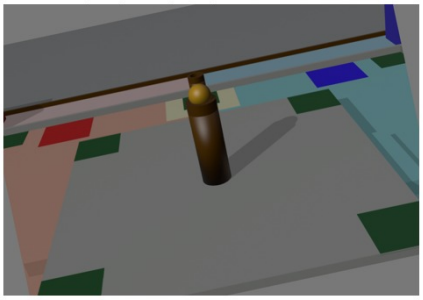
000_normal_top.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



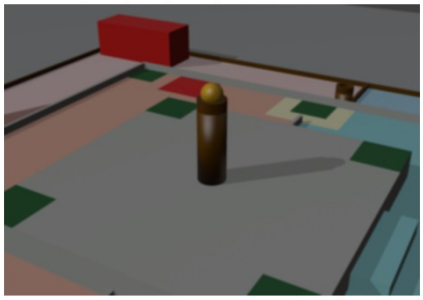
000_occlusion_top.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



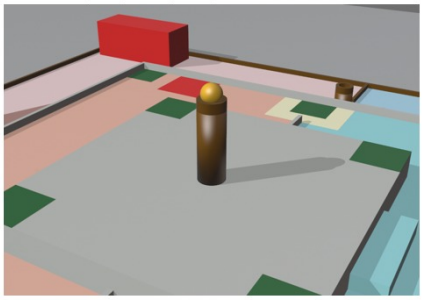
000_side_crop_top.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



000_tilt_top.png
top h=3.50m
cam XYZ=(4.50, 0.00, 3.50) m



001_blur_top.png
top h=3.50m
cam XYZ=(3.90, 2.25, 3.50) m

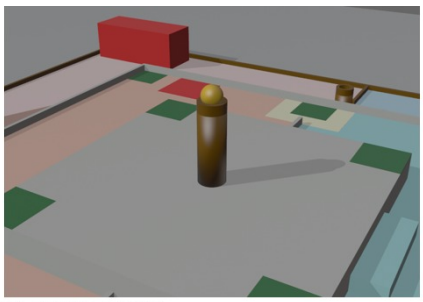


001_bright_top.png
top h=3.50m
cam XYZ=(3.90, 2.25, 3.50) m

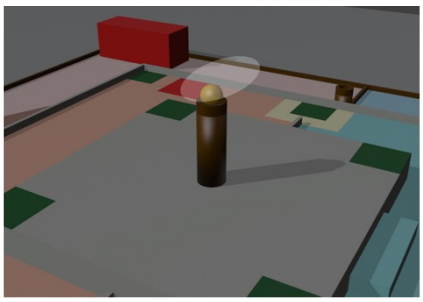
鲁棒集：192 张，第 2 页



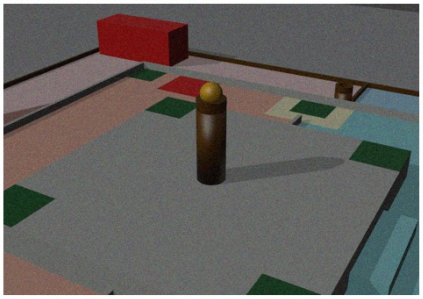
001_dark_top.png
top h=3.50m
cam XYZ=(3.90, 2.25, 3.50) m



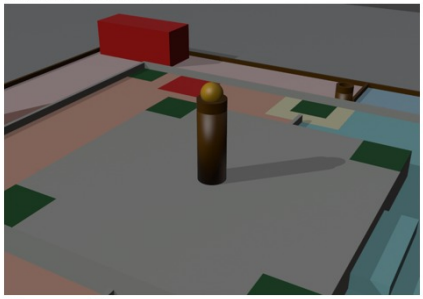
001_gamma_top.png
top h=3.50m
cam XYZ=(3.90, 2.25, 3.50) m



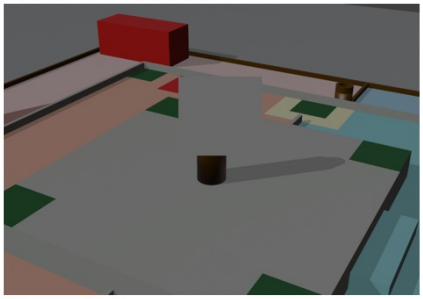
001_glare_top.png
top h=3.50m
cam XYZ=(3.90, 2.25, 3.50) m



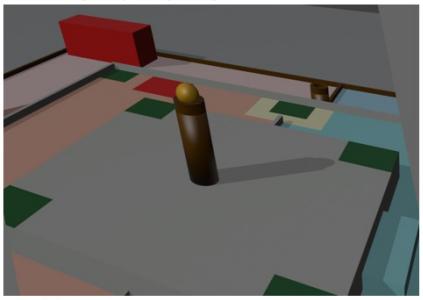
001_noise_top.png
top h=3.50m
cam XYZ=(3.90, 2.25, 3.50) m



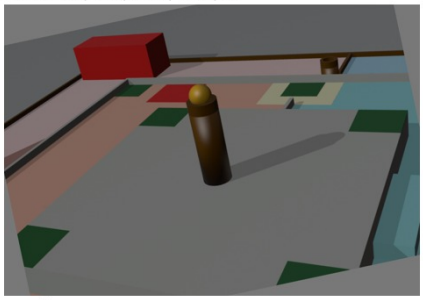
001_normal_top.png
top h=3.50m
cam XYZ=(3.90, 2.25, 3.50) m



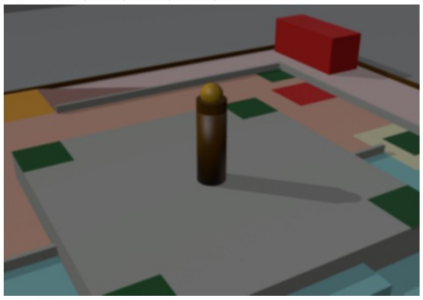
001_occlusion_top.png
top h=3.50m
cam XYZ=(3.90, 2.25, 3.50) m



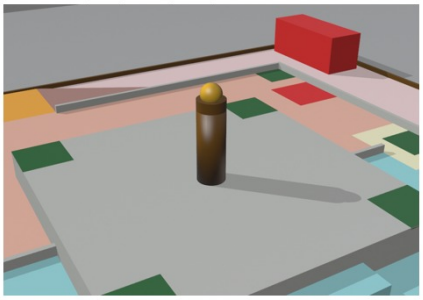
001_side_crop_top.png
top h=3.50m
cam XYZ=(3.90, 2.25, 3.50) m



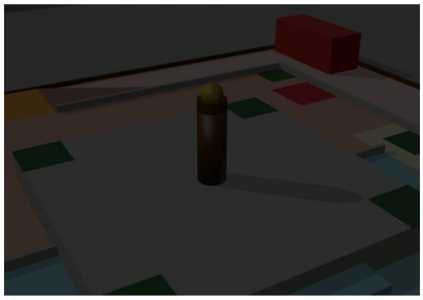
001_tilt_top.png
top h=3.50m
cam XYZ=(3.90, 2.25, 3.50) m



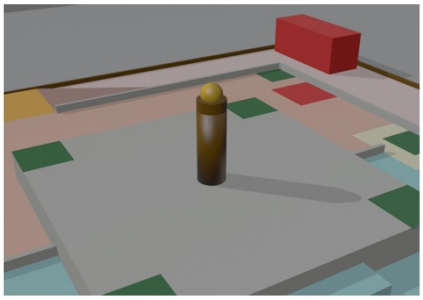
002_blur_top.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m



002_bright_top.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m

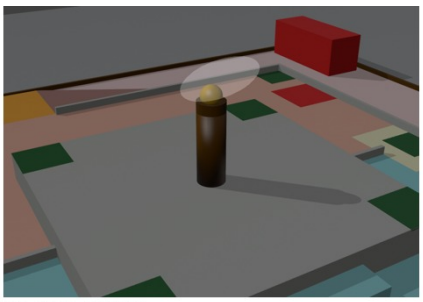


002_dark_top.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m

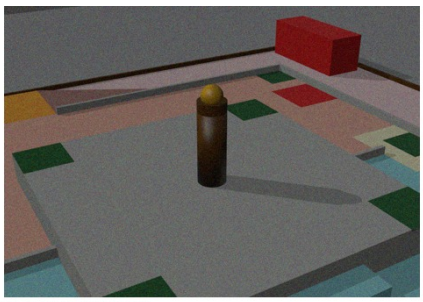


002_gamma_top.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m

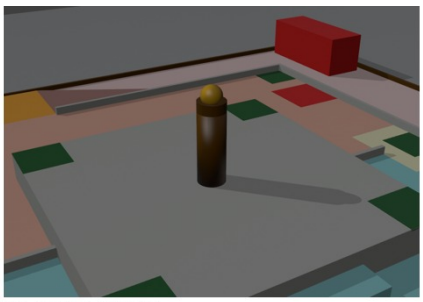
鲁棒集：192 张，第 3 页



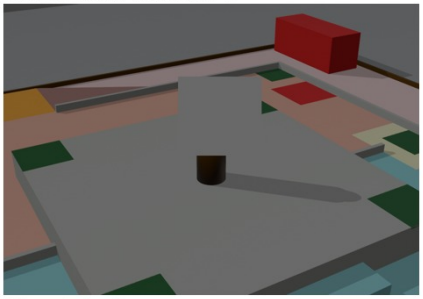
002_glare_top.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m



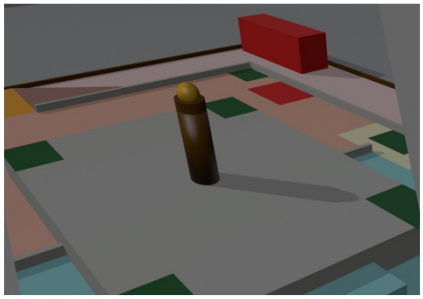
002_noise_top.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m



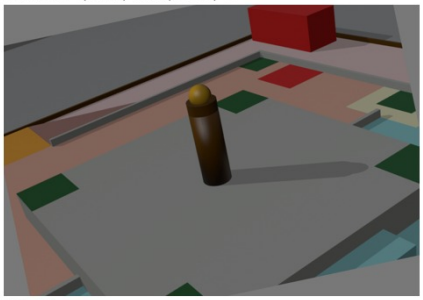
002_normal_top.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m



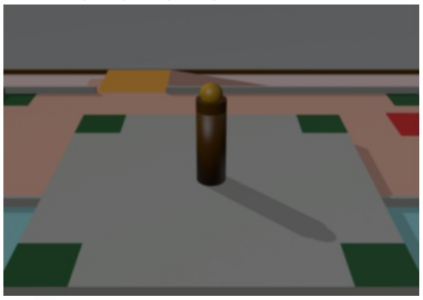
002_occlusion_top.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m



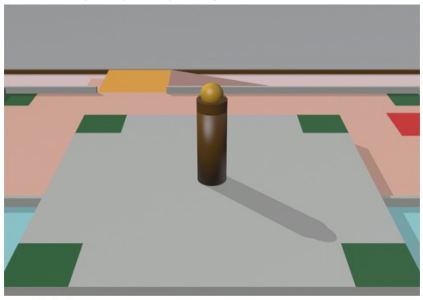
002_side_crop_top.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m



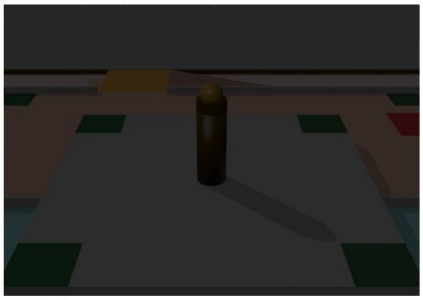
002_tilt_top.png
top h=3.50m
cam XYZ=(2.25, 3.90, 3.50) m



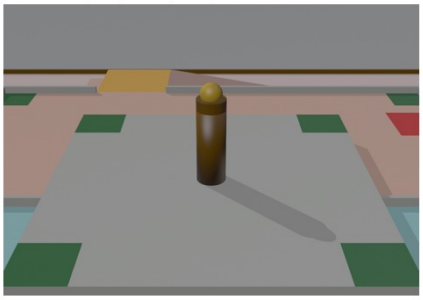
003_blur_top.png
top h=3.50m
cam XYZ=(0.00, 4.50, 3.50) m



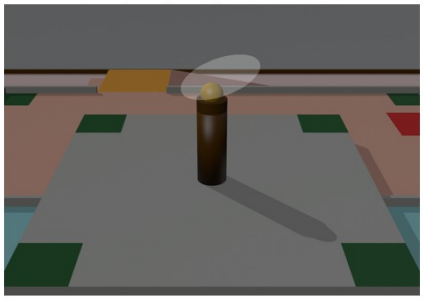
003_bright_top.png
top h=3.50m
cam XYZ=(0.00, 4.50, 3.50) m



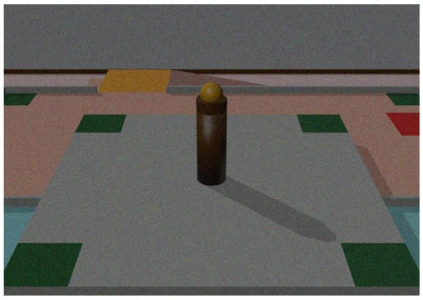
003_dark_top.png
top h=3.50m
cam XYZ=(0.00, 4.50, 3.50) m



003_gamma_top.png
top h=3.50m
cam XYZ=(0.00, 4.50, 3.50) m

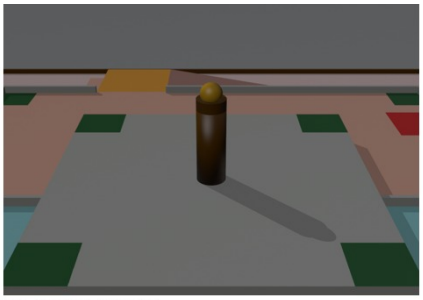


003_glare_top.png
top h=3.50m
cam XYZ=(0.00, 4.50, 3.50) m

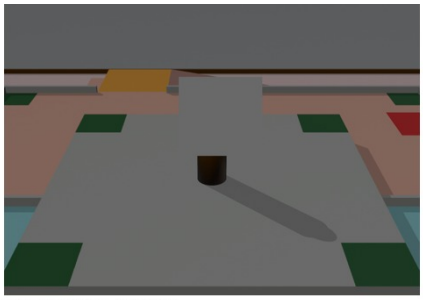


003_noise_top.png
top h=3.50m
cam XYZ=(0.00, 4.50, 3.50) m

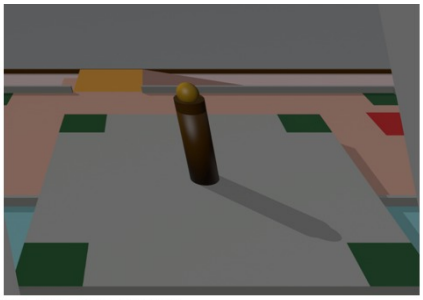
鲁棒集：192 张，第 4 页



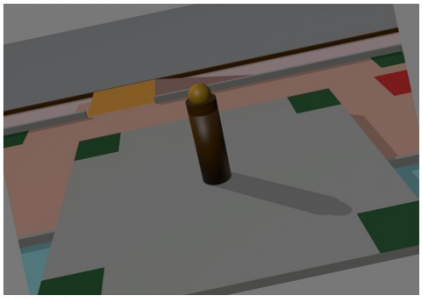
003_normal_top.png
top h=3.50m
cam XYZ=(0.00, 4.50, 3.50) m



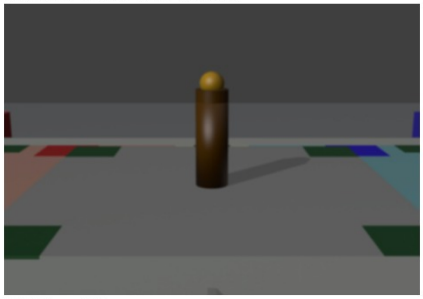
003_occlusion_top.png
top h=3.50m
cam XYZ=(0.00, 4.50, 3.50) m



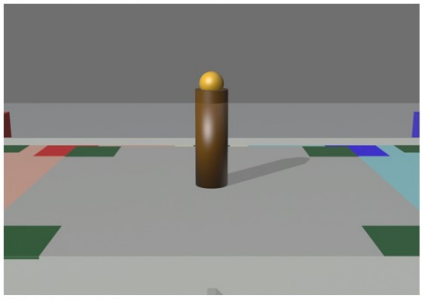
003_side_crop_top.png
top h=3.50m
cam XYZ=(0.00, 4.50, 3.50) m



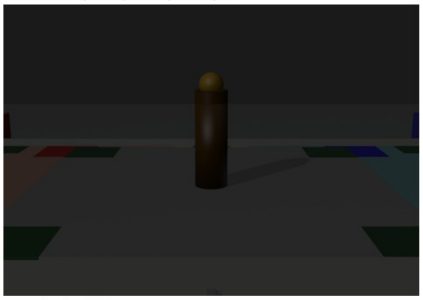
003_tilt_top.png
top h=3.50m
cam XYZ=(0.00, 4.50, 3.50) m



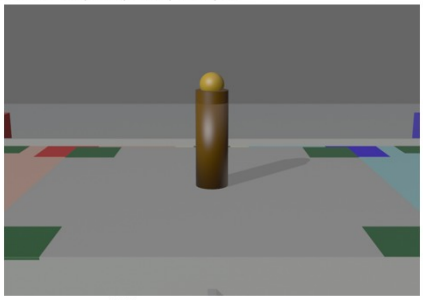
004_blur_oblique.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m



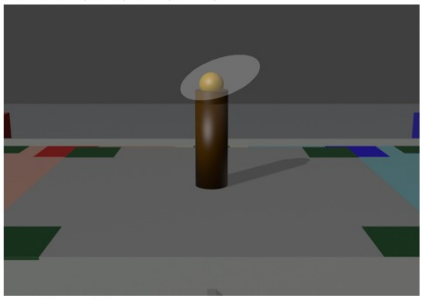
004_bright_oblique.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m



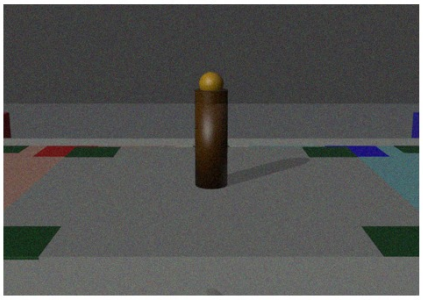
004_dark_oblique.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m



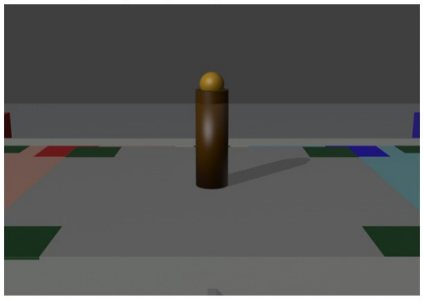
004_gamma_oblique.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m



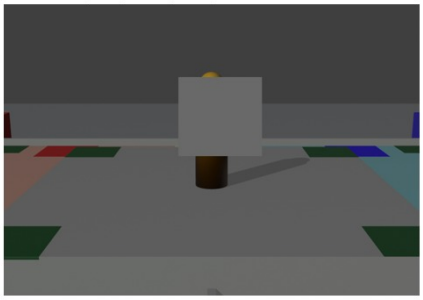
004_glare_oblique.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m



004_noise_oblique.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m

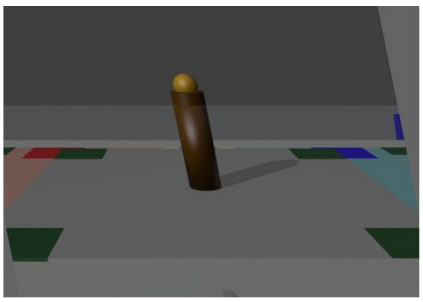


004_normal_oblique.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m

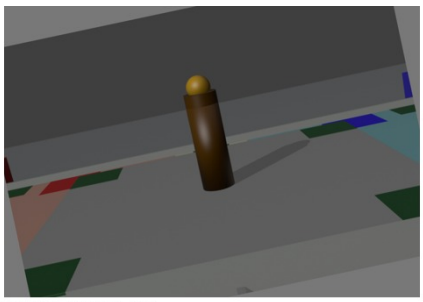


004_occlusion_oblique.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m

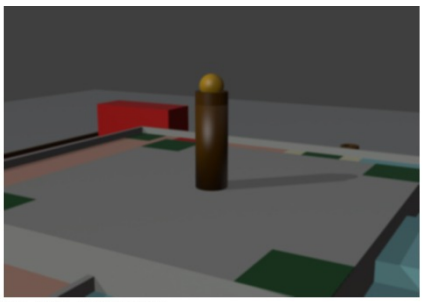
鲁棒集：192 张，第 5 页



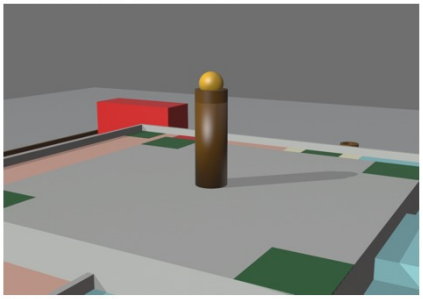
004_side_crop_oblique.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m



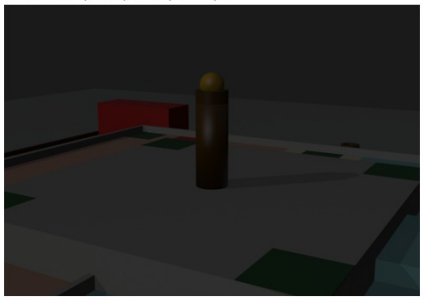
004_tilt_oblique.png
oblique h=2.20m
cam XYZ=(4.50, 0.00, 2.20) m



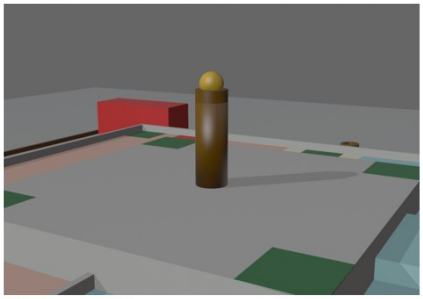
005_blur_oblique.png
oblique h=2.20m
cam XYZ=(3.90, 2.25, 2.20) m



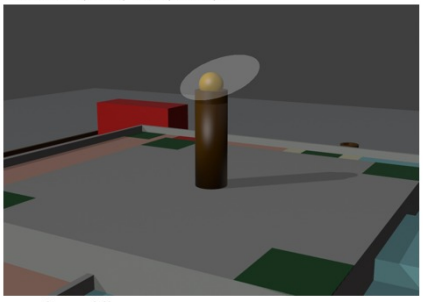
005_bright_oblique.png
oblique h=2.20m
cam XYZ=(3.90, 2.25, 2.20) m



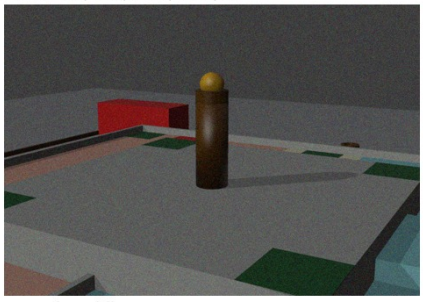
005_dark_oblique.png
oblique h=2.20m
cam XYZ=(3.90, 2.25, 2.20) m



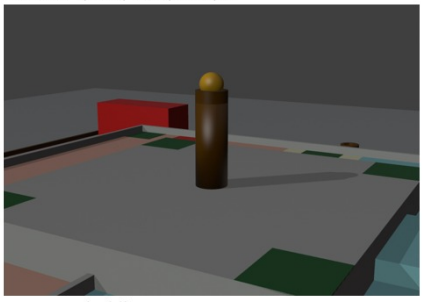
005_gamma_oblique.png
oblique h=2.20m
cam XYZ=(3.90, 2.25, 2.20) m



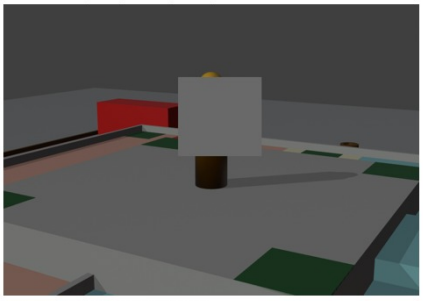
005_glare_oblique.png
oblique h=2.20m
cam XYZ=(3.90, 2.25, 2.20) m



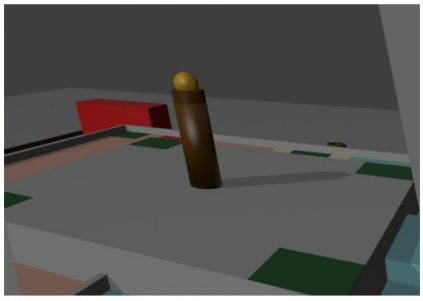
005_noise_oblique.png
oblique h=2.20m
cam XYZ=(3.90, 2.25, 2.20) m



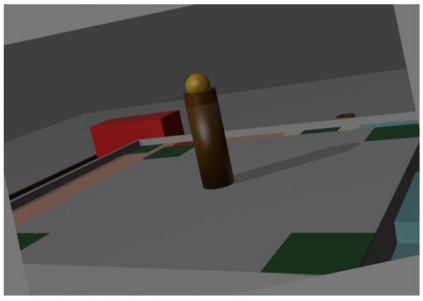
005_normal_oblique.png
oblique h=2.20m
cam XYZ=(3.90, 2.25, 2.20) m



005_occlusion_oblique.png
oblique h=2.20m
cam XYZ=(3.90, 2.25, 2.20) m

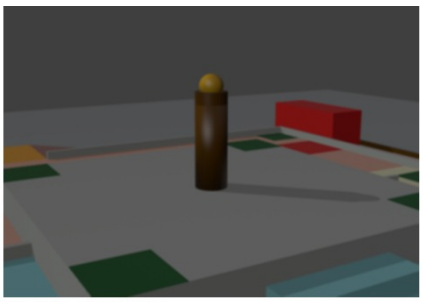


005_side_crop_oblique.png
oblique h=2.20m
cam XYZ=(3.90, 2.25, 2.20) m

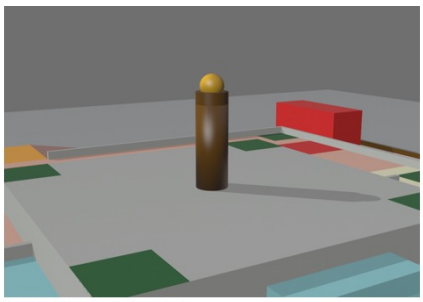


005_tilt_oblique.png
oblique h=2.20m
cam XYZ=(3.90, 2.25, 2.20) m

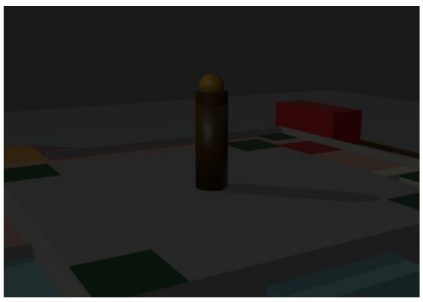
鲁棒集：192 张，第 6 页



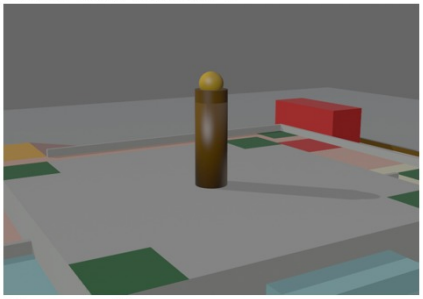
006_blur_oblique.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



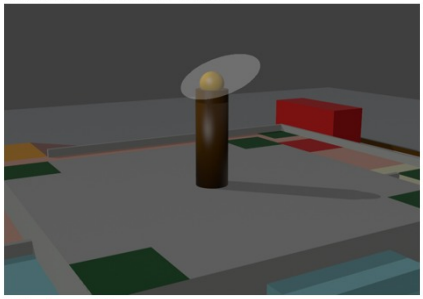
006_bright_oblique.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



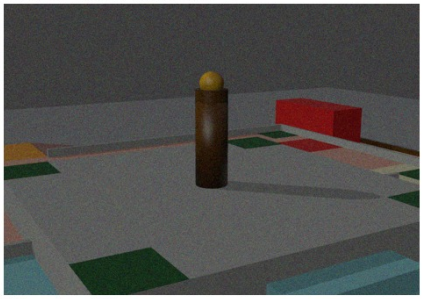
006_dark_oblique.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



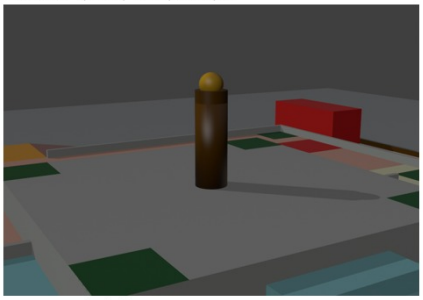
006_gamma_oblique.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



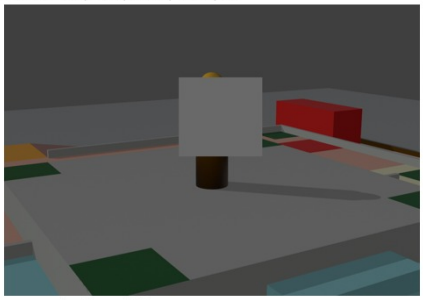
006_glare_oblique.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



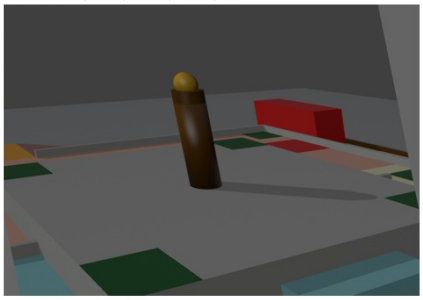
006_noise_oblique.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



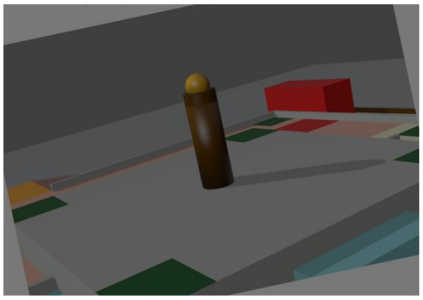
006_normal_oblique.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



006_occlusion_oblique.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



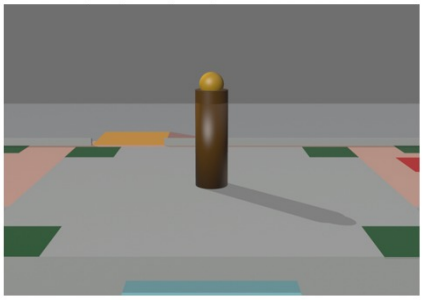
006_side_crop_oblique.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m



006_tilt_oblique.png
oblique h=2.20m
cam XYZ=(2.25, 3.90, 2.20) m

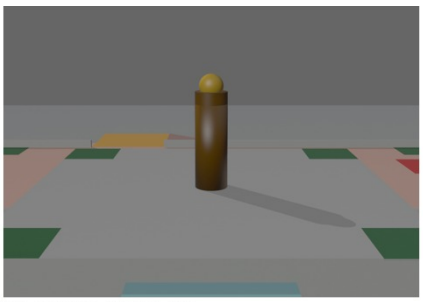


007_blur_oblique.png
oblique h=2.20m
cam XYZ=(0.00, 4.50, 2.20) m

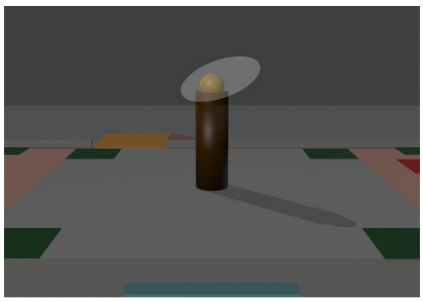


007_bright_oblique.png
oblique h=2.20m
cam XYZ=(0.00, 4.50, 2.20) m

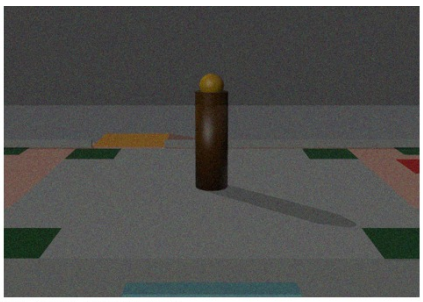
鲁棒集：192 张，第 7 页



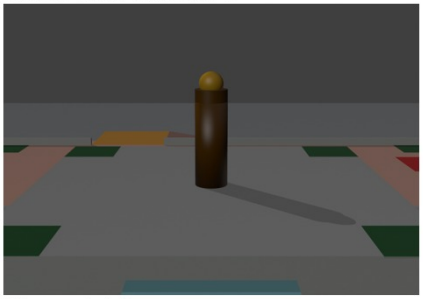
007_gamma_oblique.png
oblique h=2.20m
cam XYZ=(0.00, 4.50, 2.20) m



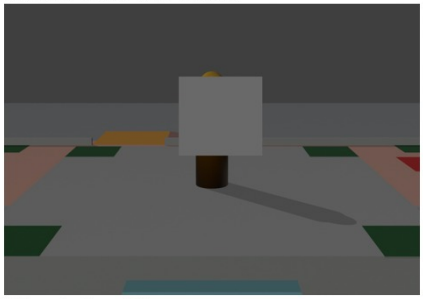
007_glare_oblique.png
oblique h=2.20m
cam XYZ=(0.00, 4.50, 2.20) m



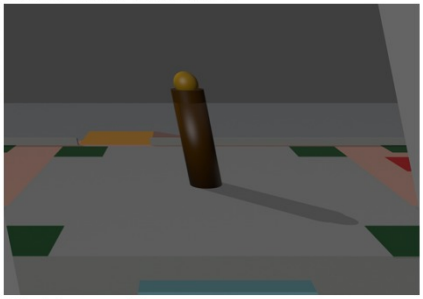
007_noise_oblique.png
oblique h=2.20m
cam XYZ=(0.00, 4.50, 2.20) m



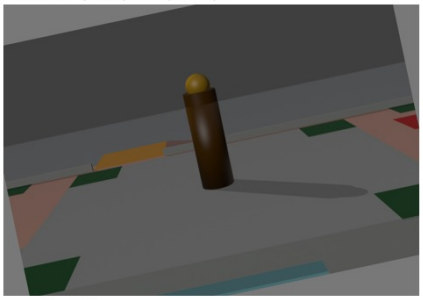
007_normal_oblique.png
oblique h=2.20m
cam XYZ=(0.00, 4.50, 2.20) m



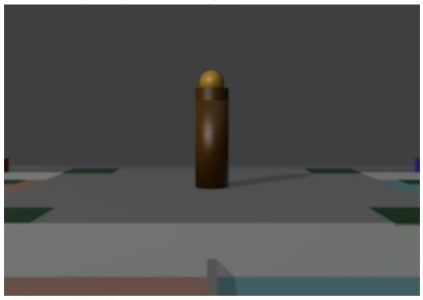
007_occlusion_oblique.png
oblique h=2.20m
cam XYZ=(0.00, 4.50, 2.20) m



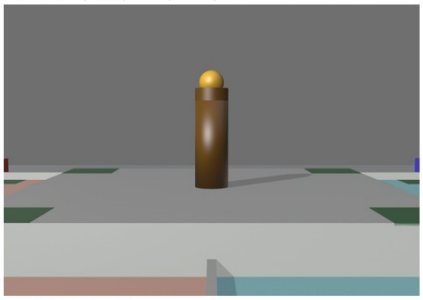
007_side_crop_oblique.png
oblique h=2.20m
cam XYZ=(0.00, 4.50, 2.20) m



007_tilt_oblique.png
oblique h=2.20m
cam XYZ=(0.00, 4.50, 2.20) m



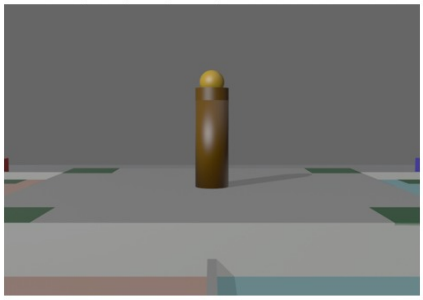
008_blur_level.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



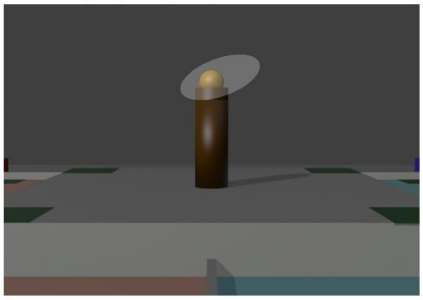
008_bright_level.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



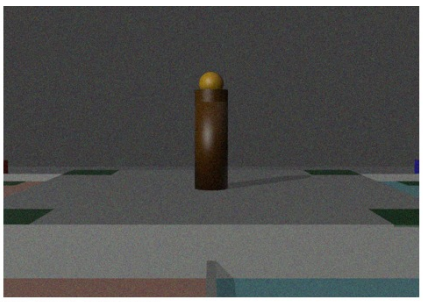
008_dark_level.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



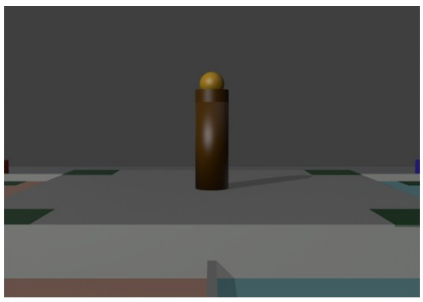
008_gamma_level.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



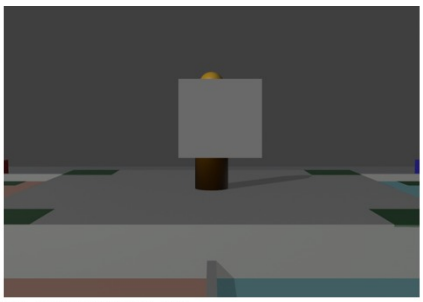
008_glare_level.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



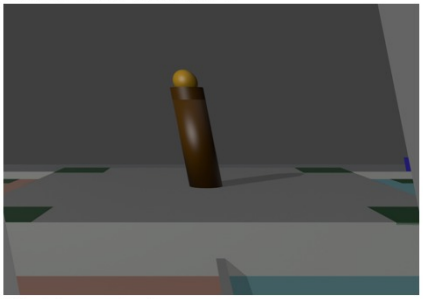
008_noise_level.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



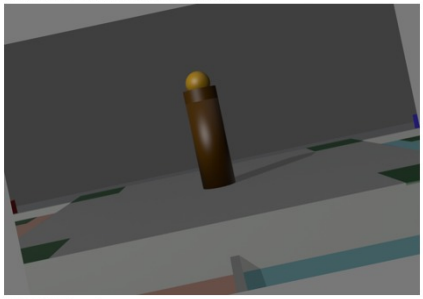
008_normal_level.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



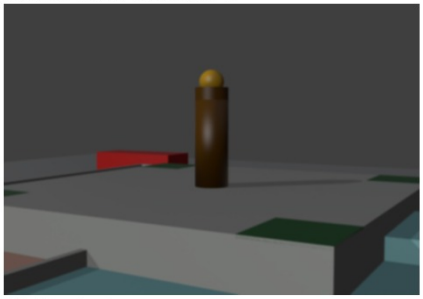
008_occlusion_level.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



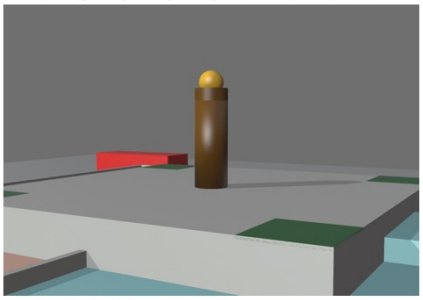
008_side_crop_level.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



008_tilt_level.png
level h=1.50m
cam XYZ=(4.50, 0.00, 1.50) m



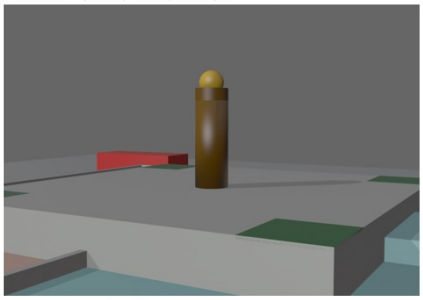
009_blur_level.png
level h=1.50m
cam XYZ=(3.90, 2.25, 1.50) m



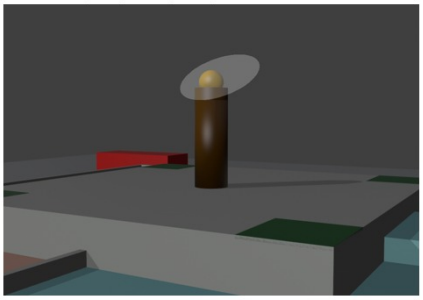
009_bright_level.png
level h=1.50m
cam XYZ=(3.90, 2.25, 1.50) m



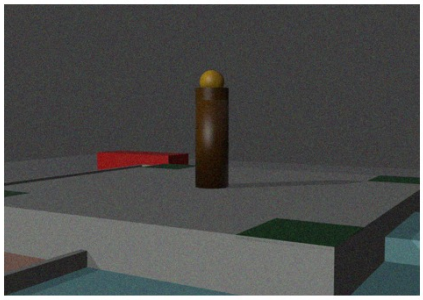
009_dark_level.png
level h=1.50m
cam XYZ=(3.90, 2.25, 1.50) m



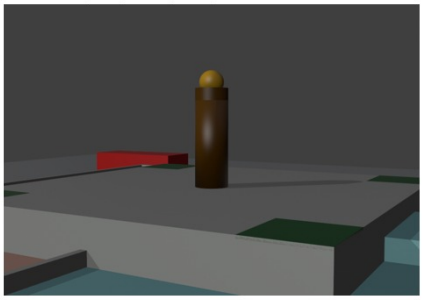
009_gamma_level.png
level h=1.50m
cam XYZ=(3.90, 2.25, 1.50) m



009_glare_level.png
level h=1.50m
cam XYZ=(3.90, 2.25, 1.50) m

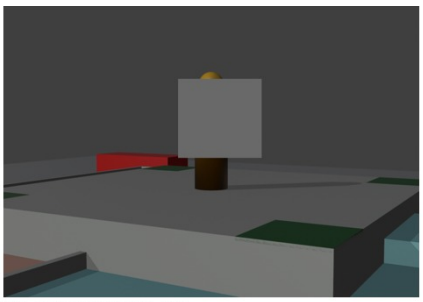


009_noise_level.png
level h=1.50m
cam XYZ=(3.90, 2.25, 1.50) m

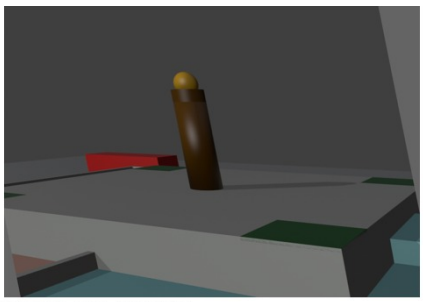


009_normal_level.png
level h=1.50m
cam XYZ=(3.90, 2.25, 1.50) m

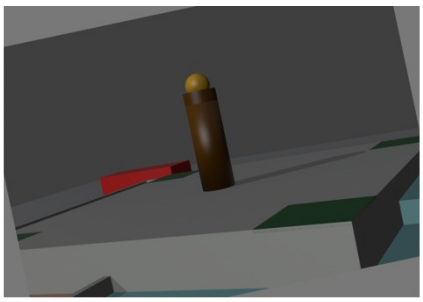
鲁棒集：192 张，第 9 页



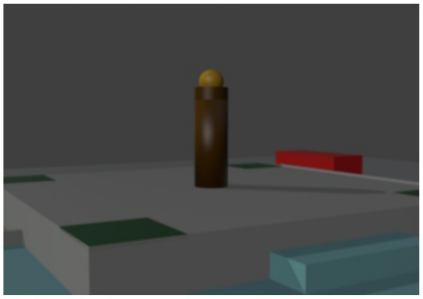
009_occlusion_level.png
level h=1.50m
cam XYZ=(3.90, 2.25, 1.50) m



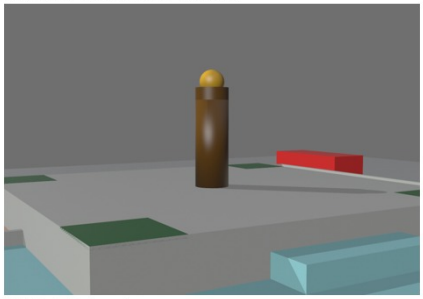
009_side_crop_level.png
level h=1.50m
cam XYZ=(3.90, 2.25, 1.50) m



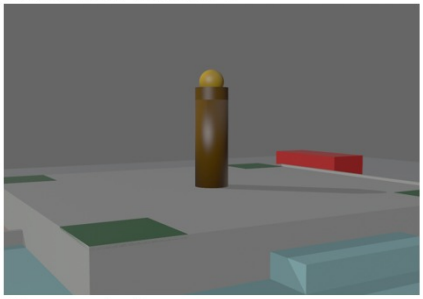
009_tilt_level.png
level h=1.50m
cam XYZ=(3.90, 2.25, 1.50) m



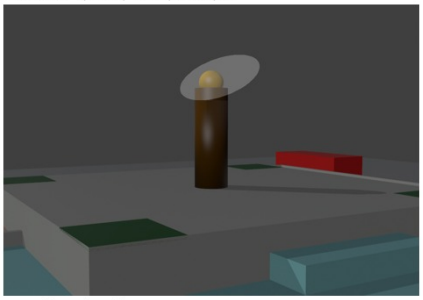
010_blur_level.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



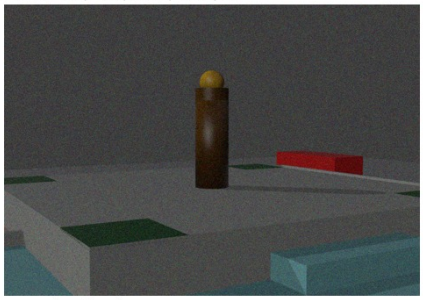
010_bright_level.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



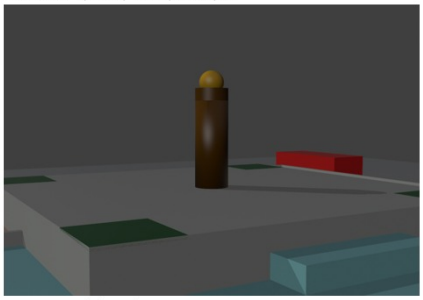
010_gamma_level.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



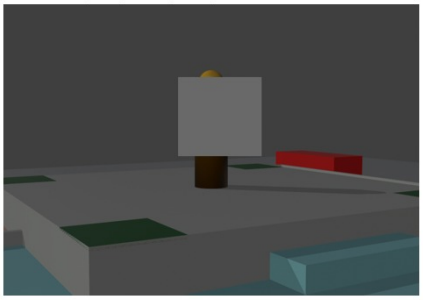
010_glare_level.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



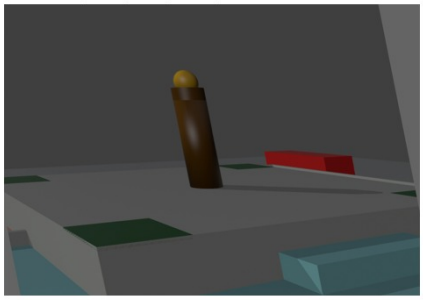
010_noise_level.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



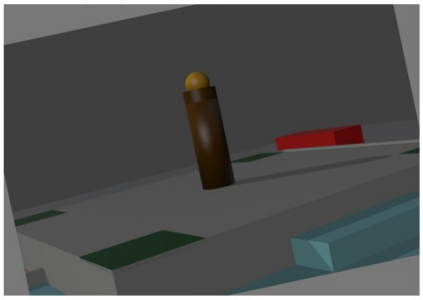
010_normal_level.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



010_occlusion_level.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



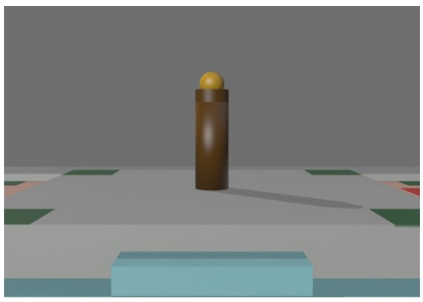
010_side_crop_level.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



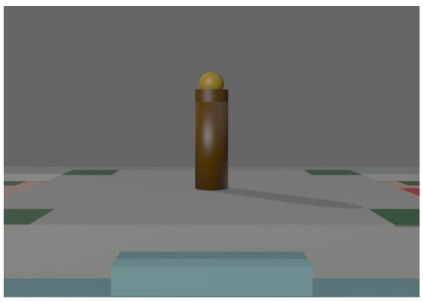
010_tilt_level.png
level h=1.50m
cam XYZ=(2.25, 3.90, 1.50) m



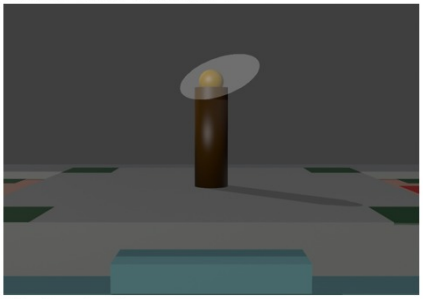
011_blur_level.png
level h=1.50m
cam XYZ=(0.00, 4.50, 1.50) m



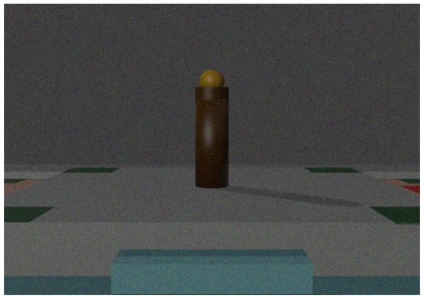
011_bright_level.png
level h=1.50m
cam XYZ=(0.00, 4.50, 1.50) m



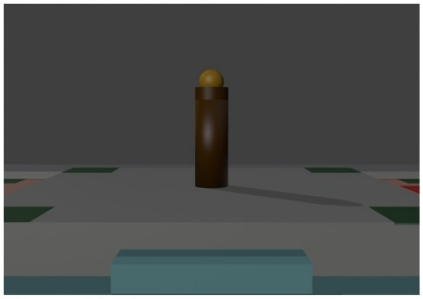
011_gamma_level.png
level h=1.50m
cam XYZ=(0.00, 4.50, 1.50) m



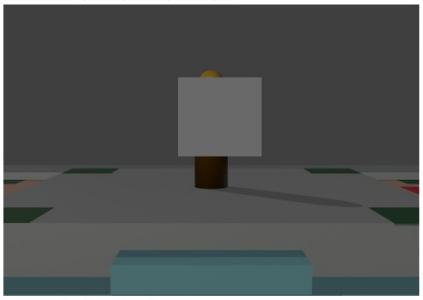
011_glare_level.png
level h=1.50m
cam XYZ=(0.00, 4.50, 1.50) m



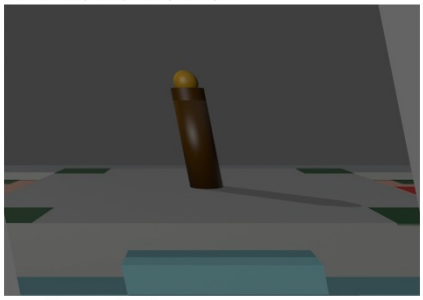
011_noise_level.png
level h=1.50m
cam XYZ=(0.00, 4.50, 1.50) m



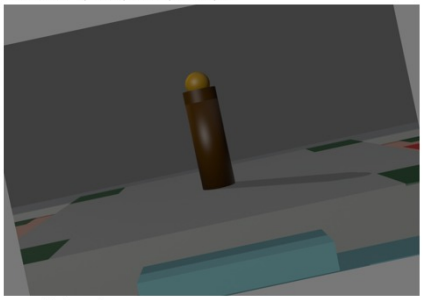
011_normal_level.png
level h=1.50m
cam XYZ=(0.00, 4.50, 1.50) m



011_occlusion_level.png
level h=1.50m
cam XYZ=(0.00, 4.50, 1.50) m



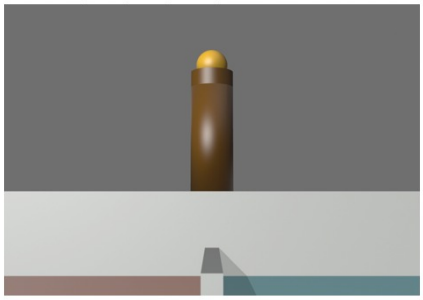
011_side_crop_level.png
level h=1.50m
cam XYZ=(0.00, 4.50, 1.50) m



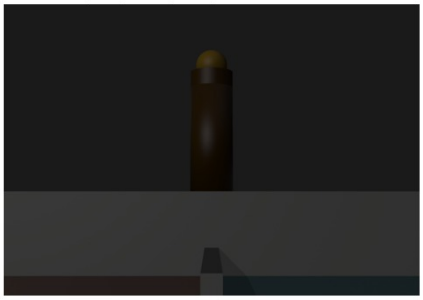
011_tilt_level.png
level h=1.50m
cam XYZ=(0.00, 4.50, 1.50) m



012_blur_upward.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m

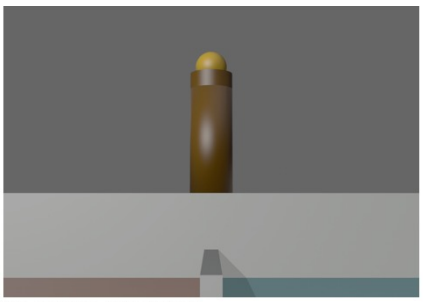


012_bright_upward.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m

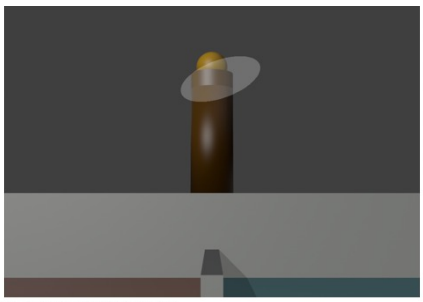


012_dark_upward.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m

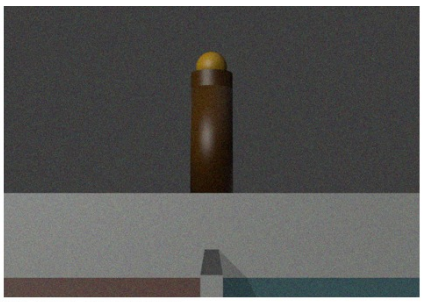
鲁棒集：192 张，第 11 页



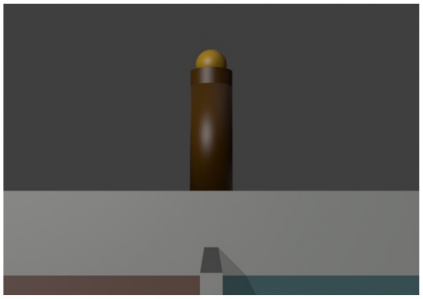
012_gamma_upward.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m



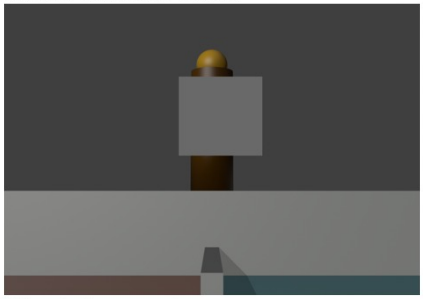
012_glare_upward.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m



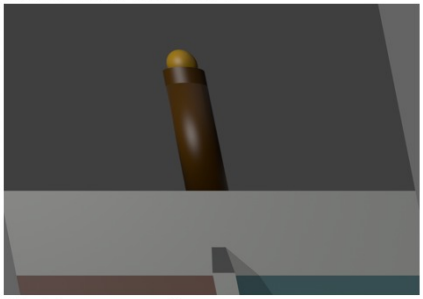
012_noise_upward.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m



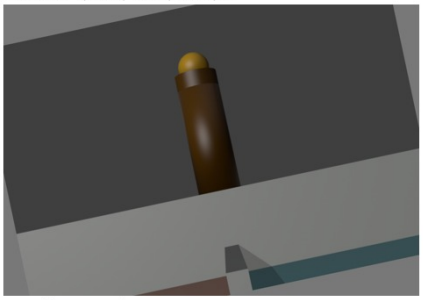
012_normal_upward.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m



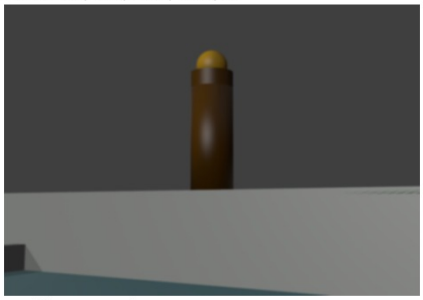
012_occlusion_upward.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m



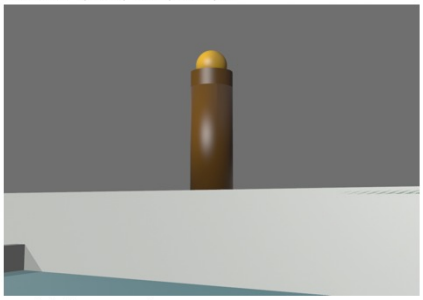
012_side_crop_upward.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m



012_tilt_upward.png
upward h=0.85m
cam XYZ=(3.50, 0.00, 0.85) m



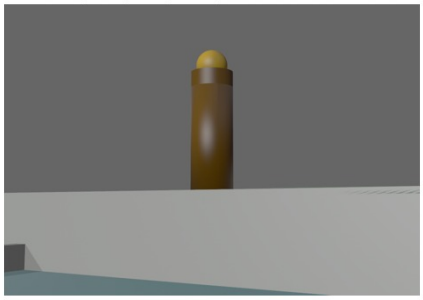
013_blur_upward.png
upward h=0.85m
cam XYZ=(3.03, 1.75, 0.85) m



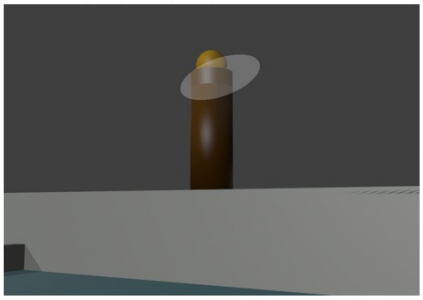
013_bright_upward.png
upward h=0.85m
cam XYZ=(3.03, 1.75, 0.85) m



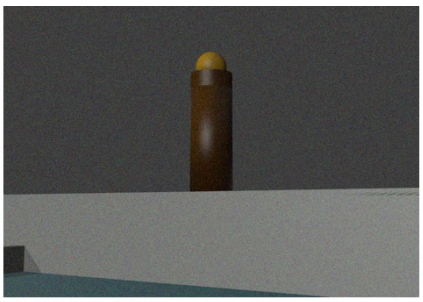
013_dark_upward.png
upward h=0.85m
cam XYZ=(3.03, 1.75, 0.85) m



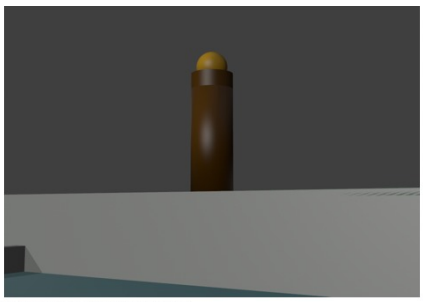
013_gamma_upward.png
upward h=0.85m
cam XYZ=(3.03, 1.75, 0.85) m



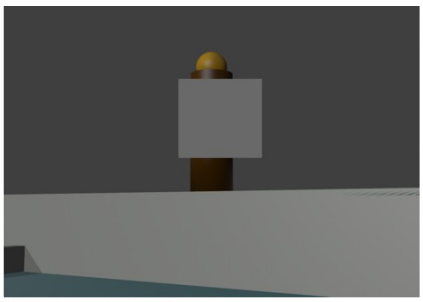
013_glare_upward.png
upward h=0.85m
cam XYZ=(3.03, 1.75, 0.85) m



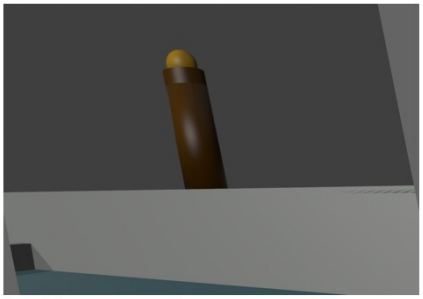
013_noise_upward.png
upward h=0.85m
cam XYZ=(3.03, 1.75, 0.85) m



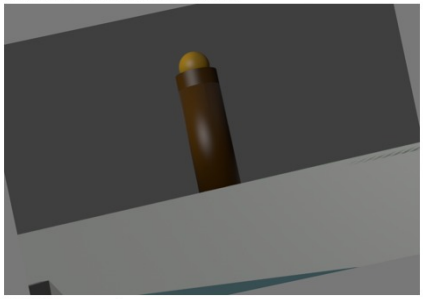
013_normal_upward.png
upward h=0.85m
cam XYZ=(3.03, 1.75, 0.85) m



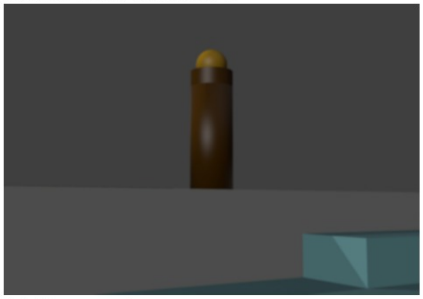
013_occlusion_upward.png
upward h=0.85m
cam XYZ=(3.03, 1.75, 0.85) m



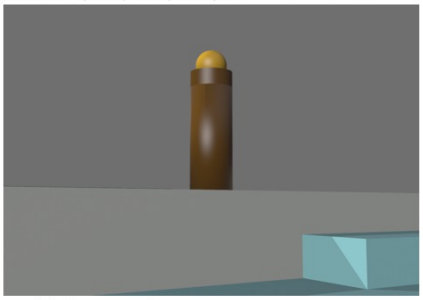
013_side_crop_upward.png
upward h=0.85m
cam XYZ=(3.03, 1.75, 0.85) m



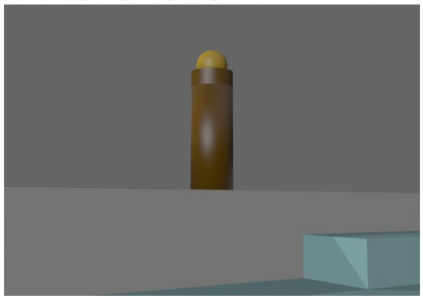
013_tilt_upward.png
upward h=0.85m
cam XYZ=(3.03, 1.75, 0.85) m



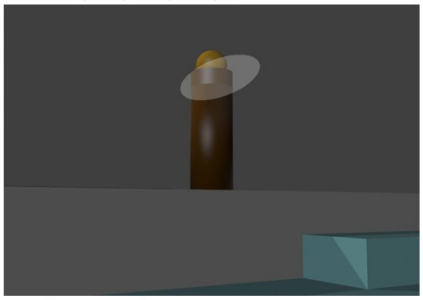
014_blur_upward.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



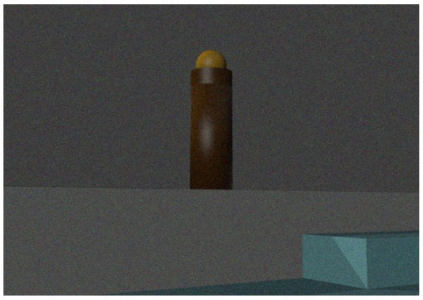
014_bright_upward.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



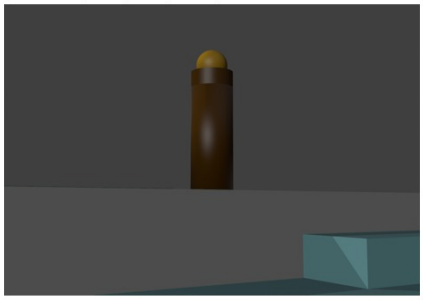
014_gamma_upward.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



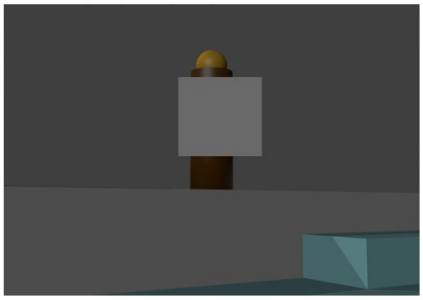
014_glare_upward.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



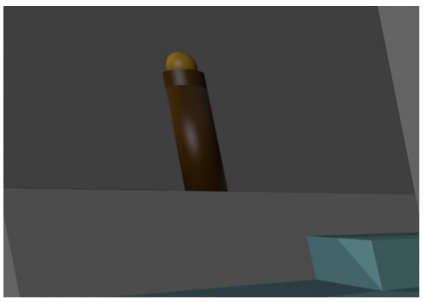
014_noise_upward.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



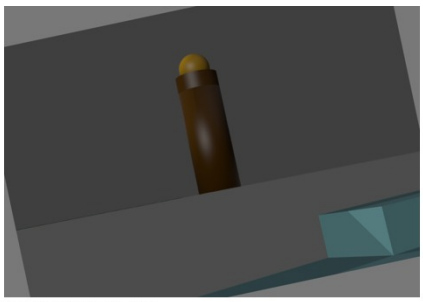
014_normal_upward.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



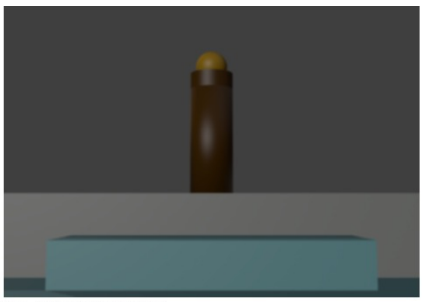
014_occlusion_upward.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



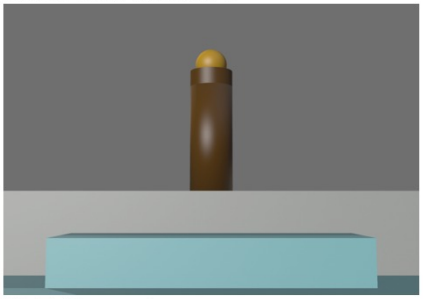
014_side_crop_upward.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



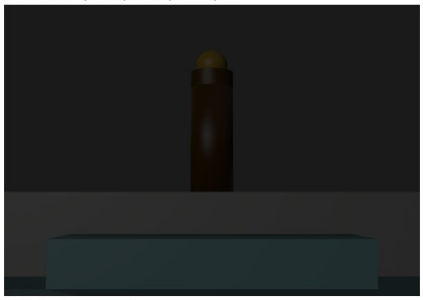
014_tilt_upward.png
upward h=0.85m
cam XYZ=(1.75, 3.03, 0.85) m



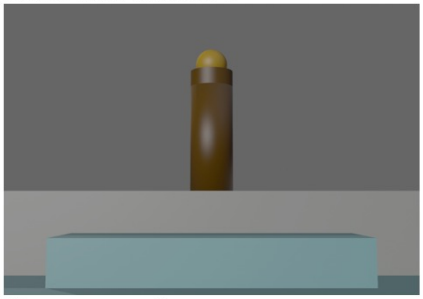
015_blur_upward.png
upward h=0.85m
cam XYZ=(0.00, 3.50, 0.85) m



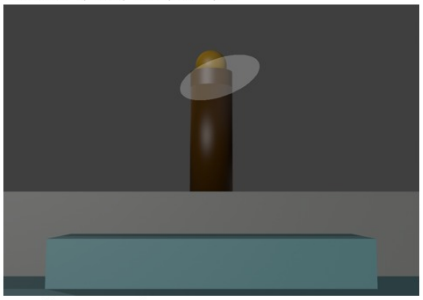
015_bright_upward.png
upward h=0.85m
cam XYZ=(0.00, 3.50, 0.85) m



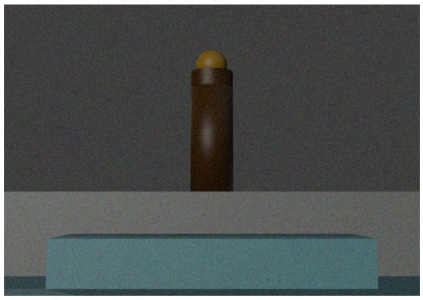
015_dark_upward.png
upward h=0.85m
cam XYZ=(0.00, 3.50, 0.85) m



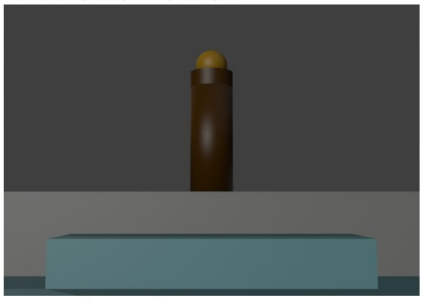
015_gamma_upward.png
upward h=0.85m
cam XYZ=(0.00, 3.50, 0.85) m



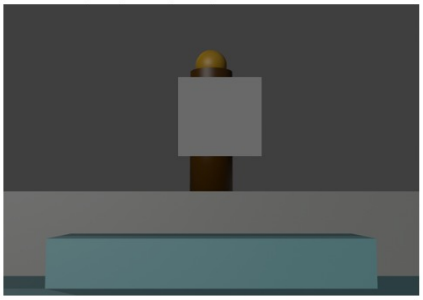
015_glare_upward.png
upward h=0.85m
cam XYZ=(0.00, 3.50, 0.85) m



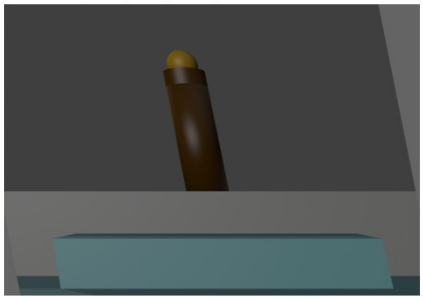
015_noise_upward.png
upward h=0.85m
cam XYZ=(0.00, 3.50, 0.85) m



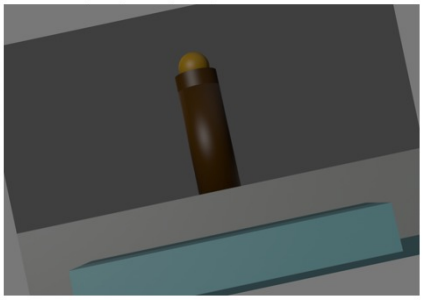
015_normal_upward.png
upward h=0.85m
cam XYZ=(0.00, 3.50, 0.85) m



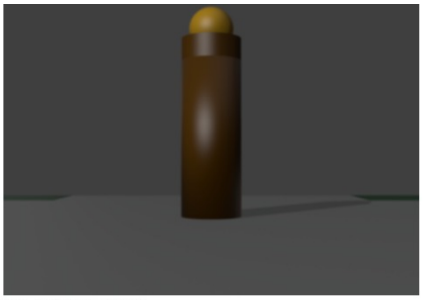
015_occlusion_upward.png
upward h=0.85m
cam XYZ=(0.00, 3.50, 0.85) m



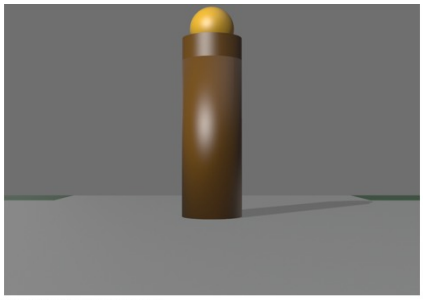
015_side_crop_upward.png
upward h=0.85m
cam XYZ=(0.00, 3.50, 0.85) m



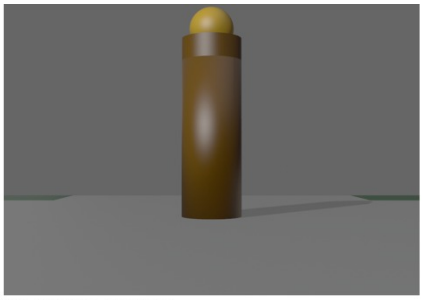
015_tilt_upward.png
upward h=0.85m
cam XYZ=(0.00, 3.50, 0.85) m



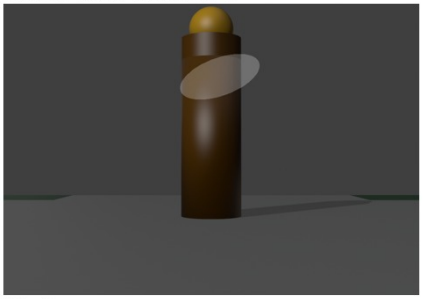
016_blur_near.png
near h=1.15m
cam XYZ=(2.50, 0.00, 1.15) m



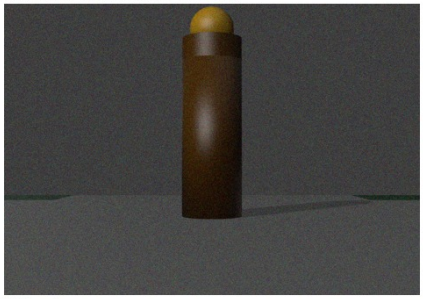
016_bright_near.png
near h=1.15m
cam XYZ=(2.50, 0.00, 1.15) m



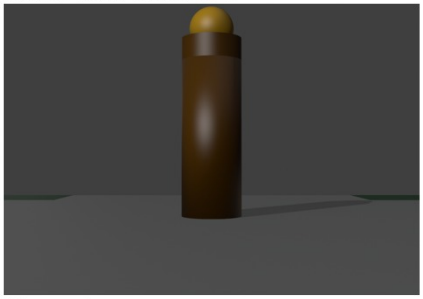
016_gamma_near.png
near h=1.15m
cam XYZ=(2.50, 0.00, 1.15) m



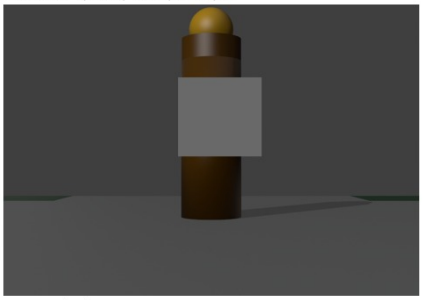
016_glare_near.png
near h=1.15m
cam XYZ=(2.50, 0.00, 1.15) m



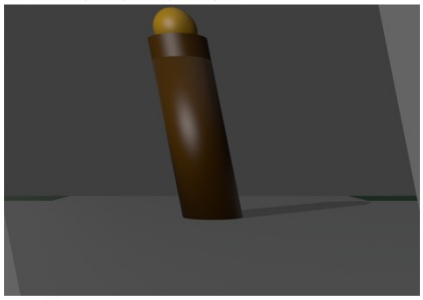
016_noise_near.png
near h=1.15m
cam XYZ=(2.50, 0.00, 1.15) m



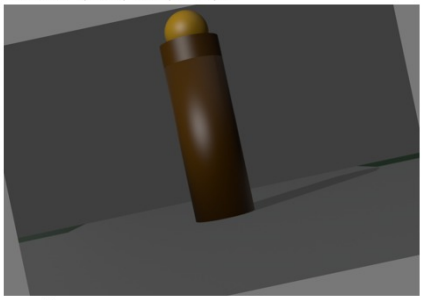
016_normal_near.png
near h=1.15m
cam XYZ=(2.50, 0.00, 1.15) m



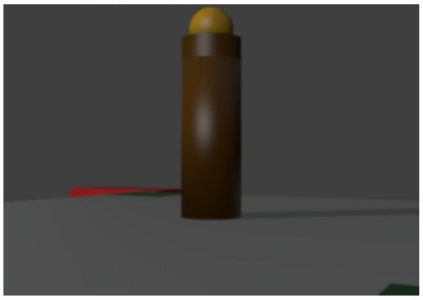
016_occlusion_near.png
near h=1.15m
cam XYZ=(2.50, 0.00, 1.15) m



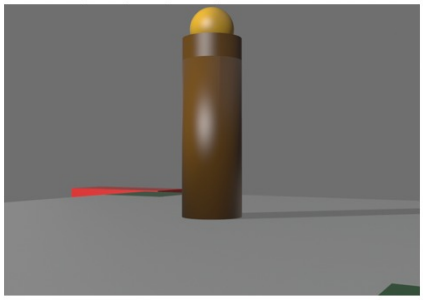
016_side_crop_near.png
near h=1.15m
cam XYZ=(2.50, 0.00, 1.15) m



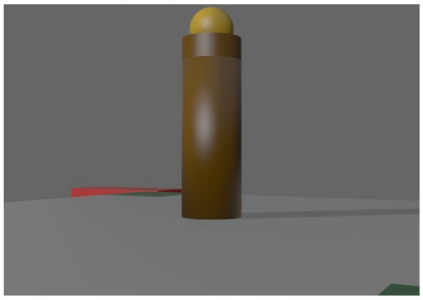
016_tilt_near.png
near h=1.15m
cam XYZ=(2.50, 0.00, 1.15) m



017_blur_near.png
near h=1.15m
cam XYZ=(2.17, 1.25, 1.15) m

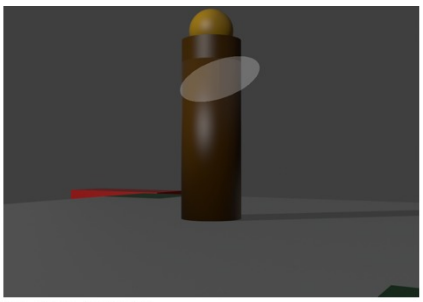


017_bright_near.png
near h=1.15m
cam XYZ=(2.17, 1.25, 1.15) m

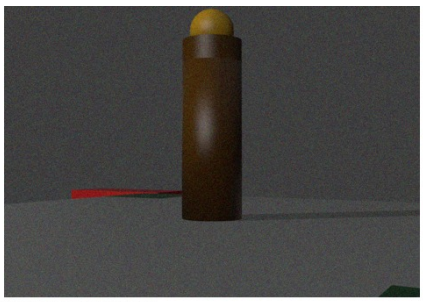


017_gamma_near.png
near h=1.15m
cam XYZ=(2.17, 1.25, 1.15) m

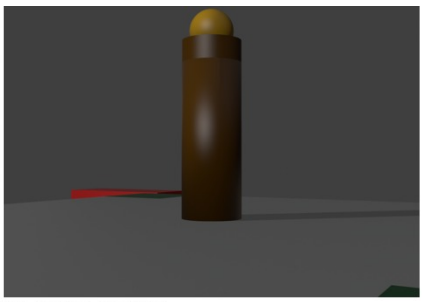
鲁棒集：192 张，第 15 页



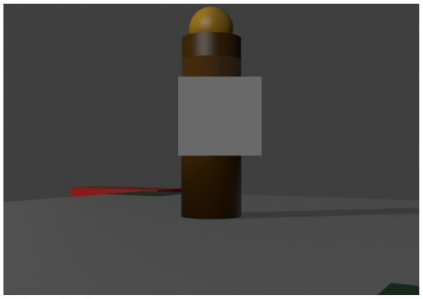
017_glare_near.png
near h=1.15m
cam XYZ=(2.17, 1.25, 1.15) m



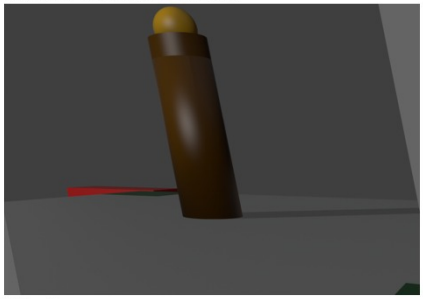
017_noise_near.png
near h=1.15m
cam XYZ=(2.17, 1.25, 1.15) m



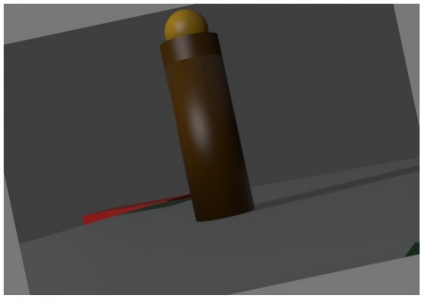
017_normal_near.png
near h=1.15m
cam XYZ=(2.17, 1.25, 1.15) m



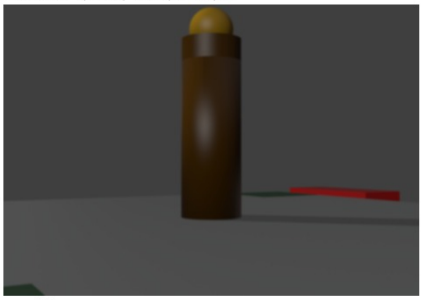
017_occlusion_near.png
near h=1.15m
cam XYZ=(2.17, 1.25, 1.15) m



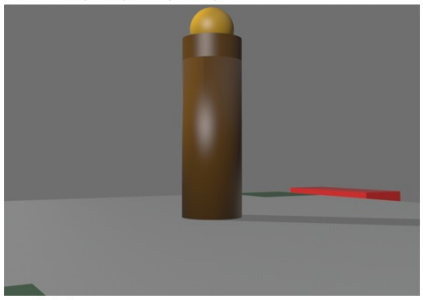
017_side_crop_near.png
near h=1.15m
cam XYZ=(2.17, 1.25, 1.15) m



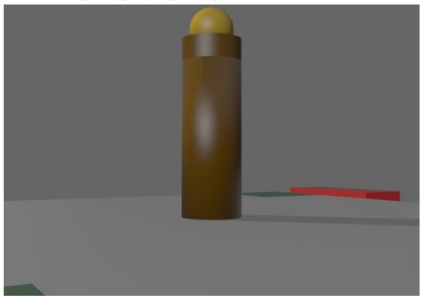
017_tilt_near.png
near h=1.15m
cam XYZ=(2.17, 1.25, 1.15) m



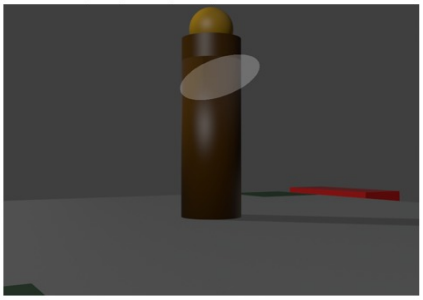
018_blur_near.png
near h=1.15m
cam XYZ=(1.25, 2.17, 1.15) m



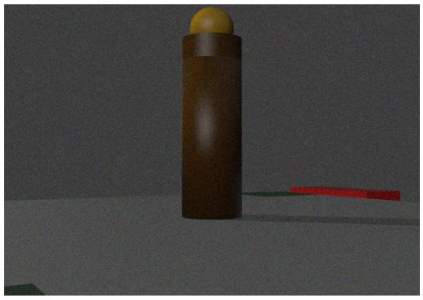
018_bright_near.png
near h=1.15m
cam XYZ=(1.25, 2.17, 1.15) m



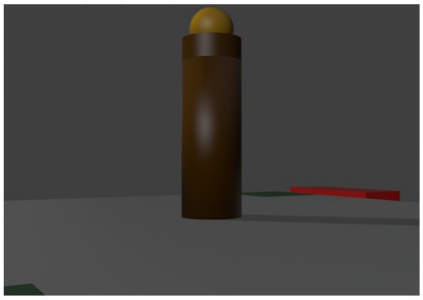
018_gamma_near.png
near h=1.15m
cam XYZ=(1.25, 2.17, 1.15) m



018_glare_near.png
near h=1.15m
cam XYZ=(1.25, 2.17, 1.15) m

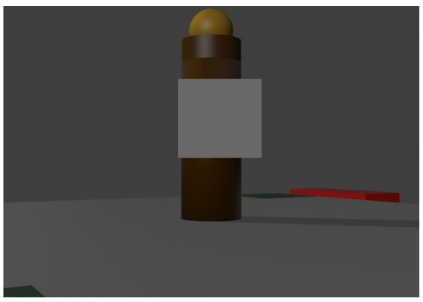


018_noise_near.png
near h=1.15m
cam XYZ=(1.25, 2.17, 1.15) m

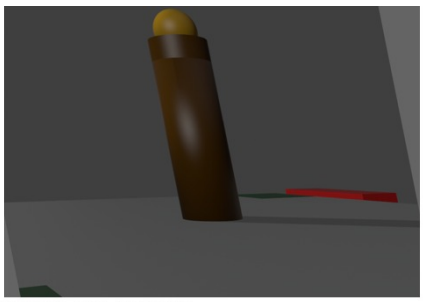


018_normal_near.png
near h=1.15m
cam XYZ=(1.25, 2.17, 1.15) m

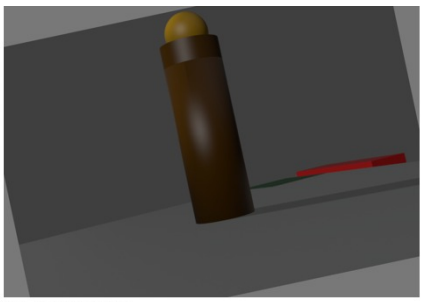
鲁棒集：192 张，第 16 页



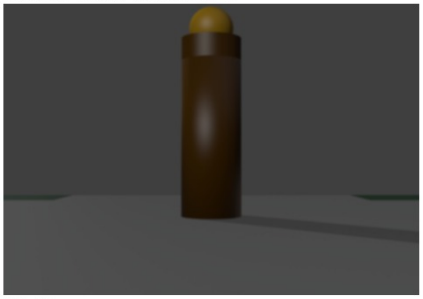
018_occlusion_near.png
near h=1.15m
cam XYZ=(1.25, 2.17, 1.15) m



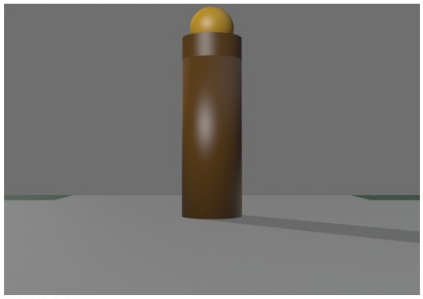
018_side_crop_near.png
near h=1.15m
cam XYZ=(1.25, 2.17, 1.15) m



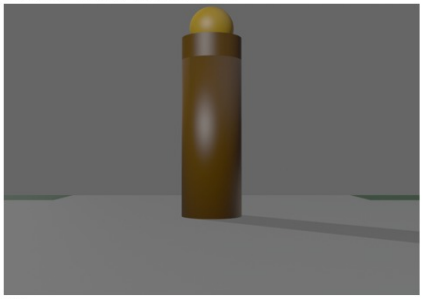
018_tilt_near.png
near h=1.15m
cam XYZ=(1.25, 2.17, 1.15) m



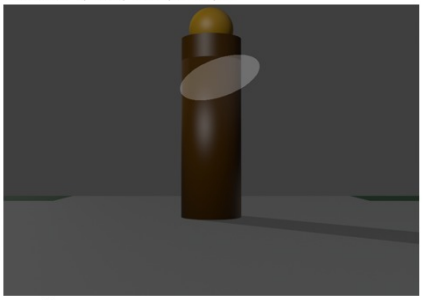
019_blur_near.png
near h=1.15m
cam XYZ=(0.00, 2.50, 1.15) m



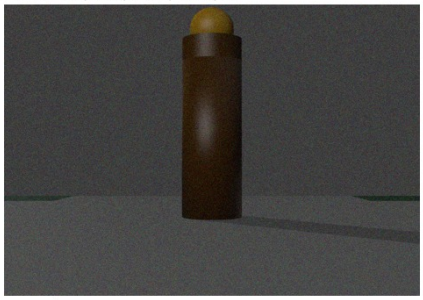
019_bright_near.png
near h=1.15m
cam XYZ=(0.00, 2.50, 1.15) m



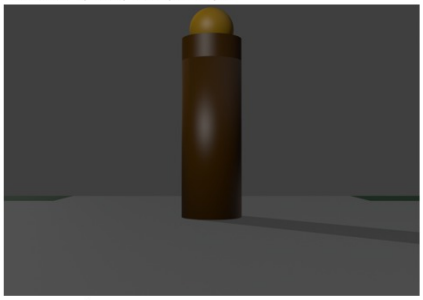
019_gamma_near.png
near h=1.15m
cam XYZ=(0.00, 2.50, 1.15) m



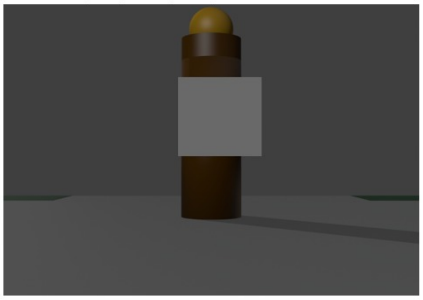
019_glare_near.png
near h=1.15m
cam XYZ=(0.00, 2.50, 1.15) m



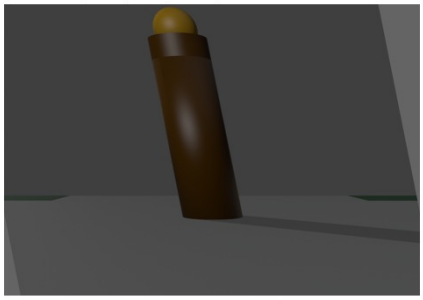
019_noise_near.png
near h=1.15m
cam XYZ=(0.00, 2.50, 1.15) m



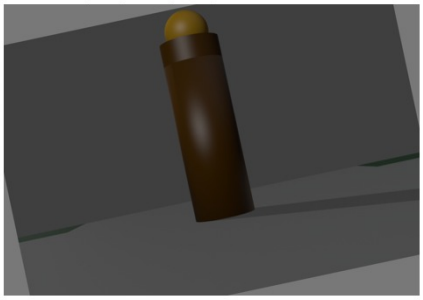
019_normal_near.png
near h=1.15m
cam XYZ=(0.00, 2.50, 1.15) m



019_occlusion_near.png
near h=1.15m
cam XYZ=(0.00, 2.50, 1.15) m



019_side_crop_near.png
near h=1.15m
cam XYZ=(0.00, 2.50, 1.15) m



019_tilt_near.png
near h=1.15m
cam XYZ=(0.00, 2.50, 1.15) m

ROS 2 功能包：ros2_ws/src/nuwa_vision。输入彩色图、对齐深度和 CameraInfo；输出 vision/ball_pose、vision/column_xy_center、vision/ball_column_state、球/柱/深度 ROI 掩膜及 debug。节点会检查 30 ms 时间戳差、相同图像坐标系和 RGB/深度尺寸。

本次 Blender 深度图由场景射线求交生成，分辨率 320×224；RGB 1000×700。文件评测仅在纵横比匹配时最近邻放大深度。实机须使用 D435i 的对齐深度流；仿真没有传感器噪声、反光空洞和深度边缘飞点，实机精度需另行验证。